

SPECTRE

Passive Signal Intelligence for Android



Receive-only observation of publicly broadcast Wi-Fi and Bluetooth LE.
Offline-first. No cloud. No account. No telemetry.
Pattern matching, visualization, alerts, and local logging.
Experimental. Use at your own risk. Not for safety-critical use.

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Notice — safety & disclaimer

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There is no guarantee that trackers, cameras, tags, access points, or any other device will be found, named, or reported. Radios that are off, cellular-only, asleep, randomized, quiet, or outside what this handset's OS exposes will not appear. Each phone has its own radios, firmware, scan quotas, and OEM battery policies, and software cannot address those limits.

Pattern matches, GPS co-travel ("Moving with you" / "possible tail"), Debrief language, and AI Export output are hypotheses — not identity, not a legal finding, and not a complete RF capture. You are solely responsible for how you use this app and this document, and for complying with local law. By using the software or this manual you accept these terms and the SPECTRE LICENSE.

Location data, if tagging is on, is this phone at hear-time — not the other radio. There is no Spectre server; stamps stay on the handset until you share them. Logs keep full coordinates even when Privacy mode masks the screen and sit reports. Debrief, Share log, AI Export (sit or one radio), and radio-detail Share as text can take that path off the phone. A TAK / CoT feed, if you turn it on, sends markers (full MAC and coordinates) onto the LAN you configured; Privacy mode pauses that feed. Online place names use the system geocoder (often the OEM / Google network), not a Spectre cloud. How you store, share, or publish those files is your responsibility.

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How this book is organized

Read chapters 4 and 5 first. Chapter 4 covers install, permissions, and the max-collection checklist (§4.5); chapter 5 maps the screens. Chapter 8 is Filters — which radios the Live display shows, including Moving with you (§8.5). Chapter 9 is Signatures — which patterns are labeled. Chapter 12 is playbooks for a question you can actually ask. The rest is reference: how the radios work, logging, field hygiene, and the stock catalog.

If you want...	Go to
Install and first five minutes	Chapter 4
Hear as often as the phone allows (battery is the cost)	§4.5
What each tab and button is	Chapter 5 (screenshots of the Live display, Filters, Signatures, Settings, detail)
Fewer radios on the Live display, not quieter rows	Chapter 8, then Live display → Display in §5.3
Who is walking or driving with me	§8.5 (how the test works) and §12.2 (the playbook)
Turn a pattern family on or off	Chapter 9
Read cleartext BLE bytes (temp, Remote ID, ...)	§9.6 Decode fields
Build my own decode map from a payload spec	§9.6.1–§9.6.5
Overlay radios on ATAK / TAK (CoT feed)	§5.8 (configure) and §12.15 (sit)
Followed / new arrival / hunt / cameras	Chapter 12
A word you do not recognize	Appendix glossary

Screenshots. Phone figures in this book were captured on a Galaxy A54 with **Settings → Privacy mode** on. MAC tails are ****:*:*** so this PDF does not publish full addresses, and GPS last-fix on detail is masked. Logs on the phone are unchanged. Turn Privacy mode off when you need the full MAC or coordinates on screen.

1. Introduction & Background

1.1 Purpose of Spectre

I built Spectre as a personal tool to look at what Wi-Fi access points and Bluetooth LE ads my phone was able to pick up, so I could better understand what devices were being used around me. It is passive: it only listens. There is no dongle, no account, and no backend server. I wanted something that would work offline in the field.

I wanted a modern interface that was easy to use; a display flexible enough that I could customize a view for the job at hand; a filtering engine so I do not have to look at everything; an extensible signature library so I can identify as many radio sources as possible and add new ones on the fly; and reports of what was seen.

I have been using it and iterating on it for a while, and it has been useful enough that I thought I would share it.

This is a hobby — something I do for fun in my spare time. There is no Spectre backend. There are no ads. Everything lives on the phone. You install an APK (sideload).

If you spot an error, something stupid, or have a feature idea — in the App or the documentation — please open a GitHub issue. This is how we make it better. I hope you find it as useful as I have. I look forward to hearing how it goes.

You do not need radio-school jargon to run it. This manual uses the app's own words and spells them out. A section number (§5.5, §12.8) jumps to more detail. Read chapters 4 and 5 first; use chapter 12 when you have a question in the field.

The Notice after the cover is not optional. Spectre is experimental, it misses radios, and it is not a safety system.

What you will see — and what you will not

On the Live display, the glanceable mark on each row is the **circle with a class glyph** — Finder tags, Phones / PCs, Audio, Home IoT, and so on. Unmatched radios show a question mark. The same glyphs sit on Filters class chips and on By class headers. There are still two radio kinds. On the Live display they are a small **Wi-Fi** or **Bluetooth** icon at the start of the subtitle (the small second line), not two-letter AP/LE tags and not the row's main mark:

- **AP (Wi-Fi access point).** A radio that is beaconing a network name (SSID): a home router, a café hotspot, a mesh node, a phone in hotspot mode, some cameras and IoT gadgets that advertise Wi-Fi. A laptop or phone that is only *joined* to someone else's Wi-Fi does **not** appear. Stock Android cannot show those clients.
- **LE (Bluetooth Low Energy advertiser).** A gadget that is broadcasting BLE packets: headphones, a phone, an AirTag-class tracker, a speaker, a watch, a lot of unnamed BLE rows. Spectre listens; it does not pair or connect for detection.

Display → Subtitle None hides that second line (and the radio-kind icon with it) so more rows fit; the class glyph and the title stay. Detail spells the kind in words (Wi-Fi access point / BLE advertiser). The scan notification still says “N Wi-Fi · N BLE · N signatures.” §5.3, §5.4.

The number on the right of a list row is **RSSI** in dBm — how loud that radio is *at this phone*, not a tape measure. Closer to 0 is louder (for example −40 is much louder here than −90). Walls, your body, a car, and the gadget's own transmit power all move that number. Hunt (§5.5, §12.8, §12.13) uses changes in loudness, not meters.

A few words used throughout

- **Class glyph.** The circle on a Live display row. First matching signature class, or ? if unmatched. That is how you scan the list. §5.4, §9.5.
- **AP / LE.** The two radio kinds. AP = Wi-Fi access point (beaconing a network, not a client). LE = BLE advertiser. On the Live display they are the small Wi-Fi / Bluetooth icons on the subtitle, not two-letter tags and not the circle. Subtitle None drops them with that line. The two bullets above are the full meaning.
- **MAC.** The radio address. Default first line of a Live display row (Display → Title). You can move it to the subtitle, or hide the second line. BLE phones often randomize it, so a “new” MAC can be the same physical gadget with a new name tag.
- **Signature.** A named pattern Spectre looks for (AirTag, Flock, Tile, ...). A chip on a row means the pattern hit — not that you have identified a person or a serial number. Signatures always label; Filters decide what the Live display shows.
- **Filter vs log.** Filters only change what the Live display shows. The rotating log, if logging is on, still writes what was heard. Hiding AirTags on the Live display does not delete them from the file.
- **Debrief vs log.** Debrief is a sit report of radios still in memory (last 15 minutes, about 400). Share log is the file. On a drive, tap Debrief more than once — roadside clutter falls out of RAM; radios moving with you stay. §11.4.1.
- **Stale / gone.** No packet for longer than Settings → Stale after (and Display → Brief hold). The row can stay on screen dimmed. It does not mean the gadget is powered off forever.
- **Watch / bookmark.** A pip and/or spoken phrase when that signature family or that one MAC appears. Voice can say the class, the signature name, or both (Settings → What to say). Detail bookmark is one radio; it does not auto-delete when the radio leaves (§5.5, §10.1–10.2.1).

1.2 Design philosophy

- **Passive only.** Spectre listens. It does not deauthenticate, probe inject, pair, connect, or join a network as part of detection.
- **Offline-first.** Configuration, signature definitions, watchlists, and logs live in the app private directory. There is no account and no backend.
- **Privacy-focused.** Backup is disabled in the manifest. No analytics SDK is included.
- **Named patterns, not identity.** A match is a pattern hit (OUI, name, UUID, manufacturer data, or co-occurrence). It is not proof of a person, a vehicle, or a specific unit serial.
- **You can change the defaults.** First launch writes a full signature catalog. Every signature and preset can be changed or deleted afterward.
- **You can change how the list looks.** the Live display is not a fixed dump of every field. Display (the tune icon on the Live display) lets you show as much or as little of each radio as you need. Filters hide radios; Display hides fields on the row. Two people on the same sit can run two different lists. The detail page still has the full decode.

1.3 Intended use cases

Use case	What Spectre is for
Privacy awareness	See which advertised Wi-Fi and BLE sources are around you and whether any match known tracker or camera patterns.
Counter-surveillance hygiene	Notice when a named signature appears, stays, or comes back after you move. Confirm with RSSI trend and presence over time, not a single packet. After a walk with GPS tagging, Debrief's tracking section still covers the last 15 minutes, even if you change Live display view or Filters.
Field observation	Walk or sit a location, watch live strength, and export a timestamped log for later review.
Signature pattern recognition	Define a signature from an observed radio and reuse it: OUI family, SSID glob, BLE UUID, or manufacturer payload.
After-action review	Use timeline, device history, and CSV/JSONL export to reconstruct when an emitter appeared and faded.
Field playbooks	Chapter 12: followed on a walk, new arrival in a room, Hunt a BLE radio / vehicle sweep, one signature family, hide clutter, own kit vs a candidate tag, camera/ALPR sit, Extra attention (card-reader / pentest kit), then Debrief / AI Export / Share log.

Not a weapon system. Spectre is an observer. It cannot disable cameras, unlock trackers, or identify people. Treat matches as hypotheses that need corroboration (sightline, second radio, or later revisit).

1.4 Status of this software

Experimental — not for safety-critical use. Experimental software. Use at your own risk. Spectre does not guarantee that trackers, cameras, tags, or any other device will be found. Each phone has radio and OEM limits that this app cannot overcome. Do not use Spectre in any situation where safety is in question, including real-life threatening situations.

The full terms are on the Notice page immediately after the cover. The intended uses in §1.3 are experimental observation and after-action review, not personal protection or emergency decisions.

2. Research & Design Decisions

2.1 Research inputs

The default signature catalog and matching rules are compiled from public sources, not from proprietary vendor documentation. Treat every component-vendor OUI as a guess — the same chip shows up in many products.

Source class	What it contributed
IEEE MA-L registry	B4:1E:52 is registered to Flock Safety (9 May 2024). This is the only high-confidence Flock-assigned OUI in the default set.
Independent ALPR / DeFlock research	Field OUI lists for LiteOn camera radios and Silicon Labs battery packs commonly seen on Flock hardware; SSID patterns Flock-XXXXXX, FS Ext Battery, Penguin, Pigvision.
GainSec / firmware write-ups	Raven BLE service UUID range 0x3100–0x3500 and manufacturer ID 0x09C8 (XUNTONG) as stronger digital fingerprints than OUI alone.
BLE tracker conventions	Apple 0x004C Offline Finding 0x12; Samsung SmartTag FD5A / 0x0075; Tile 0x00C7 and FEED/FEDD. Chipolo and Pebblebee/moto tag are name-only Find Hub locators (stock rows, on).
Android platform docs	WifiManager scan throttling, BLE ScanSettings, permission model (location, NEARBY_WIFI_DEVICES, BLUETOOTH_SCAN), foreground-service types.

2.2 Why these detection methods

Stock Android exposes two public, legal, receive-only surfaces that a normal app can use without root:

- **Wi-Fi scan results** — BSSID, SSID (or hidden), RSSI, frequency/channel, capabilities, and on recent APIs a Wi-Fi standard / channel-width hint. Every result is an access-point-class beacon (infrastructure AP, hotspot, mesh node, IoT soft-AP). Associated clients are not in this list.
- **BLE advertisements** — address, local name, RSSI, service UUIDs, manufacturer-specific data, raw advertisement bytes, PHY/TX metadata when the stack provides them.

Those are the only two collection surfaces. Spectre does not run Bluetooth Classic inquiry (`startDiscovery()`). That API transmits inquiry packets and is a different radio from BLE advertisements. Modules that only answer Classic (HC-05 / HC-06 and similar) will not appear. §3.5, §7.2.

Those fields are enough to implement a named-pattern engine, and not enough for direction finding, station-side Wi-Fi sniffing, or decryption. The engine therefore matches on OUI / MAC prefix, name or glob, 16-bit or 128-bit service UUID, manufacturer company ID, manufacturer data prefix, radio kind, hidden SSID, and optional multi-device co-occurrence (shared OUI and/or sequential MAC tails).

2.3 Why these visualizations

Five views share the same filtered set of radios, so you can change how things look without changing what was collected. You can also change each list row: Display (the tune icon on the Live display) is where you pick identity, order, and extra facts. Filters decide *which radios* appear on the Live display; Display decides *how much of each radio* you see. Logging is not affected.

- **Classic radar** is a polar plot of RSSI, not a PPI with real azimuth. It is the fastest way to see “how many, how close, which signature.” Angle is a stable hash of the MAC so the same radio stays in the same slice. The bright sweep line leads; the shaded fan trails behind it. Radar labels are not Title/Subtitle; they stay short signature / name guesses.

- **Strength list** is the working view for triage. Default row is MAC over Name + type, RSSI on the right, optional bars and signature chips. Title line, Subtitle line (including None), Sort, Brief hold, Frequency under RSSI, first/last, and the extra switches all live on Display. Vendor is on detail, not the list. Show as little as a one-line name, or as much as name + MAC + bar + sparkline + ages.
- **Timeline** answers “when did it appear and leave” over a 15-minute presence strip. A continuous on-air radio grows one solid bar; a real dropout (~45 s) then a return starts a new segment. Same Title/Subtitle as the list.
- **Hybrid** keeps the ranked list and adds a full-width RSSI sparkline on a fixed -30 to -100 dBm grid (packet order, newest on the right) plus the same trend mark.
- **By class** is an outline of the same filtered set: every class A–Z by name, then signatures A–Z, then radios. Show all (default) keeps empty classes so the list does not jump; Collapse empty hides zeros. Those chips sit at the top of the list and scroll away with it. Tap a class, then a signature, then a radio for detail. Unmatched is last. Radio rows use the same Display switches as the strength list. A radio that matched two classes is counted in both; the header still says how many unique radios are on the Live display.

2.4 Trade-offs: Android phone vs dedicated sniffer hardware

Dimension	Android handset (Spectre)	Typical ESP32 sniffer
Radios	Independent Wi-Fi (2.4/5/6 GHz) and BLE 5.x; both can run at once.	Usually one 2.4 GHz radio; Wi-Fi and BLE must time-slice.
Wi-Fi visibility	Access points only. No promiscuous STA / probe capture on stock Android.	Promiscuous mode can see stations, probes, and hidden-SSID clients.
Scan rate	OS-throttled. Foreground + location helps; background is limited.	Channel hop under operator control.
BLE	Low-latency observer, extended ads, manufacturer + service data.	Observer possible; less RAM for history.
Persistence	Large local store, export via share sheet, notifications.	microSD / LittleFS; limited UI.
Direction	None (no AoA API).	None unless extra hardware is added.
Legal / TOS surface	Uses public scan APIs; still subject to local law.	Same receive-only constraint if the firmware does not transmit.

Spectre takes the phone’s advantages — dual radios, 5 GHz APs, notifications, unlimited rule counts, history — and does not pretend to have the ESP32’s promiscuous Wi-Fi view. If an investigation needs station-side frames, use a dedicated sniffer in parallel and treat Spectre as the BLE + AP overlay.

3. Limitations

3.1 Android platform restrictions

- **Wi-Fi scan throttle.** Android rate-limits `WifiManager.startScan()` to about four scans per two minutes. Spectre therefore spaces requests at about 30 / 40 / 55 seconds (high / balanced / saver) and tracks the quota so it does not burst then black out. Wi-Fi is always a batch-then-wait radio — there is no continuous AP stream on stock Android. Settings → Faster Wi-Fi AP scans can drop that to about 8 s, but only after Developer options → Wi-Fi scan throttling is off and Spectre has read that OS switch (Android 11+). Until then the stock cadence is the ceiling. A miss between batches is a miss for signatures: an interesting AP (Cradlepoint IBR*, AirLink OUI, UniFi, Cisco, a hidden fleet SSID) is only labeled if a scan ran while you were in range. On a drive that window is short. §7.1.1, §10.3.1.
- **Location services must be on.** Without system Location enabled, BSSIDs and often BLE addresses are hidden or randomized further. Fine location is required even though Spectre is not a mapping app.
- **NEARBY_WIFI_DEVICES and BLUETOOTH_SCAN** are required on Android 12/13+. Spectre does not set the `neverForLocation` flag, because that flag would strip the identifiers the signature engine needs.
- **OEM battery managers** (notably Samsung One UI) will freeze background radios unless the app is exempted. Settings → Allow background usage, then Unrestricted battery. Some phones (Samsung among them) do not open onto Unrestricted — tap Allow background usage to click through and select it.
- **Samsung BLE suspend.** Long low-latency unfiltered scans are parked after a few minutes. Spectre recycles the scanner on a duty cycle (~70 s on, a few seconds rest) and restarts if ads go silent for ~18 s. The subtitle may read BLE cycling or BLE parked · restarting.
- **No Wi-Fi stations.** Every Wi-Fi row is an access point (or a device beaconing like one: phone hotspot, mesh node, camera/IoT soft-AP). A phone, laptop, or camera that is only a client on someone else's network does not appear.
- **No promiscuous or monitor mode.** Stock Android does not expose an 802.11 monitor interface to ordinary apps. Spectre cannot put the Wi-Fi chip in promiscuous mode, cannot capture raw frames, cannot follow a BSSID on one channel, and cannot see management frames the OS scan does not already return. A USB sniffer or rooted/custom stack is a different tool.
- **No cellular RF.** Spectre does not scan LTE/5G. A camera that only talks on the carrier network is invisible even if you are standing under it.
- **Dense crowds.** Phones rotate BLE addresses. Spectre caps the live set at 400 radios (hard ceiling 900) and drops unnamed BLE after about three minutes so the heap does not collapse. Debrief reads that same map, so a few kilometers of city driving can already have replaced the start of the trip in RAM (§11.4.1).

Wi-Fi detections are access points. The small Wi-Fi icon at the start of a Wi-Fi row's subtitle is literal (the circle is still the class glyph). Stock Android only reports scan results from radios that are beaconing an SSID (including hidden-SSID APs, which show with an empty name and the HIDDEN flag). Spectre cannot see associated stations, wildcard probe requests, or a device that is on Wi-Fi but not advertising a network. A phone in hotspot or Wi-Fi Direct group-owner mode is an access point and will show; the same phone joined to a café network will not. Station-side work needs a dedicated sniffer.

3.2 No direction finding

The phone does not expose angle of arrival. Radar angle is a hash of the MAC address so a given radio keeps a stable seat on the plot; it is not north, not left, and not a bearing. Distance from center is RSSI, which correlates only loosely with range — walls, body attenuation, and TX power dominate. Hunt (§5.5, §12.13) uses that same RSSI as a closer/further needle and, if you hold the phone against your body and turn, a crude heading from body-block. That is not DF and not a compass.

3.3 Dependence on active broadcast

A radio that is powered down, in a sleep interval longer than the stale window, or advertising on a PHY/channel the stack is not reporting will not appear. Many Flock and similar ALPR units are cellular-first: they uplink over LTE/5G and only expose Wi-Fi during provisioning, a tech visit, or as a backup. Newer quiet installs may beacon never. Absence of a match is not absence of a camera. BLE trackers often advertise at a low duty cycle — they flicker in and out of the live set. See §7.6.1.

3.4 Battery

High performance uses BLE `SCAN_MODE_LOW_LATENCY` (recycled about every 70 s) and asks for a Wi-Fi scan about every 30 s — the fastest cadence that stays under the OS four-scans-per-two-minutes cap. Balanced is ~40 s Wi-Fi / `SCAN_MODE_BALANCED`; Battery saver is ~55 s / `SCAN_MODE_LOW_POWER`. Faster Wi-Fi AP scans (Settings), only while Developer options has Wi-Fi scan throttling off, asks about every 8 s and uses more battery and heat; BLE cadence is unchanged. Keep screen on (Settings) holds the display while Spectre is in front so Samsung does not park BLE for screen-off. The scan notification is required so Android will keep the scan service running. It is not optional decoration.

3.5 What Spectre cannot do

- Intercept, decrypt, or read payload traffic.
- Deauth, jam, pair, or otherwise act on a target.
- Identify a person, a vehicle plate, or a specific camera serial from RF alone.
- Guarantee that an OUI match is Flock hardware. Only B4:1E:52 is IEEE-assigned to Flock Safety; other prefixes are component vendors used in many products.
- See associated Wi-Fi clients, hidden-SSID stations, wildcard probe requests, or frames addressed to a Flock MAC. Wi-Fi on Spectre is AP beacons only.
- Enter promiscuous or 802.11 monitor mode, capture raw frames, or lock the phone onto one channel. The stock API is `startScan()` batches of APs.
- Hear a camera's LTE/5G backhaul. Cellular is out of band for this app.
- Hear Bluetooth Classic (BR/EDR). Spectre does not run Classic inquiry. A Classic-only serial module (HC-05 / HC-06) will not appear. Hobby BLE serial matches BLE advertisement names only. §7.2, §12.14.
- Build a true RF map with headings.
- Run without the operator granting location, nearby-devices, and notification permissions.
- Guarantee that a tracker, camera, or any other device is present or absent. A miss is normal. Do not treat Spectre as a safety system.

Experimental — not for safety-critical use. Experimental software. Use at your own risk. Spectre does not guarantee that trackers, cameras, tags, or any other device will be found. Each phone has radio and OEM limits that this app cannot overcome. Do not use Spectre in any situation where safety is in question, including real-life threatening situations.

Confidence discipline. Treat B4:1E:52 plus a Flock-* SSID or Raven UUID 0x3100–0x3500 as high confidence. Treat LiteOn / Silicon Labs / Raspberry Pi OUIs as low confidence unless a name, UUID, or manufacturer ID corroborates. Component OUIs ship in millions of unrelated devices.

4. Getting Started

This chapter is the install and first-run path. End users get `Spectre.apk` from the Spectre APK GitHub repository, then sideload it.

4.1 Installation

Spectre is not on the Play Store; you install an APK yourself (sideload). Package name `app.spectre`. Minimum Android 10 (API 29); this build targets API 35. Tested on a Galaxy A54; meant for most Android 10+ phones. The app does not self-update, so keep a copy of the APK you trust.

From the Spectre APK repository (usual path). Download `Spectre.apk` from that repo. If you downloaded it on a computer, copy it to the phone (USB, Drive, Files). On the phone, open Files or Downloads and tap the APK. If Android blocks it: Settings → Apps → Special app access → Install unknown apps (Samsung: Settings → Security and privacy → Install unknown apps) and allow the app that opened the APK (Files, Chrome, Drive, ...), then tap the APK again. Play Protect may warn that the app is not from Play — expected for a sideload. Install anyway only if you trust this file. If the repository lists a SHA-256, check it against the file you downloaded.

From a computer (adb). With USB debugging on:

```
adb install -r Spectre.apk
```

When install finishes, open Spectre from the launcher. Leave the persistent “Spectre scanning” notification — Stop on that notification ends collection.

4.2 Required permissions

Permission	Why Spectre asks
Location (fine + coarse)	Android will not return BSSIDs or useful BLE addresses without it. Tag detections with GPS (on by default) requests live GPS/network updates while scanning (last-known older than 30 s is ignored) and stamps log rows, Moving with you, and Debrief.
Nearby Wi-Fi devices (13+)	Wi-Fi scan on current Android versions.
Bluetooth scan + connect (12+)	BLE observer. Connect is required by the stack to read names on some devices; Spectre does not create ACL connections for detection.
Notifications (13+)	Persistent scan status and watchlist alerts.
Vibrate	Watchlist haptic.
Foreground service (location + connected device)	Keep both radios scanning when the UI is not in front.
Internet (install-time)	Online place names (Debrief and AI Export, system geocoder) and, if you turn it on, the TAK / CoT UDP feed (§5.8). Spectre has no account and does not call a Spectre server.
Ignore battery optimizations	Optional. Requested from Settings so OEM killers do not freeze the service.

4.3 First launch

- 1 A Disclaimer page appears once. Read it. Check **I have read this and I agree**, then Continue. Scanning does not start until you do. It does not show again unless you clear app data.
- 2 Grant the permission screen. Spectre will not start radios until the required set is complete.
- 3 A foreground notification **Spectre scanning** appears. Leave it; dismissing via Stop ends collection.
- 4 On first run the app writes `files/config.json` and loads the stock signature catalog, presets, bookmarks, and Settings listed in the Appendix. Later launches reload that file. Restore default signatures & presets writes it from the stock catalog again. Settings → Export signatures shares the catalog as JSON, without logs, GPS, or filters (§5.7, §9.3).
- 5 The Live display opens on the last view mode (default: Strength list). The header shows live counts as three small icons: Wi-Fi access points, BLE advertisers, and on-air signature matches (the same hub icon as the Signatures tab). A radio hint may follow those numbers.
- 6 Turn on system Location and Bluetooth if either is off. The Live display header and Settings show radio hints (Wi-Fi next Ns, waiting on OS, BLE cycling). Keep screen on is enabled by default. If you will leave the app: Settings → Allow background usage, then Unrestricted battery. Some phones (Samsung among them) do not open onto Unrestricted — tap Allow background usage to click through and select it.

Stay scanning. After first launch, open Settings in Spectre and turn on Allow background usage, then Unrestricted battery. Some phones do not open onto Unrestricted — tap Allow background usage to click through and select it. Keep screen on is enabled by default so BLE is not parked on screen-off while you watch; turn it off when you pocket the phone. Also disable any “Put unused apps to sleep” entry for Spectre. The full max-collection checklist (permissions, OEM battery, High performance, Faster Wi-Fi) is §4.5.

Out of the box. First launch and Restore load the stock catalog. Every row labels when its rules hit. These Settings ship on: Keep screen on, Tag detections with GPS, Online place names, Watchlist alerts, and Beep. Voice on watched signature ships off. Stock bookmarks (alert on a new match) are on the Extra attention rows: Axon, WatchGuard Video, Ray-Ban / Meta glasses, Snap Spectacles, Fieldy, Plaud Note, Hobby BLE serial, Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, Porkchop, Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc, and roadside / public camera + ALPR (Flock, Penguin, Pigvision, FS Ext Battery, Genetec AutoVu, Rekor, Motorola Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua). Privacy mode, the shade notification, and the TAK / CoT feed stay off. Chip colors follow class (§9.5).

4.4 The first five minutes

- 1 Confirm the header is counting (Wi-Fi, Bluetooth, and signatures icons with numbers). If both radio counts stay at 0, Location / Wi-Fi / Bluetooth are probably off at the system level — turn them on, wait ~30 s for the first Wi-Fi batch.
- 2 You should be on the Live display, Strength list. Each row is one radio. The circle is a class glyph (unmatched = ?) — that is the glanceable mark. AP = Wi-Fi access point and LE = BLE advertiser; they still mean those two radio kinds (§1.1). On the Live display they are a small Wi-Fi or Bluetooth icon at the start of the subtitle, not two-letter tags and not the title. Default first line is the MAC (SemiBold). Default second line is that icon, then Name + type (SSID / advertised BLE name / a type guess such as Apple · AirTag), then rand/gone. Vendor is not on the list — open detail. The number on the right is RSSI in dBm (loudness here, not meters; -50 is louder than -90).

- 3 Tap the tune / chevron (top right). That is **Display**: how the list looks for this job. View, Sort, Brief hold, Title line, Subtitle line, then switches for RSSI bars, signature names, Frequency (channel · MHz under RSSI), first/last. Close it when the list looks the way you want. In a crowded plaza, set Subtitle to None and turn the extras off — you still have every radio; you just see less of each. Filters hide radios; Display hides fields. §5.3.
- 4 Tap a row. That is device detail: identity, “What this looks like,” Signature family (whether this radio’s on-air ID is a catalog candidate), signal, decode, optional Decoded fields on BLE when that signature has a map (§9.6), bookmark (watch this MAC), Hunt on BLE, Create signature from device. Custom name is hidden on a random / privacy MAC. Back returns to the Live display.
- 5 Bottom bar, left to right: Live, Filters, Signatures, Reports, Settings. The Live tab is the Live display in this book. Filters trims *who* the Live display shows (presets, class Show only / Hide these, Show only selected / Hide selected signatures). Display trims *how much of each row* you see. Signatures is the pattern catalog (tap to edit, bookmark to beep). Reports is Debrief, AI Export, Share log, Save log, Reset / clear log. Settings is intensity, GPS tagging (on), Keep screen on (on), Online place names (on), logging options. TAK / CoT is off until you turn it on (§5.8).
- 6 Do not expect every gadget in the room to appear. Phones on café Wi-Fi without hotspot, cellular-only cameras, and sleeping tags will not. That is a phone limit, not a broken install. Chapter 12 is what to do once the list makes sense.

4.5 Max collection (battery is the cost)

§4.3 is enough to see radios. This page is the walkthrough if you want the phone to scan as often as stock Android will allow — a sit, a walk, or a drive — and you accept heat and drain. Pocket carry and overnight bag are the other way: Battery saver and §13.2. A USB pack is more useful than arguing with Adaptive battery if you need this for hours.

High performance already ships on, as do Keep screen on, GPS tagging, and logging. What this checklist adds is grant every permission, stop the OEM from parking the scan, leave the screen on while you watch, and (optional) Faster Wi-Fi AP scans for signed access points at speed. It does not lift stock Android limits — no monitor mode, no Wi-Fi clients, no Classic Bluetooth, no direction finding. It only stops the phone from going to sleep on the radios.

4.5.1 System radios

Android will not give Spectre Wi-Fi BSSIDs or useful BLE without Location, even if Spectre’s own switches are on. Do this in the phone’s Settings, not only inside Spectre.

- 1 **Location ON**, and **Precise / fine** (not Approximate). High-accuracy / GPS + Wi-Fi + mobile (Samsung: Location → Location method, or Location services). Tag detections with GPS needs a live fix or the path stays 0.
- 2 **Wi-Fi ON**. Spectre only hears access-point beacons. The radio must be up.
- 3 **Bluetooth ON**. BLE advertisements only. Classic (HC-05 / HC-06) still will not appear.
- 4 If the phone has **Wi-Fi scanning** and **Bluetooth scanning** under Location / Improve accuracy / Location services, turn those on too. They let the OS scan for location; they do not replace Wi-Fi and Bluetooth being on.
- 5 Turn **Battery saver / Power saving** off while you collect, or exclude Spectre. System-wide saver can throttle scans even when Spectre is Unrestricted.

4.5.2 Permissions Spectre asks

First screen: **Spectre needs the radios** → Grant permissions. Spectre will not start scanning until the required set is complete. Wording varies by Android version. If you tapped Deny, grant later at Android Settings → Apps → Spectre → Permissions. Precise Location is required; Approximate is not enough.

Grant this	What it is for
Location — Precise (While using the app)	Wi-Fi scan results and BLE addresses. The “Spectre scanning” notification is a foreground service, so While using is enough; Allow all the time is fine if the phone offers it. Spectre does not request background-only location.
Nearby Wi-Fi devices (Android 13+)	Lets Spectre run Wi-Fi AP scans.
Nearby devices / Bluetooth scan + connect (Android 12+)	BLE observer. Connect is required by the stack to read some names. Spectre does not create a Bluetooth connection for detection.
Notifications (Android 13+)	Persistent “Spectre scanning” status, and optional watchlist alerts. Allow the scan notification; Stop on that card ends collection.

Battery exemptions are not on that first screen. They are Settings → **Allow background usage** and **Unrestricted battery** (§4.5.3). Vibrate is install-time. Internet is only for Online place names (on by default); Spectre has no account.

4.5.3 Stop the OEM from parking the scan

OEM battery managers will freeze BLE when the screen locks or when they decide Spectre is unused. Names differ by phone. Do these after first launch, then leave the scan notification alone.

- 1 In Spectre: Settings → **Allow background usage**. Opens Spectre’s Battery page; turn on that Android switch. Then Settings → **Unrestricted battery**. Pixel / stock usually shows Unrestricted / Optimized / Restricted on that page — select Unrestricted. Some phones (Samsung among them) do not open onto that choice. If you only see Allow background usage, tap that row (the words, not the switch) to click through, then select **Unrestricted**. Each Spectre switch follows that Android grant when you return.
- 2 Android Settings → Apps → Spectre → Battery (or App battery usage). Select **Unrestricted**. If that choice is not on the first screen (Samsung among others), tap Allow background usage to click through and select it.
- 3 Samsung: Android Settings → Battery → **Background usage limits** (or Put unused apps to sleep). Do not sleep Spectre. If it is on a Sleeping list, remove it. Never sleeping apps can include Spectre if you want it locked there.
- 4 Leave Spectre in Recents. Swiping it out (or Stop on the scan notification) ends the service. Home leaves scanning running.
- 5 Keep the **Spectre scanning** notification. It is how the radios stay up when the UI is not in front.

Keep screen on while you watch. Spectre Settings → Keep screen on ships on. Leave it on while the Live display or Hunt is in front so Samsung does not park BLE when the display blanks. Turn it off when you pocket the phone if you still want the notification scanning and you do not need low-latency BLE. Unrestricted background is not the same switch — it does not hold the screen, and it does not lift Wi-Fi or BLE scan quotas.

4.5.4 Spectre Settings for this sit

- 1 **Scan intensity: High performance** (ships on). BLE low-latency, recycled about every 70 s. Wi-Fi about every 30 s until you add Faster Wi-Fi. Balanced and Battery saver are the compromise; not this page. §10.3.
- 2 **Faster Wi-Fi AP scans** (optional, off until you do this). For signed APs on a drive. (1) Settings → About phone → tap Build number until Developer options exist. (2) Settings → Developer options → **Wi-Fi scan throttling** → **Off**. (3) Spectre → Settings → Faster Wi-Fi AP scans → On. About every 8 s AP batches instead of ~30 s. More battery and heat than High performance alone. Spectre will not flip the switch while the OS is still throttling. Turn the Spectre switch off when the drive is over. §7.1.1, §10.3.1.
- 3 **Keep screen on** — on while you are looking at the Live display / Hunt.
- 4 **Tag detections with GPS** — leave on if you want Moving with you, Debrief distance, or log lat/lon. High-accuracy Location. Path stays 0 until a live fix.
- 5 **Write to disk / logging** — leave on if you want Share log, Signature candidates, or a file after the 15-minute Debrief window. Turn off only if the plaza is so dense the UI feels late.
- 6 **Stale after** — default 45 s is fine for a busy list. For slow tags, 90–120 s so they do not flicker to gone between advertisements.

4.5.5 Check it is actually scanning

- 1 The shade has **Spectre scanning** with Wi-Fi / BLE / signatures counts. If that card is gone, the service is not running — open Spectre again.
- 2 Live display header counts are not both stuck at 0. If they are: Location, Wi-Fi, and Bluetooth at the system level (§4.5.1), then wait ~30 s for the first Wi-Fi batch (~8 s if Faster Wi-Fi is in effect).
- 3 Header hint is not **BLE parked · restarting** while you watch. If it is: Keep screen on, Unrestricted background, do not sleep the app. §13.2.
- 4 If Faster Wi-Fi is on, Wi-Fi next Ns is a short count. If the header reads **Wi-Fi fast scan needs Developer options**, the OS throttle came back on — fix Developer options, then return to Settings.
- 5 The phone will get warm. That is the radios and the screen, not a crash. Plug in or use a pack for a long sit.

What this will not fix. Quiet or cellular-only cameras, phones on café Wi-Fi that are not hotspots, sleeping tags, and randomized BLE that walks out of range still miss. A full permission set and Unrestricted battery do not turn a phone into a monitor-mode radio. Empty list ≠ safe. §3, Notice.

5. App Navigation & Interface Overview

This chapter maps the screens. If you have not launched the app yet, start with §4.4. The first bottom tab is labeled **Live** on the phone; this book calls that screen the **Live display** so it is not confused with live GPS, the in-memory radio set, or a recording. The circle on a Live display row is a class glyph. AP (Wi-Fi access point) and LE (BLE advertiser) are the two radio kinds (§1.1); on the Live display they are a small Wi-Fi or Bluetooth icon on the subtitle — same meanings, not two-letter tags. RSSI is loudness at this phone in dBm (closer to 0 is louder, not a distance). Filters change who the Live display shows; Display (§5.3) changes how each row looks. The log still records. Pause freezes the Live display while the radios keep running; Filters and Settings still update, and the new filter applies when you run the Live display again.

5.1 Main layout

One screen, five tabs, and a device-detail page on top. The bottom bar is Live, Filters, Signatures, Reports, Settings. Filters sits next to the Live display because the next step from that picture is usually to trim it. Signatures is the pattern catalog; Reports generates Debrief, AI Export, and Share or Save log. The top bar is always SPECTRE plus a live count line: three small icons with numbers (Wi-Fi access points, BLE advertisers, and on-air signature matches — the same hub icon as the Signatures tab). A radio hint may follow — Wi-Fi next 27s, Wi-Fi waiting on OS, BLE cycling, or BLE parked · restarting — and is the first thing to ellipsize if the bar is tight. The scan notification still spells the same three counts (“N Wi-Fi · N BLE · N signatures”). Export lives on Reports (Debrief, AI Export, Share / Save log), not on the Live display.



Fig. 1 — Live display (hybrid). Header is SPECTRE plus Wi-Fi / Bluetooth / signatures icons with counts. Circle is the signature class (unmatched = ?). Title is the MAC; Privacy mode hides the last three octets. A Wi-Fi or Bluetooth icon on the subtitle is radio kind (access point vs BLE advertiser). The number is RSSI (loudness here, not meters). Sparkline is RSSI over packets; RSSI bars off. Tune (top right) is Display. Bottom Pause freezes the picture. Privacy mode is on: last three MAC octets show as ***:***:***.

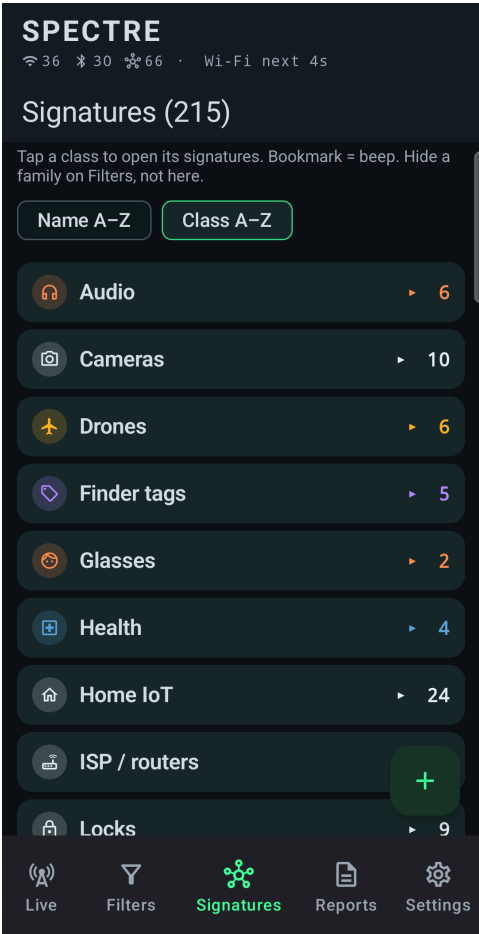


Fig. 2 — Signatures. Class A–Z outline (Health, Home IoT, ...). Tap a class to open it. Name A–Z is the alphabet. Bookmark = watch (beep and/or spoken class). Hide a family on Filters, not here. The + button adds a blank signature. Privacy mode is on: last three MAC octets show as ***:***:***.

5.2 Bottom navigation

Tab	Function
Live	Live display — the on-screen picture: radar, list, timeline, hybrid, or By class. Not a recording. Tune (top right) opens Display. Tap this tab again to Pause (radios still scan and log; Filters still apply when you run again); tap to run the list. From another tab, this item just navigates here. Double-tap SPECTRE to jump to the top. New detections only adds Mark seen / Reset seen above the tabs. Moving with you adds Start over.
Filters	Which radios appear. Order: presets, radios, Moving with you, New detections only, Signatures only, Named radios only, Hide Fast Pair account-key, class Show only / Hide these, Show only selected signatures, Hide selected signatures, RSSI / name / OUI. Signatures still label and can beep. See §8 and Figs. 3–4.

Tab	Function
Signatures	The pattern catalog. Title shows how many signatures are loaded (stock plus any you added). Each row shows the class glyph, and a hexagon when that row has a Decode fields map (§9.6). Name A–Z or Class A–Z (classes start collapsed; tap to open). Tap a row to edit (rules, color, Decode fields on BLE). Bookmark = watch (beep and/or spoken class). No matching on/off — hide on Filters. + adds a blank signature. Fig. 2, §9.6.
Reports	Debrief (text / PDF), AI Export, Signature candidates, Share log, Save log, Reset / clear log. Debrief is the last 15 minutes in memory (cap ~400), not the log. Signature candidates mines the rotating log for unmatched families. On a long drive tap Debrief more than once (§11.4.1). GPS / place names / logging on-off stay on Settings.
Settings	Theme, Keep screen on, Privacy mode (MAC tails on screen and sit reports; pauses TAK), scan intensity, GPS tagging, TAK / CoT feed (off), place names, logging, beep and/or voice / optional shade card, Test alert, Named radios, battery exemption, export / import signatures, restore defaults. Row layout is Live display → Display, not here. TAK: §5.8.

What you want	Where
Fewer radios on the Live display	Filters (chapter 8)
Less text on each row	Live display → Display (§5.3)
Stop seeing a family on the Live display	Filters → Hide these (class) or Hide selected (one family)
Hide MAC tails on the screen	Settings → Privacy mode
Put radios on an ATAK map	Settings → TAK / CoT feed (§5.8, §12.15)

5.3 Display inspector (Live display)

The Live display is not a fixed dump of every field. You can fit it to the job — a crowded plaza, a sit, a hunt, or copying a MAC. The tune / chevron icon (top right on the Live display) opens **Display**, where you choose how much of each radio to show. Close it when the list looks the way you want. Your choices stay until you change them.

Filters hide radios. Display hides fields. Filters (§8) decide *who* is on the Live display. Display decides *how each row looks*. Subtitle None does not drop a radio — it only drops the second line so more rows fit. Vendor, payload, and the full decode stay on detail. Logging is unchanged. Two operators on the same sit can run two different rows.

View, Sort, Brief hold, Title line, and Subtitle line are full-width dropdowns, stacked; on/off items are switches. Radar labels are not Title/Subtitle — they stay a short signature name or type guess.

Identity — what each row says

- **Title line** — First line of each list, hybrid, or timeline row. It stays bold even when it is a MAC. **MAC address** (default) puts the radio address on the title so you can copy it or hunt by BSSID without opening the second line. **Advertised name** is the SSID or BLE local name, or <hidden> / unnamed LE if there is no name (no type guess). **Name + type** uses the advertised name if there is one, otherwise the same guess as the detail page (for example Apple, Inc. · AirTag or Find My accessory). That “Apple, Inc.” is a type guess. IEEE vendor is on the detail page, not the list. Title cannot be None. Radio kind never sits on this line — the Wi-Fi / Bluetooth icon leads the subtitle.

- **Subtitle line** — Second line, small monospace. Always starts with a small **Wi-Fi** icon (access point) or **Bluetooth** icon (BLE advertiser). Then Advertised name, **Name + type** (default), or MAC, plus r and / gone. **None** hides the second line so more rows fit; rand/gone move onto the title. The kind icon does not move onto the title, so that line stays a readable MAC or name and the row can collapse. On Name + type or Advertised name, unnamed BLE is written unnamed (the Bluetooth icon already marks LE — not “unnamed LE” twice). Vendor is never on the list row — IEEE board/chip vendor is on detail.

View and linger

- **View** — Classic radar, Strength list (default), Timeline, Hybrid + sparklines, By class. Same filtered set; only the picture changes. Sort orders the list / hybrid / timeline and the radios under a signature in By class. Radar still plots by RSSI radius (a seat on the ring, not a stack). §6.
- **Brief hold** — Off (Stale after only), or hold 10 / 30 / 60 s after the last packet. The row stays at last-heard RSSI, rank, and radar ring; values do not decay. A new packet updates them. On-screen linger is the longer of Stale after and Brief hold. Filters → New detections only uses this same time as the minimum stay after the last packet.

Sort — order of the list

Display → Sort is the order of radios on Strength list, Hybrid, Timeline, and under a signature in By class. It does not hide radios (Filters do that) and it does not change the log or Debrief. The default is **Strongest (avg 30s)**. The eight choices:

Sort	What it does	Use when
Strongest signal	Loudest last packet at the top. The number on the right is that instant RSSI.	Instant triage, pick a Hunt target, a plaza where average would hide a sudden loud tag.
Strongest (avg 30s)	Rank by the average of RSSI samples in the last 30 s (last packet if there are none). Extra “avg N” under the RSSI. Radar rings use that average too. This is the default.	Sitting Wi-Fi that jumps every scan; a tag you are walking toward without the list jumping.
Newest heard	Most recent packet at the top, then strength.	Who just spoke. A fixture that keeps advertising stays high.
Newest alert	Most recent watchlist alert (beep / voice / flash) at the top, then last-heard. Radios that have not alerted this session sink. A second alert on the same radio jumps it back to the top. Those radios keep a phosphor bell on the list.	Watchlist sit: the last thing that piped, then the one before.
Newest arrival	First-seen this session, newest at the top, then last-heard. A radio that has been here an hour sinks even if it is loud.	Who showed up latest, not who is loudest now.
New at bottom	First-seen, oldest at the top; new rows append; gone rows drop. The list sticks to the bottom unless you scroll up.	A sit or drive that you read like a log. New at bottom is the only sort that pins the end.
Name A–Z	Alphabetical on the Title line, then MAC. Title defaults to MAC, so this is MAC order until you change Title.	Find a known SSID, copy an address, compare two lists.
Signatures first	Any signature match above unmatched, then by strength (same strength rule as last Strongest choice).	Pattern hits without turning on Filters → Signatures only — unmatched radios stay, they just sink.

Picking Strongest signal or Strongest (avg 30s) sets the list back to a strength order. The other six keep the last strength rule only as a tie-break where they use it (newest heard, signatures first). Newest alert ranks by watchlist

flash time, not RSSI. Sort is not a filter: a weak named tag is still on the list, just not at the top unless you chose Signatures first, Newest arrival, or Newest alert.

Extra facts — turn on only what the job needs

- **RSSI bars** — Switch, on by default. Track plus fill from last-heard strength (the same last packet as the number, not the 30 s average).
- **Signature names** — Switch, on by default. Independent of bars. List, hybrid, and timeline show up to three chips when a device matches more than one signature. Radar uses only the first match as the blip label. Detail lists every match. Off still matches; it only hides the chips.
- **Frequency** — Switch, off by default. Channel and MHz sit under the RSSI on the right so a long name does not cut them off (for example ch6 · 2437MHz). BLE advertisements usually have no frequency.
- **First / last seen** — Switch, off by default. Age since first packet this session and age since the last packet, ticking once a second.

Default row (Strength list, Sort Strongest avg 30s, Title MAC, Subtitle Name + type, bars on, signatures on, Frequency off, first/last off, Brief hold 10 s): class glyph in a circle (unmatched = ?); first line MAC; second line the Wi-Fi or Bluetooth icon, then name or type guess, then rand/gone; live RSSI, trend, and avg N on the right; optional bar and chips. A bookmarked new hit flashes the row for one second when the watchlist fires (pip and/or spoken class). After that, a phosphor bell stays on the chip row for the rest of the session.

Fit the row to the job

Mission	Typical Display
Learn the neighborhood	Leave the defaults: MAC over Name + type, bars on, signature names on. Strength list, Sort → Strongest (avg 30s).
Plaza / list too dense	Subtitle None. Bars, Frequency, and first/last off. Strength list, not Hybrid. You still see every filtered radio — just the MAC (the radio-kind icon is gone with the subtitle, so the title stays readable).
Copy a MAC / BSSID	Already the default (Title MAC). Set Subtitle to None if you want more rows on screen. Pause, then read or long-press.
Wi-Fi channel sit	Frequency on. Title can stay MAC. Sort strongest. Bars on if you want a glanceable loudness strip.
Hunt / follow one BLE	Hybrid + sparklines. Keep Title MAC. First/last on. Frequency off (BLE rarely has it). Then open Hunt from detail.
Sit — who came and went	Timeline. Sort newest arrival or new at bottom. First/last on. Title is already MAC if you will write addresses down.
Signatures only, quieter list	Signature names on. Subtitle None (MAC only). Pair with Filters → Signatures only (§8), not instead of it.

These Display controls are not repeated on Settings. New detections only is not on this card — it is a Filter; its Mark seen / Reset seen buttons sit above the tabs (§5.3.2). If you were scrolled to the top, new stronger rows push the list down. If you have scrolled away, the list does not snap. Double-tap the SPECTRE title / count line to scroll quickly to the top (about 140 ms, linear).

5.3.1 Pause the Live display

On the Live display the bottom item is a toggle, not a second “go to top.” While the list is updating it reads **Pause**. Tap it and the Live display freezes: same rows, same RSSI, same chips, and the header shows SPECTRE · PAUSED. Wi-Fi and BLE keep scanning and the log still appends — pause freezes the picture, not the radios. The bottom item then reads **Live** (play); tap it to run the list again. From Filters / Signatures / Settings the same item only navigates back; it does not un-pause until you tap the Live display while already on that tab.

Filters and Settings still update while paused. Chip taps, Show only / Hide these, and Restore still take effect; the frozen Live display does not change until you run it again. The pause banner says so. Use pause when the list is moving too fast to tap a row. Open detail from a paused row and you get that frozen observation even if the radio has since gone stale on the live set. Use it so you can read what was on the screen, then resume. While New detections only is on, Mark seen / Reset seen stay above the tabs and still work while paused — Mark seen absorbs the frozen list, not radios that arrived after the freeze.

5.3.2 New detections on the Live display

The switch is on Filters → New detections only. While it is on, the Live display shows two extra controls. They are not on Display and they are not on the Filters tab.

- **New only** — a one-line hint under Display (and under the Pause banner if frozen). Shows New only, New only · N hidden, or New only · learning sitting Wi-Fi.
- **Mark seen / Reset seen** — a bar just above the tab bar, always reachable without scrolling the list. Hidden when the filter is off.

On first turn-on, Spectre snapshots what is already here (plus sitting Wi-Fi through the next scan). Already-seen only grows on that first snapshot or on Mark seen. Mark seen adds what is on the Live display now; while Pause is on, that is the frozen list, not radios that arrived after the freeze. Reset seen clears already-seen to zero so those radios can show as new again. Clearing the log does not reset this. Randomized BLE addresses still look new. A new radio stays while it is heard, then at least as long as Display → Brief hold after the last packet.

5.3.3 Moving with you on the Live display

The switch is on Filters, under Radios to show. New detections only sits under it. Signatures only, Named radios only, and the class chips are further down. Turning it on starts a follow test on all radios. It clears Signatures only, Named radios only, and class Show only so those filters do not empty the list. Hide these stays if you were hiding a bag tag. The Moving with you preset at the top replaces the whole filter the same way. While it is on, the Live display shows Follow · path N m and a **Start over** button above the tabs (same bar as Mark seen / Reset seen if New detections only is also on). Start over clears the operator GPS path and every radio’s GPS trail. The live list and the log are not wiped. Path goes back to 0 m; walk or drive ~50 m again to see who comes back. A tag in the bag should reappear; house APs should not. Turning the filter off does not clear the path. After the switch is on you can add Show only again if you really want only finder tags that co-travel.

The test itself is not a fixed radius. GPS is stamped on your phone at the moment a radio is heard, and “still here” is allowed to grow with how fast you have been moving — a sidewalk tag is held to about a house-length; a tag in the car is not dropped between advertisements just because you covered hundreds of meters in a few seconds. Wi-Fi in the car is given a longer hold than BLE, because Android scans access points less often than it delivers advertisements. The full checklist, the speed table, and why a house AP at the door still fails are §8.5.

What you see on the Live display is only the current picture, not a recording. Reports → Debrief (text or PDF) still includes the last 15 minutes of tracking (possible trackers with you / possible tail) from radios in memory (cap ~400; §11.4.1). Radios you only passed are omitted. That is true whether you are on radar, list, timeline, hybrid, or By class, and whether Moving with you or any other filter is on. GPS tagging must have been on, and you must have moved, or the report says the following test was not run. §11.4, §12.2.

5.4 Status indicators

- **Class glyph** in the circle: signature class of the first match (Finder tags, Phones / PCs, Audio, ...). Unmatched radios show a question mark. The same glyphs sit on Filters class chips and on By class headers.
- **Radio kind** (Wi-Fi / Bluetooth icon) at the start of the subtitle: Wi-Fi access point versus BLE advertiser. AP means the radio is beaconing a network (or acting as a hotspot / soft-AP), not that it is a Wi-Fi client. Detail spells those words. Subtitle None drops the icon with the second line so the title stays a MAC or name.
- **Colored chips**: matched signature names when Signature names is on. List / Hybrid / Timeline show up to three. Radar labels the blip with the first match only. Four or more matches: extra names are on the detail page.
- **Decode hexagon**: inside a signature chip, in that chip's color, when *that* signature has a Decode fields map (§9.6). Dual-chip radios mark only the mapped name(s). Catalog check — the Live display does not parse the bytes. Values are on detail. Extra attention "!" is its own chip. A phosphor bell means this radio already alerted this session. Display → Signature names off hides the names and the hexagon; "!" and the bell still show.
- **>> > = < <<**: RSSI trend from recent packets (much stronger / stronger / steady / weaker / much weaker). Green approaching, red fading. Blank until a few samples exist.
- **New only · N hidden**: Live display hint while New detections only is on. Mark seen / Reset seen sit above the tabs.
- **Follow · path N m**: Live display hint while Moving with you is on. Start over (above the tabs) clears the GPS path and trails.
- **new Ns**: age since first seen, when New detections only is on.
- **rand**: locally administered / randomized address bit is set.
- **gone** (list) or a dim blip (radar): last packet older than the longer of Stale after (default 45 s) and Brief hold. Held Wi-Fi APs and BLE recycle do not count as gone.
- **first / last**: optional line when First / last seen is on. Age since first packet this session and age since the last packet (last now if it just spoke).
- **chN · MHz**: under RSSI when Frequency is on. BLE ads usually have none.
- **Scan notification**: "N Wi-Fi · N BLE · N signatures", updated about every 2.5 s. Home leaves scanning running. Swiping Spectre out of Recents (or Stop on the notification) ends the service so it does not keep alerting in the background.
- **Watchlist**: pip and/or spoken phrase on media volume, row flash, optional jump. Beep and Voice are independent (Settings). Voice can say the class, the signature name, or both; not Hunt. Fires once when a watched radio first appears (or returns after gone). After the one-second flash, a phosphor bell stays on that list row (list, hybrid, timeline, By class) for the rest of the session — not Extra attention "!". Newest alert sorts by the same event. On By class the matching class and signature open so the row can flash, and the jump keeps those headers on screen when the radio is close enough. On radar the blip pings (expanding rings and a bright core) for any of those cues, including voice-only; after the ping a phosphor ring stays on that blip (drawn on top). Test alert plays whatever is on. System shade card is off unless Settings → System notification is on.

5.5 Device detail

Tap a row or a radar blip. Detail is the full decode of that observation, resolved offline from packed IEEE and Bluetooth SIG tables (no network). Jargon is spelled out in plain language (for example BR/EDR not supported = BLE-only, no classic Bluetooth). RSSI is a short band (very strong / strong / medium / weak / very weak) so the page does not reflow as the number ticks.

What this looks like (top card). A cautious guess from advertised identity: GAP Appearance, Class of Device, service UUIDs, Apple/Google maker bytes, Fast Pair model ID, and any matched signature. Examples: “Most likely AirPods Pro”, “Most likely a mouse”, “Most likely an Apple AirTag / Find My tag.” Hedge words are Most likely / Probably / Could be. This is what the radio is broadcasting, not a visual ID. If none of those fields are present it says there is not enough to name a product type.

Extra attention (optional card under that guess). Only if a *matched* signature has text in Extra attention — not the editor Notes field. Stock fills it on Hobby BLE serial, Axon, WatchGuard Video, Ray-Ban / Meta glasses, Snap Spectacles, Fieldy, Plaud Note, Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, Porkchop, Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc, and roadside / public camera + ALPR (Flock, Penguin, Pigvision, FS Ext Battery, Genetec AutoVu, Rekor, Motorola Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua — those rows also ship with the bookmark on). A “!” mark on the Live display row means open detail and read that card. Pattern match, not identity, not a skimmer detector, not a safety finding. You can put Extra attention on any signature you edit.

Signature family (card above Create signature from device). Same analysis as Reports → Signature candidates (§5.6.1), aimed at **this** radio. It names the best unique on-air ID (name glob, vendor IE, service UUID, manufacturer-data prefix, or stable OUI) and counts distinct MACs that share it in the rotating log and on the air now. Packet count does not matter. Randomized addresses with no other ID, house-like names, chip-module OUIs, protocol IEs, generic UUIDs, and whole-OS company IDs (Apple 0x004C, Google, Samsung, Microsoft) are skipped — a crowd of those is not a catalog family. Pattern match, not identity.

- **Strong family** — about eight or more distinct radios share that ID. Phosphor card, big count. A catalog pattern, not this MAC.
- **Possible family** — two through seven. Amber. Thin sample — still a family, not a crowd.
- **This radio only** — no other MAC shares a unique ID, or there is nothing unique to cluster on. A signature from here will mostly tag this address.
- **Already tagged** — this radio already matches a catalog signature. The card names the match. That is **not** a stop on a second signature. A device may carry several chips (§7.4).

The usual second-tag case is a **protocol plus a place**. Stock iBeacon is Apple company 0x004C type 0x02/0x15 — a layout any vendor can send, not a store. Walk into a shop with hundreds of iBeacons and they often share one proximity UUID (wayfinding, baskets, asset tags). Create a custom signature on that UUID and the same radios dual-label: iBeacon **and** your store row. the Live display shows up to three chips; detail lists every match. Stock **Target Atrius basket** is this pattern (UUID 5993A94C-... plus service 0xB1BB) — mute generic iBeacon in a mall and the Target UUID still labels those baskets. Minew / Estimote / Kontakt.io also dual-label with iBeacon. Spectre drops the iBeacon chip only when a *non-beacon* product already labeled the radio (Sony TV, Tesla phone-key).

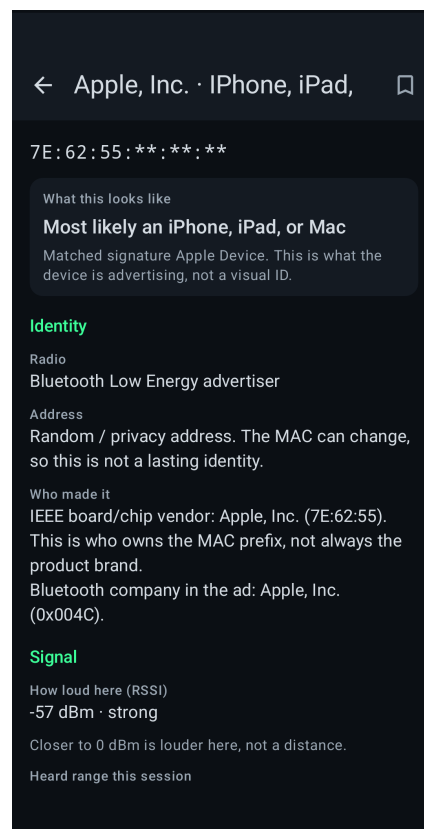


Fig. 5 — Device detail. “What this looks like” is a guess from the advertisement, not a visual ID. Privacy mode masks the MAC tail; the OUI prefix stays. Privacy mode is on: last three MAC octets show as **:***.

The card is the verdict only. **Create signature from device** still drafts this radio, including a MAC pin (§9.2), even when the card says Already tagged. For a Strong or Possible family on an *unmatched* radio, Reports → Signature candidates drafts the **shared** rule without pinning this MAC (§9.2.1). Candidates skip radios the catalog already tagged, so a floor of iBeacons will not appear there — use detail decode and Create from device, then tighten the manufacturer-data prefix to the UUID. Logging off: live count still works; the log number fills in once the rotating log has been read.

- **Identity** — advertised name, MAC, address type (public factory vs random/privacy). Vendor from IEEE OUI, MA-M, or MA-S when the address is universal; randomized BLE addresses skip the OUI table. Bluetooth Company ID from manufacturer data is named from the SIG list (~4012).
- **Signal** — RSSI plus min/max this session, claimed TX power, channel / MHz, Wi-Fi standard and channel width when the OS reports them.
- **BLE** — PHY (1M / 2M / Coded), connectable, advertising interval, decoded Flags, GAP Appearance, Class of Device (major / minor / service classes), named 16-bit service UUIDs, service data, manufacturer payload. Connectable stays Yes once any advertisement from this radio was connectable — scan responses from the same MAC are not connectable and no longer flip the line.
- **Known payloads** — iBeacon (UUID / major / minor / calibrated TX); Google Fast Pair (pairing-mode 24-bit model ID with a local name list, or account-key broadcast); Apple Continuity (AirPods/Beats model, battery, in-ear/in-case; Find My / Offline Finding; Nearby Info activity; Nearby Action; AirDrop; Handoff; Hey Siri; AirPlay; Instant Hotspot). Eddystone UID / URL / TLM / EID (service 0xFEAA) **accumulate** on this page: each frame type stays once heard, labeled on the raw line (UID, URL, TLM, EID). Frames do not replace each other. Microsoft Swift Pair / Nearby Sharing when present. Unknown 0xFF blobs stay company + hex — there is no official database of proprietary payloads.
- **Decoded fields** — BLE only. If a matched signature has a field map (§9.6), labels and values from the *current* advertisement (temperature, model, Remote ID location, ...). The heading carries the same hexagon as the Live display chip. Encrypted or short payloads stay hex; a note appears if a map exists but did not apply to this packet (Govee lights often only send a name). Matched signatures lists a hexagon next to names that have a map. Apple / Fast Pair / Eddystone / Microsoft stay in Known payloads and Maker data. Share as text, AI Export, and Debrief notable BLE print the same lines.
- **Wi-Fi AP** — SSID (or Hidden), RSN/WPA/AKM/cipher from information elements in plain language, supported rates, capability string, vendor-specific IEs (OUI + type + payload) looked up in IEEE OUI/CID.
- **Session** — first/last seen, hit count, optional GPS (operator phone at hear-time), **every** matched signature (not capped at three), raw advertisement bytes, RSSI sparkline, 15-minute presence. Bookmark (top bar) watches this radio — see below. BLE: Hunt. Signature family card. Then Create signature from device.

Decode blocks on this page (Identity, Signal, Bluetooth advertisement, services including Eddystone, maker data, Session, Signature family) keep the tallest height they have reached on this radio, so a shorter frame or a shorter guess does not yank Hunt and the share buttons around.

Opened from a paused Live display, detail is the frozen snapshot even if that radio has since gone stale. Long-press a row, timeline card, or radar blip to jump straight to the signature editor with rules pre-filled from that observation.

Custom name is behind the edit icon on the advertised-name row (or Name if there is none). It labels **this MAC** without turning Alert on. A saved name shows as a one-line caption; tap edit for the field. Save name writes it to Settings → Named radios (Alert off until you bookmark). The name shows on the Live display. Filters → Named radios only keeps only those rows. If the address looks random / privacy (IEEE local bit or Android Random type), the edit control is hidden — a name would not follow a rotation. A name you already saved still shows as a caption. Bookmark (top-right) still lets you watch this MAC, with the same rotation warning.

Bookmark (top-right) watches **this radio only** — the device key (KIND + MAC), not the signature family. Outline = not watched; filled = on the watchlist. One tap; Spectre prefills a name from the advertised name or type guess. A

saved custom name still shows as a caption under the advertised name. Rename, turn Alert on or off, or drop it on Settings → Named radios. It does **not** go away when the radio leaves the Live display or goes stale. While it is gone, nothing alerts. If the *same* MAC returns, Spectre alerts once (Settings → Watchlist alerts, plus Beep and/or Voice). Sitting detections do not fire again. Randomized BLE gets a new address: the watch stays on the old key and will not follow the rotation — that orphan stays on Named radios until you remove it or Clear all. Bookmark still works on a random / privacy address (IEEE local bit or Android Random type) — Spectre cannot tell a static random MAC from one that will rotate, and the watch will not follow a new address. Signature-row bookmarks are the other kind of watch (any match of that family); Named radios never lists those. Full alert behavior: §10.1.

Hunt (BLE only). Device detail → Hunt. A full-screen page for this advertiser. You get a large RSSI and a cue: Very Close, Closer, Further, About the same, Quiet, or Gone. Very Close is about -45 dBm or louder (the same “very strong” band as detail). Closer / Further / About the same come from a few seconds of smoothed packets (about a 3 dB step). A ring around YOU contracts on Closer / Very Close and expands on Further. Loudest this hunt and a hunt-only sparkline sit with it. Cue and hint are in fixed slots so the dBm number does not jump. Beep and Vibrate (off by default, remembered) sit under Reset / Back and tick faster as RSSI gets louder — a short click, not the watchlist chirp. Silent when Quiet or Gone. The screen stays on while Hunt is open. It uses the live BLE store, even if the Live display was paused. This is not meters, not a compass, and not a map pin on the other radio. Wi-Fi access points have no Hunt button: stock Android batches them about every 30 s (about 8 s with Faster Wi-Fi AP scans), still too slow to walk toward. Randomized BLE can vanish mid-hunt (Gone). Experimental. Field tactics: §12.13.

At the bottom of the page, under Hunt and the Signature family card: **Create signature from device**, then two shares that dump **this one radio** — not the 15-minute sit. Both use the Android share sheet. Neither is a legal identity. Treat the paste as operationally sensitive (MAC, SSID, payload, GPS).

- **Share as text** — Plain dump of the fields on this page: identity, What this looks like, Extra attention if present, signal (including recent RSSI samples), advertisement decode, session times, last fix if tagging is on, matched signatures, raw bytes, 15-minute on-air windows. Sparklines become a number list. Header: experimental, not a legal identity, stock Android radios.
- **AI Export** — Paste-ready prompt for a chat about **this radio only**. Same experimental-use disclaimer as Reports → AI Export (the model is told to repeat it and not give safety advice). Includes the Share-as-text dump plus instructions to use public registries (IEEE OUI, Bluetooth SIG company, GAP Appearance, known formats) and to cite which field supports each claim. Asks for likely product class, competing hypotheses, what Spectre could not see, and what not to conclude (owner, following, distance). If Online place names and GPS tagging are on, a geocode of the last fix may be included — that is still the *phone*, not this radio. One device; not the sit inventory.



Fig. 5b — Hunt. Cue and dBm on top; rings around YOU (in = closer, out = further); sparkline; Reset, Back, then Beep / Vibrate. Switches start off; this capture has them on. BLE only — Wi-Fi APs have no Hunt. Privacy mode is on: last three MAC octets show as *****:***:**.

5.6 Reports tab

Bottom bar, next to Settings. This tab runs sit reports, mines the rotating log for signature families, and takes the log off the phone. GPS tagging, Online place names, logging on/off, CSV vs JSONL, and rotate size stay on Settings.

- **Debrief (text) / Debrief (PDF)** — Last 15 minutes in memory (about 400 radios; hard ceiling 900). Same sit report, two formats. Opens with the hobby / as-is disclaimer (hypotheses, not identity; local law; not for safety). Not the rotating log. GPS following / planted-tracker assessment when tagging is on and you have moved. Radar vs list, Moving with you, Signatures only, and Hide selected do not change what Debrief sees. On a drive, a few kilometers of city RF can fill that cap — tap Debrief more than once (§11.4.1).
- **AI Export** — Sit-level paste-ready prompt (last 5/15 minutes). Includes the onboard Debrief verbatim, then working data, and asks a chat for statistical analysis and depth the phone report cannot do. Not a rewrite of Debrief. For a *single* radio, use AI Export on the device-detail page instead.
- **Signature candidates** — Mines the rotating log for unmatched families that share a unique on-air ID. Create signature opens a draft (shared rule, no MAC pin). Save returns you to the list and re-runs it. Offline. §5.6.1, §11.5.
- **Share log / Save log to SD card / storage...** — Share uses the Android share sheet. Save uses the system picker (SD, Downloads, USB, Drive). Line and disk counts are shown here; logging must have been on for new rows.
- **Reset / clear log** — Bottom of Reports. Deletes rotated files on the phone. Does not reset New detections already-seen.

5.6.1 Signature candidates

Use this when the Live display is full of unmatched radios and you want **families**, not a dump of every ?. It reads the rotating on-disk log (§11), not the 15-minute Debrief map. Logging must have been on. Spectre collapses rows to unique kind+MAC, **re-matches against the catalog on the phone now** (the log's fleets column is whatever matched at write time — ignore it), then clusters leftovers that share a unique on-air ID: a name glob, a Wi-Fi vendor IE, a BLE UUID, manufacturer data, or a stable OUI. A family needs two or more distinct radios. Randomized addresses with no other ID, house-like names, chip-module OUIs, and protocol IEs (WPA / RSN / P2P) are skipped — one muted line at the top says how many. Device detail runs the same test on one radio (Signature family, §5.5). Already-tagged radios are omitted here; a second store-UUID chip is Create from device (§5.5, §9.2).

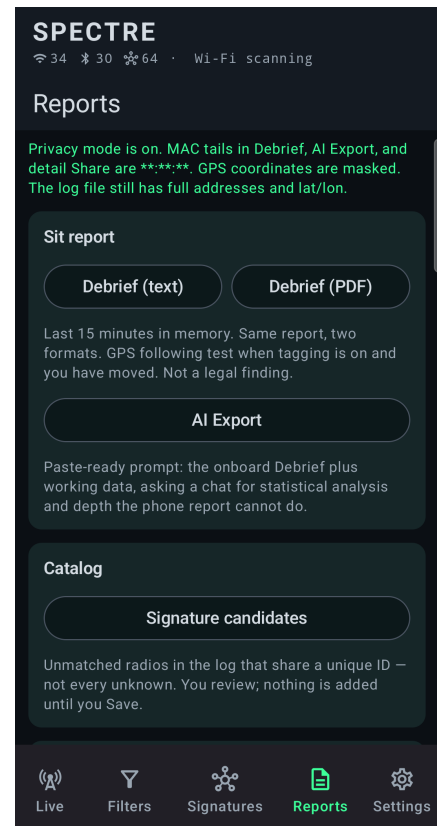


Fig. 5c — Reports. Sit report and Catalog section cards (Debrief text/PDF, AI Export, Signature candidates). Privacy mode is on. Share log / Save / Reset sit below the fold. Privacy mode is on: last three MAC octets show as ****.

Each card is one family: proposed name, class guess, radio count (number plus a Wi-Fi or Bluetooth icon), the rule in monospace, one sentence why it is not a house SSID, and a couple of examples. Privacy mode masks MAC tails on those examples. **Create signature** opens the existing editor as a draft with the **shared** rule — no MAC pin (that is Create from device, §9.2). Change the name or class, then Save. Nothing is added until you Save. The new row is **your** signature, not a stock catalog edit. Save returns you to Signature candidates and re-runs the list so that family should drop off. Cancel returns to the same list without re-running. Offline; no web lookup. Pattern match, not identity.

Vendor-IE families (a Roku-class ID on a randomized BSSID) need the `vendor_ie` column, which is only on rows written after this feature shipped. Name globs and stable OUIs still work on older log parts. Empty list: logging was off, the log is short, or everything unmatched is noise. §11.5.

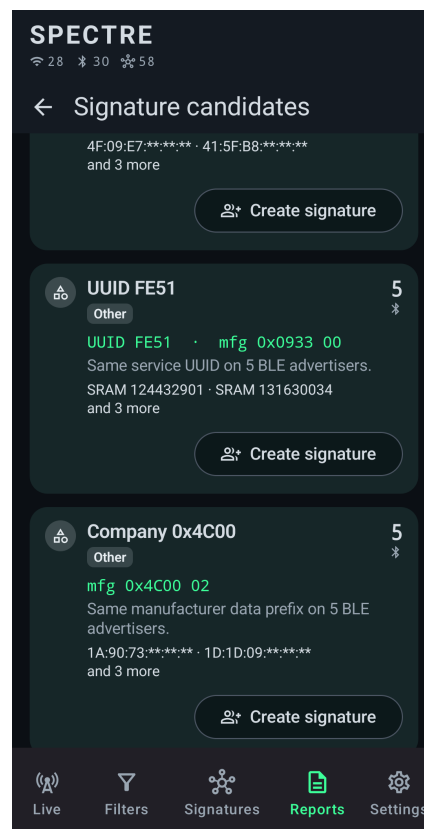


Fig. 5d — Signature candidates. Ranked families from the rotating log. Skipped randomized / house-like names. Create signature opens a draft (shared rule, no MAC pin). Privacy mode is on: last three MAC octets show as `***:***:***`.

5.7 Settings screen

Settings is radios, logging, and appearance. It is not where you hide radios on the Live display, and not where you tap Debrief. Mute a noisy OUI or name with the per-rule switches on a signature. Hide a whole family on Filters. See §7.6, §8.

- Appearance** — Dark theme. Keep screen on (on by default): holds the display while Spectre is in front so Samsung does not park BLE; turn it off when you pocket the phone. Privacy mode (off by default) hides the last three octets of every MAC on the Live display, radar, timeline, detail, Hunt, Named radios, and watchlist cards as `**:*:*`. GPS last-fix on detail and coordinates in Debrief / AI Export / detail Share become “masked”; street names are omitted from those sit reports. The OUI stays. Logs, matching, Moving with you, and saved signatures still use the real MAC and GPS. A TAK / CoT feed is paused while Privacy mode is on so full MACs and coordinates are not sent onto the LAN (§5.8).
- Radios** — Scan intensity: High performance / Balanced / Battery saver (Wi-Fi ~30 / 40 / 55 s). Faster Wi-Fi AP scans: a second switch. Spectre reads the OS Wi-Fi scan-throttle flag (Android 11+) and will not turn this on while that flag is still on. Developer options → Wi-Fi scan throttling → Off, then flip Spectre. About every 8 s instead of ~30 s. Purpose: more chances to hear an AP while it is in range so a catalog signature (OUI or factory SSID) can fire — important on a drive, when a roadside or vehicle AP may only be loud for a few seconds. More battery and heat. Header may read Wi-Fi fast scan needs Developer options if the OS switch came back on. Spectre cannot flip Developer options. §7.1.1, §10.3.1.
- Watchlist** — Watchlist alerts is the master switch (off: no beep, voice, flash, jump, or shade card; bookmarking still works). Beep and Voice are independent: pip only, spoken phrase only, or pip then phrase. Voice (off by default) can say the class (finder tags, audio, ...), the signature name (Apple AirTags, Axon, ...), or both — Settings → What to say; default is class. Not Hunt; a second hit is skipped while a phrase is being spoken. Jump to new watched detection works with beep, voice, or both. Optional system notification (off by default). Test alert plays whatever is on. **Named radios (N)** opens the list of one-MAC names and optional alerts: rename, Alert on/off, remove one, or Clear all (signature watches stay on Signatures). Stock bookmarks already watch the Extra attention families (body-cam, camera glasses, recording wearables, pentest, public-safety vehicle APs, roadside / public camera + ALPR). Unbookmark a row on Signatures if you do not want that alert. Flock LiteOn / Espressif OUIs can be noisy. Field write-up: §10.1–10.2.1.
- Tag detections with GPS** — On by default. Requests live GPS and network location updates while scanning, then stamps each hear (detail, Moving with you, Debrief, log lat/lon). Last-known older than 30 s is ignored. Path stays 0 until a live fix. High-accuracy Location. Needed for Debrief distance/following, Filters → Moving with you, and heard-here TAK pins. Advertised payload pins (Remote ID) do not need this. A Share log with tagging on contains operator coordinates.
- TAK / CoT feed** — Off by default. UDP Cursor-on-Target to ATAK / WinTAK / iTAK. Default port 10011 (ATAK CIV CoT input); host 239.2.3.1 (SA multicast). Same-phone ATAK CIV: this phone's Wi-Fi IPv4 and 10011. Other ports: 6969 (SA multicast), 8087 (many TAK servers). What to send chips: Extra attention (on), Payload location (on — this is how stock Remote ID pins, using decode field ids latitude/longitude, not a Remote ID special case), Watchlist (off), All signatures (off). Privacy mode pauses the feed. Full configuration: §5.8. Sit: §12.15.

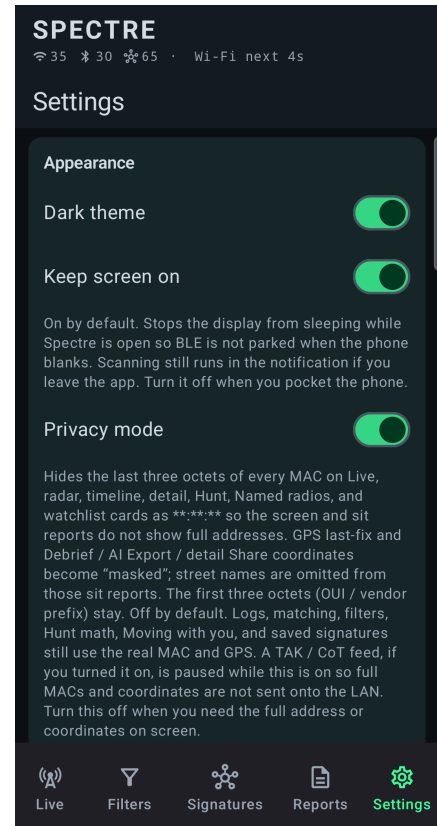


Fig. 6 — Settings → Appearance (section card, phosphor switches). Night mode is off in this shot; Fig. 6a is the same tab with Night mode on. Privacy mode is on for every screenshot in this book. MAC tails on screen and in Debrief / AI Export / detail Share are `**:*:*`. Logs stay full addresses. Live display / radar / timeline / hybrid / By class / detail / Hunt shots use Filters → BLE only so local access-point names are not printed. Privacy mode is on: last three MAC octets show as `**:*:*`.

- **Night mode** — Off by default. Red-on-black field display (cockpit / sit). Text, chips, switches, RSSI, and Hunt marks become shades of red so green and blue do not dump into a dark eye. Background stays dark. Phone brightness is unchanged. Overrides Dark theme while this is on. Restore defaults turns it off. Fig. 6a.
- **Online place names in Debrief** — On by default. Reverse-geocodes GPS stamps for Debrief and AI Export when the phone has internet (install-time INTERNET; system geocoder; no Spectre server). Offline: a note in the report, no error dialog. Turn off if you do not want streets of your path in those files. The generate buttons themselves are on Reports (§5.6).
- **Logging** — Write to disk, CSV vs JSONL, rotate size, Stale after slider (when a radio is marked gone). Line/disk counts. Share, Save, and Reset / clear log are on Reports.
- **Allow background usage** — Switch. Opens Spectre's Battery page; turn on Allow background usage so the OS may run the scan when Spectre is not in front. Follows that Android setting. Not Keep screen on.
- **Unrestricted battery** — Switch. Opens the Battery page. Select Unrestricted (not Optimized). Some phones (Samsung among them) do not open onto that choice — tap Allow background usage (the words, not the switch) to click through and select Unrestricted. Spectre follows that grant when you return.
- **Signatures — export / import** — Export signatures shares a JSON pack of the whole catalog (stock plus any you added or edited, including Decode fields). Save signatures to SD card / storage... writes the same file through the system picker. Import signatures... reads a pack from another Spectre. Same id or the same match rules are skipped, so importing twice does not clone the catalog. Extra rules on a stock row (for example a glob you added to Govee) merge onto the local row; a missing Decode fields map on that stock id is filled from the pack. A new name that already exists is imported as "Name (imported)". Watchlist, filters, settings, logs, and GPS are not in the pack. The file is spectre-signatures-YYYYMMDD.json.
- **Restore default signatures & presets** — Rewrites the catalog (stock rows, class colors, and stock Decode fields maps), stock bookmarks, the full stock filter-chip set (including chips you long-press deleted), and the default Settings switches (Keep screen on, Tag detections with GPS, Online place names, TAK / CoT off, Night mode off). This wipes custom signatures and any chips you saved. Export signatures first if you want a backup. It is not an undo for a single rule. To drop one preset chip, long-press it on Filters.

How the Live display row looks — View, Sort, Title line, Subtitle line, RSSI bars, signature chips, Frequency, first/last, Brief hold — lives on the Live display → Display, not here. New detections only is a Filter.

The Settings footer (under the version line) shows this phone's current IPv4, or none. It refreshes when you return to Settings. Same-phone ATAK CIV: that address is the TAK Host (§5.8).

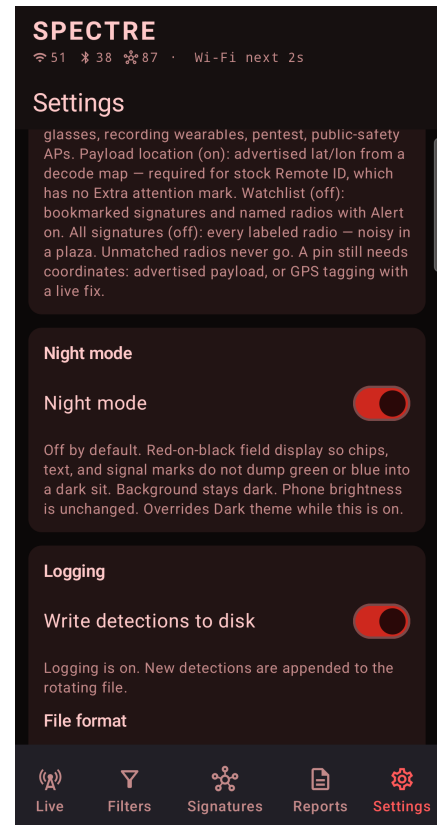


Fig. 6a — Settings → Night mode (on). Red-on-black field display. The card sits under TAK / CoT, above Logging. Phosphor switches and chips become red. Phone brightness is unchanged. Fig. 6 is the same Settings tab with Night mode off. Privacy mode is on: last three MAC octets show as **.*.*.*.

5.8 TAK / CoT feed

Off by default — this puts coordinates on the LAN. The TAK / CoT feed sends Cursor-on-Target UDP markers to whatever is listening on the host and port you set (ATAK, WinTAK, iTAK, or a TAK server). Markers include the full MAC and either this phone's GPS or an advertised payload lat/lon. Privacy mode pauses the feed so those are not sent. There is no Spectre TAK server and no account. You are responsible for who is on that network and for local law. Do not use this overlay as a safety, intercept, or targeting picture.

Settings → **TAK / CoT**, directly under Tag detections with GPS. The master switch ships off. Turning it on does not change the Live display, Filters, logging, or watchlist beeps. It is a parallel UDP feed of radios Spectre already labeled.

5.8.1 What it is, and what it is not

Cursor-on-Target (CoT) is the XML event ATAK already understands. Spectre writes a small event per radio — uid, type, a point, a callsign, and a remarks line — and sends it as a UDP datagram. The default port is 10011 (ATAK CIV's usual CoT input). Host defaults to 239.2.3.1 (SA multicast). For ATAK CIV on the same phone, set Host to the phone's Wi-Fi IPv4 and leave 10011. SA multicast is 239.2.3.1:6969. Many TAK servers use 8087. Unicast host:port is the other shape, for a TAK server or a single EUD.

- **It is** an overlay of radios Spectre heard, at a coordinate Spectre already has.
- **Heard here** means this phone's GPS at last hear (the same stamp as a log row). The other radio is somewhere in earshot, not on that pin.
- **Advertised position** means the BLE advertisement itself decoded to WGS84 (stock Remote ID Location; any custom map whose field ids are latitude / longitude). That pin is what the gadget claimed, not a Spectre DF fix.
- **It is not** a Remote ID plugin, a drone tracker, direction-finding, pairing, GATT, or Wi-Fi monitor mode. It does not join the multicast group (send-only). It does not delete markers — ATAK drops them when the CoT stale time passes (~120 s without a refresh).
- **It is not** the Live display. Filters do not shrink the feed. A radio you hid on the Live display still publishes if it matches What to send and has a pin.

5.8.2 Two kinds of pin

Every published event needs a coordinate. Spectre picks in this order:

- 1 **Advertised payload.** If this radio has a sticky latitude + longitude from a decode map, that pair is the pin. GPS tagging can be off. Stock Remote ID uses this path.
- 2 **Operator GPS (heard here).** Otherwise the last tagged hear on this radio. Settings → Tag detections with GPS must be on, and there must be a live fix (last-known older than 30 s is ignored, same as the log). GPS tagging off and no payload → nothing is sent for that radio.

Remote ID (ASTM F3411 / OpenDroneID on BLE UUID 0xFFFA) rotates message types. A Location message has lat/lon/alt; the next packet is often Basic ID with none. Spectre keeps the last *valid* Location pair on that radio for the session, so the ATAK pin does not blink off between types. The System message's op_lat / op_lon are the **pilot / operator** location in that standard — they are stored, they are *not* the aircraft pin. A 0,0 pair, a non-finite number, or a value outside ±90 / ±180 is rejected and does not clobber a previous good fix.

Heard-here is still this phone. An Axon, Flipper, or Pineapple marker at your GPS is “I heard that radio here,” not “that gadget is at this lat/lon.” Walk, and the pin follows you while the radio stays in earshot. Remote ID Location is the exception: that pin is the advertised aircraft position, which may be kilometers from you.

5.8.3 Generic field ids — Remote ID works without a special case

The publisher never asks “is this Remote ID?” It looks up stable decode field **ids** on whatever signature matched. Stock Remote ID already maps Location to `latitude`, `longitude`, and `alt_geo` (i32 LE $\times 1e-7$ degrees, protocol v2). A custom signature that uses those same ids pins advertised position the same way. There is no TAK checkbox on Decode fields. Labels can say “Aircraft lat”; the id must be `latitude`.

Field id	Role
<code>latitude</code> (aliases <code>lat</code>)	Advertised WGS84 latitude. Required with longitude for a payload pin.
<code>longitude</code> (aliases <code>lon</code> , <code>lng</code>)	Advertised WGS84 longitude. Required with latitude.
<code>alt_geo</code> (aliases <code>altitude</code> , <code>alt</code> , <code>hae</code>)	Optional HAE meters for the CoT point. Sticky with the lat/lon pair.
<code>op_lat</code> / <code>op_lon</code> (aliases <code>operator_lat</code> / <code>operator_lon</code>)	Remote ID System operator (pilot) location. Not the pin. Do not reuse these ids for the aircraft.

How to wire a custom map: Signatures → the row → Decode fields → More on the lat/lon cards → set ID to `latitude` and `longitude` (scale as the spec requires; Remote ID is i32 LE, scale $1e-7$, unit $^\circ$). Save. Settings → TAK / CoT feed on, Payload location chip on. No other TAK setup. §9.6.3.

5.8.4 Turn it on — step by step

- 1 Put the phone and the TAK client on the **same LAN** if you will use multicast (239.2.3.1). Cellular and most guest Wi-Fi will not deliver that group. Some access points filter multicast — then use unicast.
- 2 On ATAK CIV: Manage Inputs should show a UDP CoT listener (this build used 10011 on 0.0.0.0). Stock ATAK also listens to 239.2.3.1:6969 for self-SA. You do not install a Spectre plugin.
- 3 Spectre: Settings → Privacy mode **off** (the feed pauses while it is on).
- 4 Spectre: Settings → TAK / CoT feed → On. Port defaults to 10011 (ATAK CIV). Host defaults to 239.2.3.1 — for ATAK CIV on this phone, set Host to the phone’s Wi-Fi IPv4 and leave 10011. SA multicast is 239.2.3.1:6969; many TAK servers use 8087.
- 5 Leave Extra attention and Payload location on (they ship on). That is the out-of-the-box set: “!” families at your GPS, plus any radio that advertised lat/lon (Remote ID) at that advertised point.
- 6 Scan. An Extra attention radio with a GPS fix, or a Remote ID Location packet, should appear on the TAK map within a few seconds. Callsign is the signature name (or a Named radio label). Remarks start with Spectre · BLE/Wi-Fi · MAC · RSSI · heard here or advertised position.

5.8.5 Host and port

Setting	Default	What to put
Host	239.2.3.1	SA multicast on the LAN, or a unicast IPv4 / hostname for a TAK server or one EUD. Trimmed on save. Empty is ignored so a wipe of the field does not store blank.
Port	10011	UDP port 1–65535. Digits only. ATAK CIV CoT input (default). SA multicast uses 6969. Many TAK servers use 8087.

- **Multicast (default).** Spectre sends; it does not join the group. TTL is 1 (this LAN, not routed). Phone and ATAK must share a Wi-Fi (or Ethernet via a tether) that actually forwards 239.2.3.1. A hotspot that isolates clients will fail; try unicast to the ATAK device's IP.
- **Unicast.** Host = the TAK server or the EUD's address. ATAK CIV on this phone: the phone's Wi-Fi IPv4 and port 10011 (not 127.0.0.1). SA multicast is 239.2.3.1:6969. Many TAK servers use 8087. No multicast membership on either side.
- Spectre already has install-time INTERNET (Online place names). The feed uses that permission for UDP. There is no Spectre cloud; packets go only to the host you typed.
- Restore default signatures & presets resets host/port and turns the feed off.

5.8.6 What to send

Four independent chips, shown only while the master switch is on. A radio publishes when *any* selected chip matches *and* a pin exists. Unmatched radios (no signature) never go, unless you named that one MAC and Watchlist is on with Alert on.

Chip	Ships	Who is selected
Extra attention	On	Any matched signature whose Extra attention text is not empty (stock: Hobby BLE serial, Axon, WatchGuard Video, Ray-Ban / Meta glasses, Snap Spectacles, Fieldy, Plaud Note, Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, Porkchop, Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc, Flock, Penguin, Pigvision, FS Ext Battery, Genetec AutoVu, Rekor, Motorola Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua). Pin is usually heard-here (your GPS).
Payload location	On	Any radio with a sticky advertised lat/lon from a decode map. Stock Remote ID is the reason this defaults on: that row has no Extra attention mark, so without this chip it would never publish. Custom maps with the same field ids are included.
Watchlist	Off	Bookmarked signatures, and Named radios whose Alert is on. Named radios with Alert off (label only) stay off the feed.
All signatures	Off	Every labeled radio. A plaza will flood ATAK. Use it for a short sit, not a walk.

A radio can match more than one chip (Remote ID is Payload location; a bookmarked Axon is Extra attention and Watchlist). It is still one uid, one marker. When many radios qualify at once, Spectre sends at most about 24 per tick, Extra attention and payload pins first, then louder RSSI.

5.8.7 Privacy, GPS tagging, and when nothing is sent

- **Master off** (default) — no datagrams.
- **Privacy mode on** — the feed pauses even if the master is on. Settings shows a line under the switch. Spectre will not send masked MACs or masked coordinates. Turn Privacy mode off to publish; the screen can stay on Privacy for screenshots while you are not feeding TAK.
- **GPS tagging off** — heard-here pins stop. Advertised payload pins (Remote ID) still publish.
- **No live GPS** — same as tagging off for heard-here. Path 0 m on the Live display is the hint. High-accuracy Location, scanning running, wait for a fix.
- **No matching chip** — a Ruuvi, a phone, an unmatched AP does not go unless you turn All signatures or Watchlist (and bookmark it).
- **Invalid advertised coords** — 0,0, NaN, or out of range are ignored. A previous good Location pair on that radio stays.
- **Scanning stopped** — Stop on the scan notification ends the loop. Markers already on ATAK stale out in ~120 s.

Rate limit: a radio is sent when it first qualifies, when it has moved about 30 m, or when about 10 s have passed. Sitting on one corner does not spam ATAK. The CoT stale attribute is about 120 s after the last send, so a radio that left earshot drops off the map without a delete event.

5.8.8 What ATAK shows

- **uid** — SPECTRE-BLE- or SPECTRE-WIFI- plus the MAC without colons. The same radio updates in place; a rotated BLE address is a new marker.
- **type** — advertised drone-class (Remote ID / DJI class) is a-u-A-M-H-Q (unknown UAV). Everything else, including heard-here Extra attention, is a-u-G (unknown ground). Not friendly/hostile affiliation.
- **callsign** — Named radio label if that MAC is named; else an Extra attention signature name; else the first matching signature; else the advertised name; else the MAC tail.
- **point** — lat/lon as above. hae is advertised altitude when known, else the CoT unknown (9999999). ce/le are unknown.
- **remarks** — Spectre · Wi-Fi or BLE · full MAC · RSSI dBm · advertised position or heard here (operator GPS) · up to three signature names · Extra attention text when present.
- **how** — m-g (machine / GPS). Time/start/stale are UTC.

This is not a TAK data package, not a KML, and not a GeoJSON share. If the map stays empty: Privacy mode, master off, wrong LAN, AP multicast filter, or no qualifying radio with a pin. §14 troubleshooting.

5.8.9 Limits you will hit

- Stock Android does not give Wi-Fi Neighbor Awareness Networking / stuffed beacons reliably. Many Remote ID aircraft also broadcast Wi-Fi; Spectre's Remote ID row is the BLE UUID 0xFFFA advertisement. A miss on BLE is a miss on TAK.
- No DF, no range. Heard-here follows you. Advertised position is whatever the gadget encoded, including a bad GPS on the aircraft.
- No pairing, no GATT, no encrypted ads. If lat/lon only exist after a connect, Spectre will never pin them.
- A randomized BLE MAC is a new uid when it rotates. Remote ID often uses a stable module address; phones do not.
- All signatures in a plaza will load ATAK with café APs and headphones. That is the chip working. Turn it off.

- Multicast does not traverse the internet. A teammate on LTE will not see 239.2.3.1 from your phone. Use a TAK server unicast (or their VPN) if you need that.

5.8.10 Field checklist

A drone sit with ATAK open is §12.15. Extra attention overlay (body-cam / glasses / pentest) is the same switch with Payload location optional. For decode-map ids, §9.6.

6. Visualization Modes

All five views show the same filtered radios. If radar looks empty and the list does not, check the filter first. Radar draws stale devices as dim blips so sitting on the Filters tab does not blank the plot. List, hybrid, and timeline share Display → Title line and Subtitle line; radar labels stay a short signature name or type guess. Display is how you pack more or less onto each row without changing who is on the air (§5.3).

6.1 Classic radar

What it shows. A polar plot. You are the center. Each filtered device is a blip.

- Radius is last-heard RSSI. Rings are labeled -40, -60, -80, -100 dBm. Stronger signals sit closer to YOU. Brief hold keeps the blip on that ring; it does not crawl outward.
- Angle is a hash of the MAC. It is a stable seat, not a compass heading.
- Color is the first matching signature palette (class colors, §9.5), or the RSSI palette if unmatched (≥ -55 green, -55 to -70 amber, -70 to -85 orange, else red).
- Named (signature-matched) blips are larger, haloed, and labeled with the first matching signature name (list views can show three).
- A watchlist hit pings that blip for one second: two expanding rings and a bright core (the same flash as the list — pip, spoken class, or both). After the ping, a phosphor ring stays on that blip for the rest of the session (same radius as the list bell). Alerted blips are drawn on top. Distinct from the named class-color halo.
- Dim blips are last-seen positions after the stale window.
- The rotating sweep is a scope cue only. The bright line leads; the shaded fan trails. It does not scan azimuth.
- Tap a blip for detail. Long-press to create a signature.

How to interpret. A cluster of bright, close blips is a dense, loud RF neighborhood — typically the room you are in plus nearby APs. A named halo that walks inward on successive looks is something getting stronger, not necessarily closer in a straight line. If every blip is dim, the radios have not been heard within the stale window; check that scanning is still running and Location is on.

When it is most useful. Walk-throughs, briefings (“how busy is this plaza”), and watching whether a named signature is in the current bubble without reading MACs.

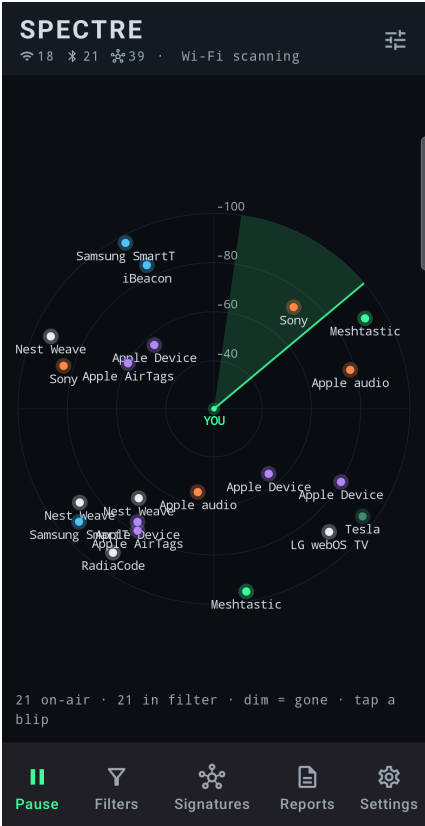


Fig. 7 — the Live display, Classic radar. YOU at the center. Radius is RSSI (not meters). Angle is a hash of the MAC, not a compass heading. Named signatures are labeled. Privacy mode is on: last three MAC octets show as `***:***:`.

Strengths	Weaknesses
Instant density and relative loudness. Signature color pops.	No real bearing. Overplotting in crowded 2.4 GHz spaces. Labels omitted when more than 24 unnamed devices are up.
Same filter as the list, so Signatures only becomes a clean threat scope.	Easy to over-read radius as meters. Indoor multipath will jump a blip in and out.

6.2 Strength-ranked live list

What it shows. Devices in Display → Sort order (default Strongest averaged over 30 s; every option is in §5.3). This is the view Display was built for: one row per radio, as dense or as spare as you set. Each row: class glyph in a circle (unmatched = ?); first line from Title line (default MAC); second line from Subtitle line (Wi-Fi or Bluetooth icon, then Name + type plus rand/gone). Subtitle None hides that line and moves rand/gone onto the title — the kind icon goes with the subtitle so the title stays a MAC or name. Vendor is not on the list — open detail. When Frequency is on, channel · MHz sit under the RSSI on the right. Optional >>/<< trend, RSSI bar, first/last ages, and compact signature chips (up to three; further matches on detail). A hexagon in a chip means that signature has a Decode fields map; the values are on detail (§5.4, §9.6). A bookmarked new hit flashes the row for one second when the watchlist fires. A phosphor bell stays on the chip row for the rest of the session. Fit-the-row recipes: §5.3.

How to interpret. Work top-down unless Sort is New at bottom (then new rows append). A sudden new row near the top is something loud that just appeared (strongest / newest sorts). A name of <hidden> is a hidden SSID AP. On the subtitle, unnamed BLE is unnamed (no advertised name and no useful decode — the Bluetooth icon already marks LE). Title Advertised name still says unnamed LE. Apple, Inc. · AirTag... on the subtitle is Name + type, a type guess. IEEE vendor is on the detail page. The number on the right is RSSI.

When it is most useful. Default working view. Triage, long-press to create signatures, confirm a filter, or pick a device to watch. Thin it (Subtitle None, extras off) in a plaza before you start hiding radios with Filters.



Fig. 8 — the Live display, Strength list. Class glyph, MAC title (Privacy mode), Wi-Fi / Bluetooth icon on the subtitle, RSSI, optional bar / Frequency / first-last. Tune (top right) is Display. Privacy mode is on: last three MAC octets show as ****.*.

Strengths	Weaknesses
Readable identifiers. Best place to act (tap / long-press / watch). You can change how much each row shows.	No spatial intuition. Easy to miss a weak named device if you only look at the top.
Gone rows stay visible so you can still inspect history. Subtitle None packs more rows without dropping radios.	Very dense environments still produce a long scroll — then use Filters, not only Display.

6.3 Timeline view

What it shows. One card per filtered device heard in the last 15 minutes, in Display → Sort order. Title and subtitle follow the same Display lines as the list, including the class glyph and the Wi-Fi / Bluetooth icon on the subtitle. Under that identity is a time strip: left = 15 minutes ago, right = now. Solid segments are on-air windows; gaps are real dropouts. The number is current RSSI, not an average over the bar; Frequency, if on, sits under it. The clock ticks about once a second so a sitting radio's bar grows toward the right until it fills the window.

How to interpret. A continuous growing bar is a fixture (home AP, or a phone that keeps advertising). Wi-Fi scan waits do not count as dropouts — sitting APs stay one bar. A new segment appears only after the device is marked gone (longer of Stale and Brief hold) and then heard again. A short tick that repeats is a low-duty advertiser. A bar that starts when you turned a corner and ends when you left is geometry, not “the device left the universe.” After 15 minutes of continuous presence the bar is full width; it does not grow past the left edge.

When it is most useful. Sits, vehicle rides, and any question of the form “did this come and go with me.”

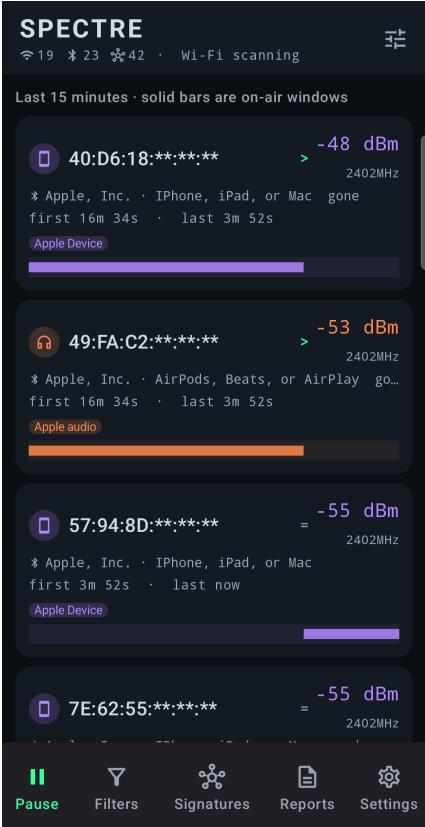


Fig. 9 — the Live display, Timeline. 15-minute presence bar under each radio. Solid = on-air. MAC tails hidden (Privacy mode). Privacy mode is on: last three MAC octets show as ***:***:***.

Strengths	Weaknesses
Makes appearance / disappearance obvious. Good for log correlation.	Fixed 15-minute window. No multi-hour strip in-app (use the exported log).
Independent of radar's RSSI-radius metaphor. Same Title/Subtitle as the list.	UUID still waits for detail. Channel · MHz only if Frequency is on.

6.4 Hybrid view

What it shows. The strength list (same Title/Subtitle, Sort, and extra-fact switches) plus a full-width sparkline of recent RSSI samples (up to 40 points kept per device) and the same >> / << trend mark. Scale is fixed: top = −30 dBm, bottom = −100 dBm. A faint 10 dB grid with a left-hand scale (−30 / −50 / −70 / −100) and four vertical columns. A dot marks the newest packet. There is no banner on this view; the grid is the legend. In a plaza, drop back to Strength list and Subtitle None before you blame the sparkline for the clutter.

How to interpret. Left = older packets, right = newest, up = stronger. A line high on the strip is loud. A line near the bottom is weak. A flat line at one height means the radio is sitting still at that strength — that is the usual Wi-Fi picture between 30 s scans. The line only climbs or drops when RSSI actually changes. After 40 samples the waveform shifts left. The chevron is the glanceable answer: >> much stronger (~+8 dB), > stronger (~+3 dB), = steady, < weaker, << much weaker. A sawtooth is often a duty-cycled BLE advertiser, not motion.

When it is most useful. Following one or two candidates while still seeing the rest of the field. Better than radar for “is this getting louder.”

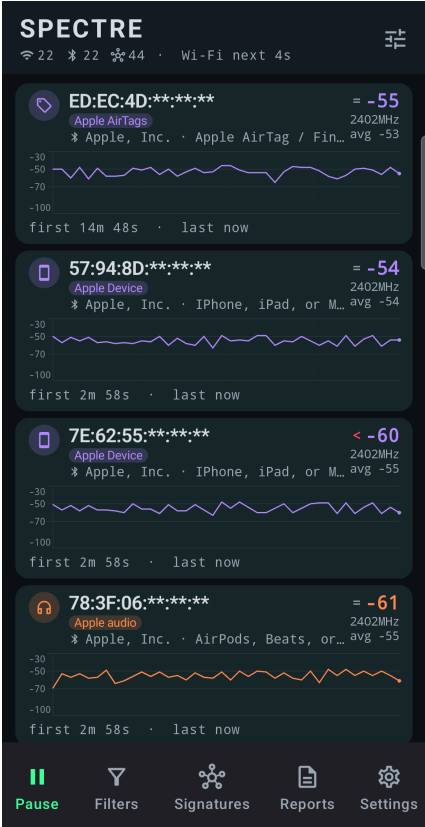


Fig. 10 — the Live display, Hybrid. Strength list plus RSSI sparkline (fixed −30 to −100 dBm). RSSI bars off so the sparkline is the strength picture. Fig. 1 is this view too. Privacy mode is on: last three MAC octets show as ***:***:***.

Strengths	Weaknesses
Combines identity and trend without leaving the list. Fixed scale so a sitting radio is a flat line at a real dBm height.	Still no bearing. Detail has a larger copy of the same plot.
Same actions as the list (tap / long-press).	Still no bearing.

6.5 By class

What it shows. An outline of the same filtered Live display. Every signature class (Audio, Wearables, Cameras, ...) is a row with the class glyph, A–Z by class name. Show all (default) keeps empty classes dimmed at 0 so the list does not jump; Collapse empty hides those zeros. The chips sit at the top of the list and scroll off as you go down — they are not sticky. Tap a class to list the signatures in it (A–Z by name; a hexagon on the signature row means that row has a Decode fields map); tap a signature to list those radios (same class glyph, Title/Subtitle, RSSI bars, signature chips, Frequency, and first/last as the strength list); tap a radio for the existing detail page. Back returns to the outline. Unmatched is last. This is a Display view, not a Report — Debrief is still the 15-minute snapshot on Reports.

How to interpret. Counts are radios, not packets. A dual-chip radio (Tapo on a TP-Link OUI, for example) appears under each class it matched; the header still reports unique radios on the Live display, and “N in more than one class” when that happens. Filters still apply: Show only Cameras → Cameras has the hits and the other classes sit at 0 (unmatched is 0 too) unless Collapse empty is on. Tapping a class does not turn on Show only. Pause, New detections, and Moving with you work the same as the list. A watchlist hit opens that class and signature so the row can flash (and jumps so those headers stay on screen when the radio is close enough, if Jump to new watched detection is on). A phosphor bell stays on that radio row for the rest of the session.

When it is most useful. A sit where you want “how many of each family” before you pick a radio. Plaza noise is the Unmatched count.

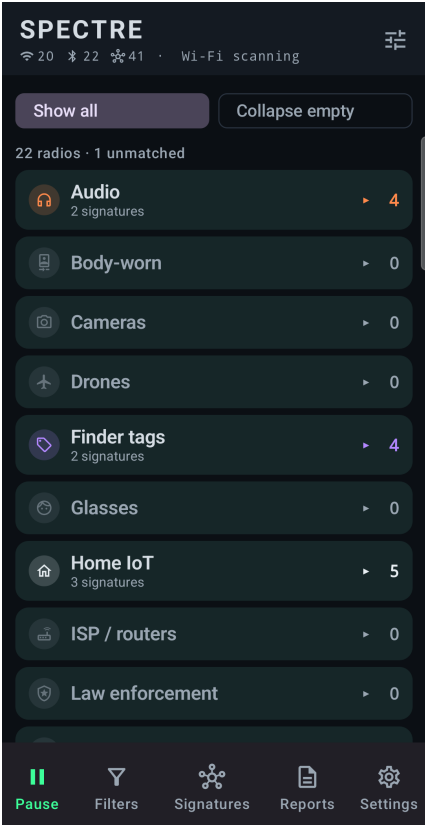


Fig. 6b — the Live display, By class. Class glyph on each header (unmatched = ?). Show all keeps empty classes; Collapse empty hides zeros. Open a class, then a signature, then a radio. Radio rows use the same class glyph and Display switches as the list. Privacy mode is on: last three MAC octets show as ***.**.*.

Strengths	Weaknesses
Class totals without walking the list. Same radios as Live display.	No RSSI plot. Dual-class radios appear twice if you expand both.
Tap through to the same detail page. Display extra facts apply to the radio rows.	A Cameras filter leaves other classes at 0 — tap All traffic first for the whole plaza.

7. Detection Methods

7.0 How collection fits together

This chapter is how Spectre hears radios on stock Android. You can skip the API names and still run the app; they are here so a technical reader can see what the phone will and will not expose.

One pipeline: **hear** (this chapter) → **match** named signatures (§7.3–7.4, §9) → **filter** which radios appear on the Live display (§8) → **show and log** (the Live display, watchlist, CSV/JSONL) → **report** (Debrief / AI Export / Signature candidates, §11). Filters never change what was heard, and Display never changes the filter. Debrief ignores the Live display filter and uses the last 15 minutes of memory. Signature candidates reads the rotating log, not that RAM map.

Two surfaces only: Wi-Fi **access-point** scan results, and BLE advertisements. No cellular, no Wi-Fi clients, no Bluetooth Classic inquiry, no direction finding.

7.1 Wi-Fi passive scanning

Spectre registers for `SCAN_RESULTS_AVAILABLE_ACTION` and a `ScanResultsCallback`, and calls `WifiManager.startScan()` on a quota-aware interval. Each `ScanResult` becomes an observation of kind `WIFI` with BSSID, cleaned SSID, RSSI, channel derived from MHz (2.4, 5, and 6 GHz bands), hidden-SSID flag, and extras (capabilities, and on API 30+ `wifiStandard` and `channelWidth`).

A successful scan returns every AP in one batch; the OS then requires a wait. High performance asks about every 30 s, balanced 40 s, saver 55 s. If `startScan` is refused, Spectre backs off (up to 45 s) and shows Wi-Fi waiting on OS. Cached results are not treated as a fresh batch. The last AP set is held on screen during the wait, and empty result lists are ignored. Signature matching only runs on radios from those batches. An AP you drove past between scans is not labeled, not logged as a new hear, and does not beep — even if its OUI is in the catalog. Faster Wi-Fi AP scans (§7.1.1, §10.3.1) is the optional way to shrink that gap after the OS throttle is off.

7.1.1 Faster Wi-Fi AP scans (optional)

Stock Android is the bottleneck, not Spectre's matcher. A Cradlepoint IBR, an AirLink OUI, a UniFi vendor IE, a Cisco BSSID, or a hidden fleet SSID is only tagged when a `startScan()` batch actually contains that BSSID; between batches the catalog is idle. On a sit that is usually fine: the same house and campus APs come back every 30 s. On a drive it is the difference between catching a vehicle or roadside AP and missing it entirely.

How far you roll between batches, approximate:

Speed	Stock High performance (~30 s)	Faster scans (~8 s)
25 mph (city)	about 335 m	about 90 m
45 mph	about 600 m	about 160 m
65 mph (highway)	about 870 m	about 230 m

A patrol or fleet gateway (Cradlepoint IBR600/IBR1100/IBR1700, Sierra Wireless AirLink, some Compex / Novatel / Utility Inc radios) is often only in RF range for one short pass — a few hundred meters at speed, sometimes less with body, glass, and other cars in the way. Hidden SSIDs still expose the MAC, so the OUI rule is the usual hit, but the phone still has to scan while that MAC is audible. One stock batch every 600–870 m can skip the whole pass. Batches every ~8 s give several chances to hear the BSSID, run the signature, show the chip, and (if that row is bookmarked) beep. That is the point of the switch: interesting APs by signature, sooner, while you are moving.

It does not make Wi-Fi continuous. Each scan is still a full AP list, then a wait. It does not see Wi-Fi clients. It does not change BLE. It is not Hunt (Hunt stays BLE-only because Wi-Fi will never be packet-by-packet). RSSI on a drive is

still not meters. Keep Tag detections with GPS on if you want that hear on the path and in Debrief.

How to turn it on: Android 11 or newer. Settings → About phone → tap Build number until Developer options exist. Developer options → **Wi-Fi scan throttling** → Off. Return to Spectre → Settings → Faster Wi-Fi AP scans → On. Spectre reads `WifiManager.isScanThrottleEnabled()` before it will accept the switch; if the OS is still throttling, the switch stays off and a dialog explains, with Open developer options. Spectre cannot change the OS switch. If you turn OS throttling back on, the Spectre switch may stay saved on but scans return to ~30/40/55 s and the header can read Wi-Fi fast scan needs Developer options. Come back to Settings and the caption updates on resume.

While it is actually in effect, Spectre skips the four-scans-per-two-minutes quota and asks about every 8 s. If `startScan()` starts returning false, the existing backoff still applies (up to 45 s) so a hostile OEM limit does not spin. Battery and heat go up; use it for the drive or the first minutes on a corridor, not an all-day pocket sit. Battery saver intensity still applies to BLE. Turn Faster Wi-Fi AP scans off when you no longer need the extra AP batches.

Every WIFI observation is an access-point-class emitter: a router, extender, mesh node, phone hotspot, or IoT/camera radio that is advertising an SSID. The Live display labels these AP. A device that is only associated as a station — a phone on café Wi-Fi, a laptop, a camera talking to its own AP — is not in the scan results and will not appear, even if it is a foot away. Hidden-SSID APs still appear (empty name, HIDDEN flag) because they are beaconing; hidden-SSID *clients* do not.

This is not promiscuous capture. Spectre never opens a monitor interface and never sends probe requests beyond what the OS scan already does. There is no continuous Wi-Fi search on a stock Galaxy. If the job needs stations, probes, or frames to a specific MAC, use a dedicated sniffer in parallel.

7.2 Bluetooth Low Energy scanning

A single `BluetoothLeScanner` runs in the foreground service with a match-all `ScanFilter` (so Samsung does not treat it as an unfiltered screen-off scan). Ads are queued with drop-oldest backpressure; the service batches them instead of starting a coroutine per packet.

Intensity	BLE ScanSettings	Wi-Fi interval
High performance	SCAN_MODE_LOW_LATENCY, CALLBACK_TYPE_ALL_MATCHES, MATCH_MODE_STICKY. Recycled ~70 s on / ~2.5 s rest. If the OS parks the scan (no ads ~18 s), rest 8–12 s and continue in BALANCED until it recovers.	30 s *
Balanced	SCAN_MODE_BALANCED, same match policy, longer on-period (~180 s)	40 s *
Battery saver	SCAN_MODE_LOW_POWER, recycle about every 20 minutes	55 s *

* *Wi-Fi column is the stock OS-safe wait. About 8 s instead while Faster Wi-Fi AP scans is in effect (Developer options throttle off + Settings switch). §7.1.1, §10.3.1.*

Each advertisement yields address, local name, RSSI, service UUIDs (including keys from service data), the first manufacturer ID and a short manufacturer hex, and extras (primary PHY, txPower, advertise flags when present). Spectre does not connect, pair, or read GATT characteristics for detection. Do not bounce the scanner every few seconds — that is what produced 100+ start/stop cycles and a dead BLE list on Samsung.

This is BLE only. Spectre never calls `BluetoothAdapter.startDiscovery()` (Classic BR/EDR inquiry). That scan transmits, typically pauses BLE on the same adapter, and is how HC-05 / HC-06 serial modules show up in Android Bluetooth settings. They will not appear on the Live display. A BLE advertiser may still *claim* dual-mode in its flags or Class of Device; that is self-description, not a Classic scan. §3.5.

7.3 Identifiers the matcher uses

Field	Typical origin	Notes
OUI / MAC prefix	First 1–6 octets of the address	OUI is 24-bit. Full MAC is more specific but fragile on randomized BLE.
Name / SSID	Wi-Fi SSID or BLE AD local name	Case-insensitive substring or glob (* and ?).
Service UUID	BLE AD type 0x02/0x03/0x06/0x07 or service data keys	16-bit values are compared as both short and Bluetooth-base 128-bit forms.
Manufacturer ID	BLE AD type 0xFF company identifier	Apple 0x004C, Samsung 0x0075, Tile 0x00C7, XUNTONG 0x09C8, etc.
Manufacturer data prefix	Bytes after the company ID	AirTag Offline Finding uses 0x12 as the first payload byte.
Radio kind	WIFI or BLE	Do not OR this alone or the signature matches every radio of that type.
Hidden SSID	Empty SSID on a Wi-Fi result	Matches the class of hidden APs. Combine with a full BSSID (AND) to follow one radio.
Vendor IE OUI	802.11 element 221 when the OS returns IEs	LiteOn 00:80:19 / 00:0A:EB on Flock. Many phones strip IEs from scan results.
Co-occurrence	min peers + window + cluster-by-OUI / sequential MAC	Optional on a custom signature. Stock Unknown Signature does not use clustering.

7.4 How a named signature is applied

- 1 Each new or updated sighting is stored by key `KIND:MAC`.
- 2 Every signature is evaluated. If match-any is on, one hitting rule is enough. If it is off, every enabled rule must hit.
- 3 Signatures with min-peers, cluster-by-OUI, or sequential-MAC run a second pass over currently live devices inside the peer window (default 60 s).
- 4 A device may match several signatures. Chips show up to three names; detail lists every match; the log joins all names with +. A store iBeacon UUID on top of stock iBeacon is the usual dual-label (\$5.5, \$9.2).

Re-matching also runs when you save a signature or toggle a rule, so a newly created signature labels devices already in memory without waiting for the next packet.

7.5 In-memory set size and crowded rooms

Every phone in a plaza can rotate its BLE address, so “unique devices” grow without bound if every MAC is kept. Spectre caps the live map at 400 radios, dropping the oldest unnamed devices first. Unnamed BLE older than about three minutes is evicted; named / signature-matched devices stay up to 15 minutes after last seen. In a dense environment you still see a full list, but it is the strongest and most recent 400, not every random MAC that walked by. Debrief and AI Export read this same map, not the log — on a drive the start of the trip can already be gone from RAM (§11.4.1). Logging also samples in a flood (new radios, or every 25th hit, at most 16 lines per batch) so the disk mutex does not freeze the UI. Radios still inside Brief hold are not dropped by the crowd cap.

7.6 Per-rule enable

Each rule on a signature has its own on/off switch in the editor. Off keeps the rule in the list but it does not match. Custom and built-in signatures work the same way.

Typical field use: open **Flock Safety Cameras**, turn off the LiteOn / Espressif OUI rules, leave B4:1E:52 and the Flock-* name globs on. Match-any still means one enabled rule is enough. Penguin and Pigvision are name-only — if you switch every name rule off, they will not match until you turn one back on. Unknown Signature is name-only (ESP_*, ANDROID-, DIRECT-, UNIT-).

Stock Android cannot see wildcard probe requests from a hidden Flock STA, or frames addressed to a Flock MAC. See §7.6.1 for what a Flock/camera match is worth in the field.

7.6.1 Real-world limits of Flock / ALPR / camera matching

A colored chip that says Flock Safety Cameras is a **pattern match on a public broadcast**, not a visual identification, not a serial number, and not proof that a Flock pole is at your GPS pin. Use it as a cue to look with your eyes, then write what you actually saw.

The phone's radios are the limit, not just the signatures. Spectre is a stock-API listener on a phone. It cannot put Wi-Fi in promiscuous or 802.11 monitor mode, cannot capture raw frames or probe-only stations, cannot lock onto one channel, and cannot hear LTE/5G at all. If the camera is quiet on Wi-Fi and BLE, this phone will not invent a detection. A dedicated sniffer (external adapter / monitor-mode device) is a different collection system — Spectre does not replace it.

Cellular-first and quiet radios. Current Flock-style and similar ALPR/camera poles often treat the carrier modem as the primary path. Wi-Fi is not a standing beacon for the neighborhood; it is a commissioning or fallback radio. After install it may be powered down, hidden, or up only when a tech is on site. BLE, if present at all, is usually a short-range maintenance advert, not a continuous billboard. Drive-by during the week of installation can light up Flock-* SSIDs; the same block six months later can be RF-silent on the bands this phone can see while the cameras are still up and recording. **No chip does not mean no camera. A quiet pole is the expected mature state.**

The camera may not be on Wi-Fi at all. LTE/5G backhaul with the Wi-Fi radio off is normal. Spectre only sees access-point beacons. A camera that is only a modem, or only a client on a city SSID, does not appear.

You never see the camera as a station. Stock Android does not report associated clients, hidden-SSID stations, or probe-only devices. You cannot watch a Flock radio join a network. You only see it if that radio is *beaconing* an AP (including a hidden SSID AP, which shows as <hidden> plus BSSID — the name rules will not fire until the SSID is visible).

Only one OUI is actually Flock's. IEEE MA-L B4:1E:52 is registered to Flock Safety. Treat that, especially with a Flock-* SSID, as high confidence. The other ~28 prefixes in the catalog are LiteOn, Espressif, and similar module vendors. Those chips ship in printers, plugs, toys, and cameras that are not Flock. An OUI-only hit on 3C:71:BF (Espressif) is a weak hypothesis. Silicon Labs OUIs on FS Ext Battery are the same story: lots of unrelated IoT.

Names help, and they lie. Flock-ABCDEF during provisioning is a strong name. A substring “Flock” on an unrelated SSID is weaker. Penguin / Pigvision are name-only and low uniqueness. Optional catalog rows (Verkada, Axis, Hikvision, ...) are also mostly names, ship **off**, and will label any AP that chose that word. Do not write “Hikvision camera” in a log because an SSID contained Hikvision.

Raven is a different radio. Acoustic ShotSpotter / Raven sensors are not the ALPR camera. Stronger digital fingerprints are BLE UUID 0x3100–0x3500 and manufacturer 0x09C8 (XUNTONG), plus OUI D4:11:D6. The name “RAVEN” alone is easier to collide with.

Vendor IEs are often missing. LiteOn 00:80:19 / 00:0A:EB in element 221 is a nice corroboration when the OS includes IEs. Samsung scan results frequently omit them. A silent vendor-IE switch is normal, not proof the AP is clean.

Geometry is not identity. RSSI is not meters. Radar angle is a hash of the MAC, not a bearing. The GPS on a detection is the *phone's* fix at hear-time. A loud Flock-* SSID means the AP is nearby enough to decode a beacon, not that the pole is at that coordinate. Walk, watch RSSI, and look up.

Absence and presence are both easy to over-read. Duty cycle, OS Wi-Fi throttle (~30 s batches), 5/6 GHz APs that drop while 2.4 GHz remains, hidden SSID, cellular-first / quiet radios, and a disabled signature all produce an empty list. Conversely, a LiteOn OUI plus a generic camera SSID in a retail park is usually someone else's kit. Bookmark the signature if you want a beep; still confirm visually before you call it infrastructure. A sit that finds nothing on a known camera corridor is still a valid sit — it means this phone heard no matching *broadcast*, not that the corridor is empty.

How to talk about a match. High: B4:1E:52 and/or SSID Flock-* / FLCK, or Raven UUID 0x3100–0x3500 / mfg 0x09C8. Medium: FS Ext Battery by name, or Flock name without the IEEE OUI. Low: LiteOn / Espressif / Silicon Labs OUI alone, Penguin/Pigvision name, optional catalog name-only cameras. Never: “this is a Flock camera” from RF without a visual or the IEEE OUI (or a clear Flock-* SSID). Pattern match ≠ plate, person, or serial.

7.7 Offline assigned-number databases

Vendor and UUID names are not fetched at runtime. Spectre ships a packed binary (assets/lookups/radiodb.bin, binary-searched) built from the current official listings: IEEE MA-L / MA-M / MA-S / CID CSVs and the Bluetooth SIG Assigned Numbers YAML (company identifiers, GAP appearance, 16-bit service UUIDs). Class of Device is decoded from the Core Assigned Numbers bitfields. Rebuild with `python3 scripts/build_lookups.py`. Randomized MACs skip OUI lookup. Beacon interval is not an Android ScanResult field and is omitted when the stack does not provide it.

7.8 Advertisement payload decode

After the standard AD header, manufacturer-specific bytes are the vendor's private format. Spectre decodes layouts that are public specs or well-documented field research, and shows those fields above the raw hex on detail:

- **iBeacon** — Apple type 0x02 / 0x15: UUID, major, minor, calibrated TX.
- **Google Fast Pair** — service 0xFE2C. Three-byte model ID = pairing mode (named from a local well-known list). Longer payloads = already-paired account-key bloom filter (show/hide UI).
- **Apple Continuity** — company 0x004C TLV stream: Proximity Pairing / AirPods (0x07), Find My (0x12), Nearby Info (0x10), Nearby Action (0x0F), AirDrop, Handoff, Hey Siri, AirPlay, Instant Hotspot.
- **Eddystone** — service 0xFEAA. UID, URL, TLM, and EID frames are kept separately on detail once heard (labeled on the raw line). They do not overwrite each other as the beacon rotates.
- **Microsoft** — company 0x0006 Nearby Sharing / Swift Pair device class when the beacon type is recognized.

There is no official catalog of arbitrary 0xFF payloads. Rebuild the IEEE/SIG name tables with `python3 scripts/build_lookups.py`. Fast Pair product names are a curated local list, not Google's full partner catalog.

8. Filtering System

Filters are how you stop staring at every radio in a plaza. They sit on the second tab, next to the Live display, because the usual loop is look at the list, then trim it. Nothing on this tab turns the radios off. Debrief, AI Export, and the on-disk log ignore the Live display filter *and* which view is open (radar / list / timeline / hybrid) — they still see the 15-minute memory / the file. If the list goes empty, undo the last switch before you assume the area is clean.

Do not reach for Filters first when the list is merely *busy*. Live display → Display can hide the second line, bars, Frequency, and first/last without dropping a radio (§5.3, §12.11). Use this tab when you actually want fewer radios on the Live display.

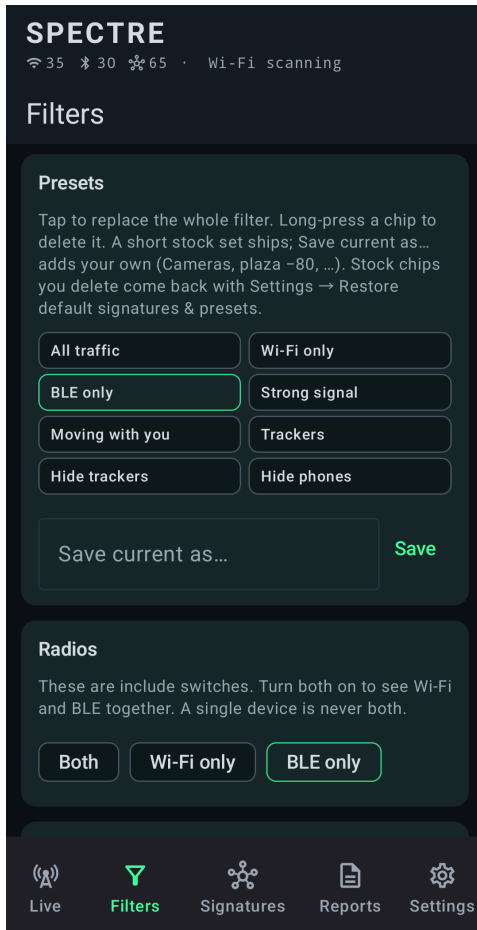


Fig. 3 — Filters, top. Presets section (eight stock chips, two across) then Radios. A tap replaces the whole filter. Save current as... is under the chips. BLE only is selected here. Privacy mode is on: last three MAC octets show as **:**:**.

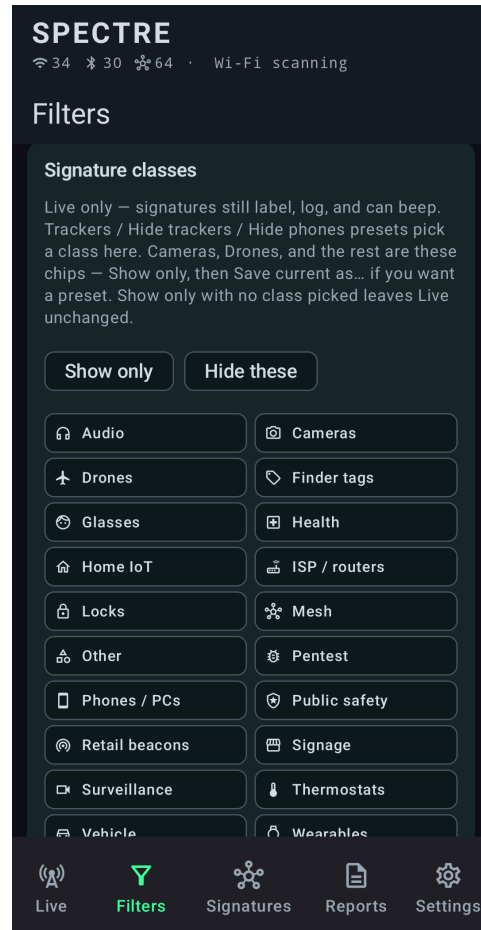


Fig. 4 — Filters, Signature classes. Show only / Hide these plus two-across class chips (Health, Public safety, ...). Show only selected / Hide selected open a list of families further down. Privacy mode is on: last three MAC octets show as **:**:**.

8.1 Architecture

Hearing radios and showing them are separate jobs. Spectre always records what it hears; the filter only changes what the Live display (radar, list, timeline, hybrid, By class) shows. The log still writes every observation, including radios the current filter would hide, so you can thin the list without punching holes in the file.

On the Filters tab, top to bottom: presets, radios to show, Moving with you, New detections only, Signatures only, Named radios only, Hide Fast Pair account-key, signature classes (Show only / Hide these plus class chips), Show only selected signatures, Hide selected signatures (class A–Z lists open under those switches), RSSI / name / OUI, Extra filter logic (AND/OR). Reset filter clears every clause, including class picks, remembered show/hide picks, and New detections only (the already-seen set is cleared).

On the tab, in the same order as Figs. 3–4:

- **Radios to show** — Both, Wi-Fi only, or BLE only. A device is never both.
- **Moving with you** — GPS co-travel, loud radios only (most GPS-stamped samples at about –75 dBm or stronger). Needs Tag detections with GPS (live updates, not a stale last-known) and ~45 m of path. A house AP you meet at the door does not count; a bag or car tag does. “Still here” grows with your recent speed and is longer for Wi-Fi than for BLE (§8.5). A second phone usually will not: BLE MAC rotation starts a new radio with an empty trail. The switch starts a follow test on all radios (clears Signatures only / Show only / Named radios only; Hide these stays). Always AND. Live display → Start over clears the path and trails, not the log. Debrief still writes the last-15-minute tracking section even if this is off.
- **New detections only** — hide radios already here. Always AND. Mark seen / Reset seen sit on the Live display (§5.3.2). First turn-on snapshots what is on the air. Randomized BLE addresses still look new. All traffic or Reset filter turns this off and clears already-seen.
- **Signatures only** — hide radios that match no signature. While class Show only or Show only selected signatures is narrowing the Live display, this is already true: the switch stays on and disabled until you turn those off.
- **Named radios only** — hide radios that do not have a custom name (Settings → Named radios). Alert can still be off. Not the same as Signatures only. A random / privacy MAC will not follow a rotation. the Live display shows a Named radios only strip while this is on.
- **Hide Fast Pair account-key** — drop plaza Fast Pair chips that are already paired, when Fast Pair is the only signature on the row. Pairing-mode stays (chip Fast Pair pairing, subtitle pair). A Pixel that also matched Google still shows. This is not Hide selected Fast Pair, which removes the whole family including pairing-mode. Always AND. What the two payloads mean: §9.5.
- **Signature classes** — Live display only; signatures still label. **Show only** keeps radios of the class chips you pick. **Hide these** drops those classes and leaves the rest, including unmatched radios. Show only with no class picked leaves the Live display unchanged. Class chips sit two across with the same glyphs as the Live display. Trackers / Hide trackers / Hide phones presets pick a class here; Cameras / Drones / Surveillance and the rest are these chips. Class vs color: §9.5.
- **Show only selected signatures** — only radios matching the families you pick stay on the Live display. The list opens under the switch, grouped Class A–Z (tap a class to open its signatures, same outline as the Signatures tab). Empty list = no extra include. Separate picks from Hide selected.
- **Hide selected signatures** — hide one family (your bag AirTag) without hiding the whole Finder tags class. The list opens under the switch, grouped Class A–Z (tap a class to open). Empty list = hide nothing. Picks are remembered if you turn hide off and on again.
- **RSSI / name / OUI** — last RSSI ≥ slider (default –100 dBm = off); substring of name or MAC; substring of MAC or vendor string.

Class Show only is how you get “just trackers” or “just cameras” on the Live display. Hide these (or Hide selected) only takes a family off the Live display — it still labels, still logs, still appears in Debrief, and can still beep if

bookmarked.

8.2 AND vs OR

AND (default): every active clause must pass. Use this for a focused sit (“BLE + signatures only + RSSI $\geq$ -70”).

OR: radios, Signatures only, Named radios only, class Hide these, Hide selected, Moving with you, and New detections only still apply. Among the optional clauses (name query, OUI query, RSSI floor if raised above -100, class Show only), any one pass is enough. Use this when you want a broad search (“name contains Flock OR OUI contains B41E52”). AND/OR never overrides a hidden class or signature — if Hide these Finder tags is on, AirTags stay off the list even in OR mode.

8.3 Signatures only, classes, and selected families

Signatures only hides radios that match no signature, so the Live display is pattern hits of every class. If the list goes empty, nothing matching is in earshot — signatures always label; Filters only hide radios from the Live display. Class **Show only** or Show only selected signatures already hides unmatched radios, so this switch stays on and disabled until you turn those off. **Hide these** does not hide unmatched — Signatures only stays a real switch then (named hits minus the hidden classes, or the whole field minus those classes).

Signature classes group stock (and custom) rows into Finder tags, Retail beacons, Signage, Wearables, Surveillance, Drones, Pentest, Public safety, Vehicle, Glasses, Audio, Cameras, Thermostats, Locks, Health, Home IoT, ISP / routers, Mesh, Phones / PCs, and Other. Each signature has one class, set in the editor. Public safety is Axon / WatchGuard Video (body-worn and in-car) plus public-safety vehicle APs (Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc). Many of those radios are used in law enforcement; they are not exclusive to it. Government, municipal, and other corporate fleets likely run some of the same kit. Filters → **Show only** plus those chips keeps matching radios of the picked classes. **Hide these** drops the picked classes. Neither button turns labeling off — a hidden class still labels, still logs, still appears in Debrief, and can still beep if bookmarked. Stock presets that pick a class: Trackers (Show only Finder tags, BLE), Hide trackers, Hide phones (Phones / PCs). Cameras, Drones, Surveillance, and the rest are these chips — Show only, then Save current as... if you want that sit as a named preset. See §9.5 for which stock rows sit in which class.

Hide selected signatures removes the individual families you pick and leaves everything else, including unmatched radios. Use it when you want one row gone (your bag AirTag) without hiding Tile and Chipolo. Turn the switch on to see the indented list, then turn a signature on to hide it. Turning Hide selected off collapses that list and stops hiding, but the same signatures stay picked. Reset filter is what forgets the picks. Class Hide these and Hide selected can be on together.

Signatures only plus Hide these Finder tags is named hits minus bag-tag clutter. Show only Surveillance is the camera/ALPR sit. They are not locked against each other.

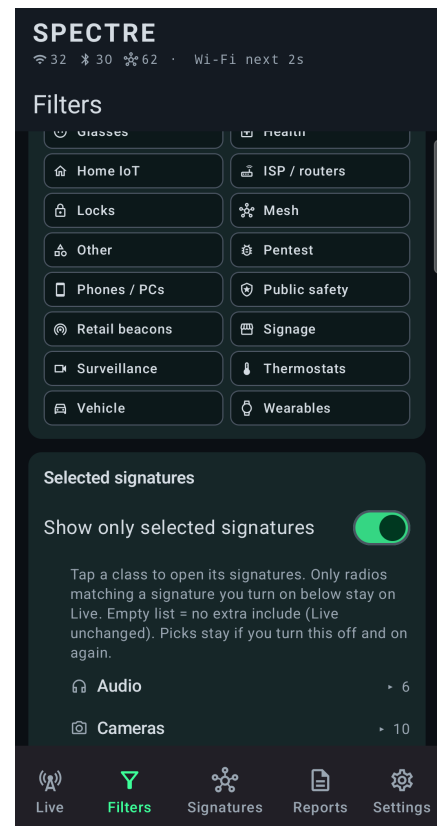


Fig. 4b — Selected signatures. Show only selected signatures on. Class chips above; under the switch, families are grouped Class A–Z (tap a class to open). Empty picks leave the Live display unchanged. Hide selected is the same idea, inverted. Privacy mode is on: last three MAC octets show as **:**:**.

8.4 Presets

The top of Filters is two-across chips. A tap **replaces the whole filter** (radios, Moving with you, New detections only, Signatures only, Named radios only, class Show only / Hide these and the class chips, Show only selected and Hide selected and their picks, RSSI floor, name/MAC and OUI queries, AND/OR). It does not change Display, GPS tagging, or the log. Stock chips are ordinary saved filters built from signature class, not a second matching engine.

Save current as... (name field + Save) appends a new chip with a snapshot of whatever is on this tab right now, including class picks, hide picks, and New detections only. It does not overwrite a built-in chip. Tap a chip to apply it (the active chip is highlighted). Long-press any chip and confirm Delete. That removes the chip from this list, not the filter currently on the Live display. Stock chips you remove stay gone across catalog updates; Settings → Restore default signatures & presets puts the short stock set back (and still rewrites the catalog). Reset filter still clears this tab even if you deleted All traffic. **Reset filter** at the bottom clears this tab (All traffic, New detections off, already-seen cleared). It is not “undo the last preset.” Applying All traffic or Reset filter also turns New detections off and clears already-seen.

A class preset only *shows* radios that already matched. If Finder tags are hidden on Filters, Trackers looks empty — turn Hide these off, or tap Trackers again. Cameras / Drones / Surveillance and the other class sits are not stock chips: pick Show only plus the class chip, then Save current as... to keep that sit.

Preset	What it sets	Notes
All traffic	Both radios, no RSSI floor, no queries, Signatures only off, class filter off, Hide selected off, Hide Fast Pair account-key off, Moving with you off, New detections off.	Show everything. First look at a block. Applying it also clears already-seen.
Wi-Fi only	BLE off.	APs, hotspots, mesh, soft-AP. No LE rows.
BLE only	Wi-Fi off.	Advertisers only.
Strong signal	RSSI floor −70 dBm. Both radios.	Drops the quiet crowd. Not a Hunt. Pair with Display → Subtitle None if the list is still busy.
Moving with you	Moving with you on. Both radios. No Signatures only, no class Show only.	Still needs Settings → Tag detections with GPS and ~50 m of path. Bag/car tag yes; house AP on arrival no. “Still here” grows with speed (\$8.5). The switch starts a follow test on all radios (it clears Show only and Signatures only). Debrief tracking is separate (\$12.2).
Trackers	BLE only + Show only Finder tags.	AirTag / SmartTag / Tile / Chipolo / Pebblebee. Not iBeacon / Minew (Retail beacons class chip). Bag tag will dominate. Matching stays on. Signatures only is implied (switch dimmed).
Hide trackers	Hide class Finder tags. The rest of the field stays.	Unmatched radios still show. Watchlist will not beep for a hidden class. Signatures only stays a real switch.
Hide phones	Hide class Phones / PCs.	Apple Device, Fast Pair, Google, Microsoft Device, Phone hotspot. Unmatched radios still show. Hide Fast Pair account-key is a narrower switch (plaza chips only).

Stock presets leave New detections only off. After you save your own chip, tap it to get that sit back in one hit (for example “plaza −80 + Hide these Finder tags”). Display Sort / Title / Subtitle are not stored in a preset — those live on the Live display → Display.

8.5 How Moving with you works

Moving with you is Spectre's answer to a simple field question: which loud radios are staying with *me* as I walk or drive, rather than radios I only meet when I arrive? It is a Live display filter, not a Hunt and not a direction finder. GPS is always your phone at hear-time. RSSI is loudness here, not meters. Pattern chips on those rows are still not identity.

Two very different radios can both be loud. A tag in the bag or a hotspot in the car is heard again and again while the phone's GPS path lengthens. A house access point is silent for the whole drive and then suddenly strong at the driveway. The filter is built to keep the first and drop the second. Passing phones on an interstate sit in between: they are loud for a few seconds and then gone. Those should flash through and leave. A radio that is actually with you should stay.

8.5.1 What has to be true before anyone qualifies

Tag detections with GPS must be on, Location must be high accuracy, and scanning must be running so the phone is giving live fixes. A last-known location older than 30 seconds is ignored. The operator path has to grow to about 45 m before the test is even willing to run — sitting still does not count. Filters shows the path meter; the Live display shows Follow · path N m while the switch is on.

Each radio then has to pass every gate below. Fail any one and it stays off this filtered list (it is still in the log, and Debrief can still discuss it).

- **Heard recently.** Last packet in the last 90 seconds.
- **Loud enough right now.** Last RSSI about -75 dBm or stronger. A quiet dip below that drops the row until it is loud again.
- **A trail, not a single ping.** At least two GPS stamps on that radio. Stamps are only added when the phone has moved about 8 m (or 18 m within 30 s), so a stationary crowd does not pad the trail.
- **The trail itself moved.** The radio's GPS path, and the box around those stamps, must each cover at least about 27 m (60 percent of the 45 m operator move). A radio heard only at the door has a tiny trail and fails here even if it is screaming loud.
- **Loud along the trail, not just at the end.** At least two-thirds of those GPS stamps must also be about -75 dBm or stronger. A radio that was quiet for most of the drive and only woke up at the house does not qualify.
- **Still here, given how fast you are going.** The last time Spectre heard that radio, the phone's GPS was somewhere. That stamp is compared with where the phone is now. How far you are allowed to have gone in between is the speed-aware part (§8.5.2).

A second phone in the car usually fails the trail gates, not the speed gate. iOS and many Android stacks rotate the BLE address, so Spectre sees a brand-new radio with an empty trail. Use a tag with a stable MAC (AirTag, Tile, SmartTag in the bag) as the confidence check that the filter is working.

8.5.2 Why the “still here” window grows with speed

The last GPS stamp is not the other radio's location. It is your phone, at the moment Spectre heard that radio. If the radio is quiet for a few seconds, the phone keeps moving. On a sidewalk those few seconds are a house-length. On an interstate they are hundreds of meters. If “still here” were a fixed 50 m circle around the phone, those two cases would look identical, and a tag in the cup holder would blink off between advertisements even though it never left the car.

Moving with you therefore asks a different question as you speed up: not “is this radio 50 meters from me right now?” but “have I traveled farther, since we last heard this radio, than this radio could reasonably have gone silent?” Recent speed is path length divided by the time on that path — an average over the follow, not a single GPS velocity. It is clamped between 0 and 40 m/s (about 0–90 mph) so a jumpy fix cannot grant kilometers of slack.

Android delivers Bluetooth advertisements more often than it scans for Wi-Fi access points. A hotspot in the car can go tens of seconds between scans; a BLE tag in the bag usually speaks every few seconds. The hold time is therefore 15 seconds for BLE and 50 seconds for Wi-Fi, then multiplied by your recent speed. A 50 m floor still wins when you are slow, so a sidewalk tag is not given a city-block of slack. For BLE that floor hands off around a fast walk or slow bicycle (about 7.5 mph). Wi-Fi leaves the floor sooner, because 50 seconds of walking is already more than 50 m. On top of that product, 25 m of GPS slack is added at every speed — enough to cover ordinary fix jitter without pretending RSSI is a tape measure.

In short: **allowed distance = the larger of 50 m or (speed × hold time), plus 25 m.** Hold time is 15 s on BLE and 50 s on Wi-Fi.

Your recent speed	BLE still-here	Wi-Fi still-here	What that feels like
Walking (~3 mph)	about 75 m	about 90 m	A house-length. The 50 m floor is still doing the work for BLE.
Fast walk / bicycle (~7.5 mph)	about 75 m	about 190 m	BLE is just leaving the floor. Wi-Fi already uses the 50 s hold.
Neighborhood (~15 mph)	about 125 m	about 360 m	A tag that skipped a couple of advertisements still counts as with you.
City street (~30 mph)	about 225 m	about 700 m	A few seconds of silence is a couple of blocks, not a miss.
Interstate (~55 mph)	about 400 m	about 1.3 km	Two seconds of driving is already ~50 m. The hold is now a handful of advertisements, or about one Wi-Fi scan.
Fast highway (~70 mph)	about 500 m	about 1.6 km	A car tag should stay. A roadside AP you passed still fails the trail-moved gates.

Those distances are how far the *phone* is allowed to have traveled since the radio's last GPS stamp. They are not a detection radius around the other device, and they do not mean Spectre thinks a tag is 400 m away. A bag tag that is still advertising will keep getting fresh stamps, so it stays near “here” in GPS terms even while the allowed window is large.

8.5.3 What still drops, even on the highway

A large still-here window does not keep passing radios on the list. The trail-moved and two-thirds-loud gates are unchanged at every speed. A phone in the next lane is loud for a few seconds. It rarely collects two GPS stamps that already span ~27 m of *your* path, and its trail goes quiet as soon as you pull away. Those rows still flash up and leave. That is the filter working. A tag in the car has been stamped all the way down the interstate, so it survives the same gates.

House access points fail the same way when you pull in: they have no long loud trail behind you. If they do show after you get home, the path was too short, GPS only started stamping at the destination, or you have not walked/driven the ~45 m the test needs. Live display → Start over, then move again.

Start over (on the Live display, above the tabs) clears the operator path and every radio's GPS trail so you can run the test again. It does not wipe the live list or the log. Turning Moving with you off does not clear the path. Reports → Debrief still writes the last-15-minute co-travel callouts whether this filter is on or off — that sit report is §11.4 and §12.2, not this switch.

This is still a heuristic. A row that stays is a loud radio whose GPS trail followed your phone. It is not proof that someone is following you, and a miss is not a clean bill. Quiet tags, rotating Find My addresses, and radios the OS did not deliver will not appear. If safety is in question, leave and get help — do not wait on this list.

8.6 Field recipes

One-line filter setups. Chapter 12 walks each strategy: setup, what you should see, and what you must not claim from the picture.

Situation	Suggested filter
Quick sweep of a block	All traffic, High performance, Strength list or Hybrid. Display defaults. Do not use Signatures only yet — you need the noise to know the neighborhood.
Long sit / vehicle	Balanced or Saver. Timeline. If the list is busy, Display → Subtitle None before you raise the RSSI floor. Leave logging on.
Focused family	Preset Trackers, or Filters → Show only plus the class chips (Public safety, Cameras, Drones, Surveillance, ...). Matching stays on.
Named hits minus clutter	Signatures only + Hide these Finder tags (or Hide selected on one family, e.g. your bag AirTag).
Drop trackers, keep everything else	Preset Hide trackers (hides the Finder tags class).
Tracker hunt	Preset Trackers (Finder tags, BLE). Watch the signatures. Battery saver is usually enough; tags advertise slowly.
SSID / OUI lead	Name query or OUI query, AND, both radios on. Do not enable Signatures only or you will hide an unmatched but relevant AP.
Who is walking with me	Settings → Tag detections with GPS (high-accuracy Location). Walk ~50 m until the path is not 0. Filters → Moving with you (preset or the switch — not Trackers then the switch). A tag in your bag/car will match. A second iPhone usually will not (MAC rotation). Walking uses a tight still-here window; a drive uses a larger one (\$8.5). After the walk: Reports → Debrief — last 15 minutes of tracking, even if you leave this filter or change view.
What just arrived	Filters → New detections only. On the Live display, Mark seen once the room is known (buttons sit above the tabs). Reset seen starts over. Hint: New only · N hidden.

9. Named Signatures & Creating Custom Signatures

9.1 What a signature is

A signature is a named bundle of match rules, a color, and optional cluster flags. It is not proof of vendor identity. Built-in only marks where a row came from; you can still edit or delete it. Each rule has its own on/off switch, so you can mute a noisy OUI or name without deleting the rule. Edit rules here, not on Settings. When the rules hit, the radio is always labeled. Display can hide the chip on the Live display; Filters hide the radio. The catalog list is Name A–Z by default. Class A–Z is an outline of classes (collapsed until you tap one) so you can open Public safety or Vehicle without scanning the alphabet.

9.2 Create from an observed device

- 1 Find the device in the Live display (any view) or open its detail.
- 2 Long-press the row/blip, or tap Create signature from device on the detail screen.
- 3 Spectre suggests a name from the advertised name, vendor+OUI, or “BLE/Wi-Fi + last octets.”
- 4 Pre-filled rules: full MAC prefix (this radio); name or glob if a name exists; up to three service UUIDs; manufacturer ID and first data byte when present. OUI is added only if nothing else except MAC was available. A “radio = BLE/Wi-Fi” OR rule is deliberately not added — that would label every radio of that type.
- 5 If the address looks randomized, the notes field warns you. Prefer name / UUID / manufacturer rules and consider deleting the MAC rule.
- 6 Edit, set match-any, color, min-peers if needed, then Save.
- 7 If you already know the BLE payload layout, open the new row → Decode fields and map the bytes (§9.6). Identity (the rules) and decode (the field map) are separate saves.

Read **Signature family** on that radio’s detail page first (§5.5). Strong family or Possible family means other MACs already share the ID — Create from device will still pin **this** MAC. Use Signature candidates (§9.2.1) when you want the shared rule instead. This radio only is when a MAC pin is the honest unmatched draft. **Already tagged is not a veto**. If the decode shows a store-wide iBeacon UUID (or another ID tighter than the stock row), keep Create from device, delete the MAC pin, and lengthen the manufacturer-data prefix to type 0x02 / length 0x15 plus that UUID (hex 0215 + 32 UUID digits) — the same shape as Target Atrius basket. The stock iBeacon row stays; your row is the second chip. The draft’s first manufacturer byte alone (02) would match every iBeacon, not this store.

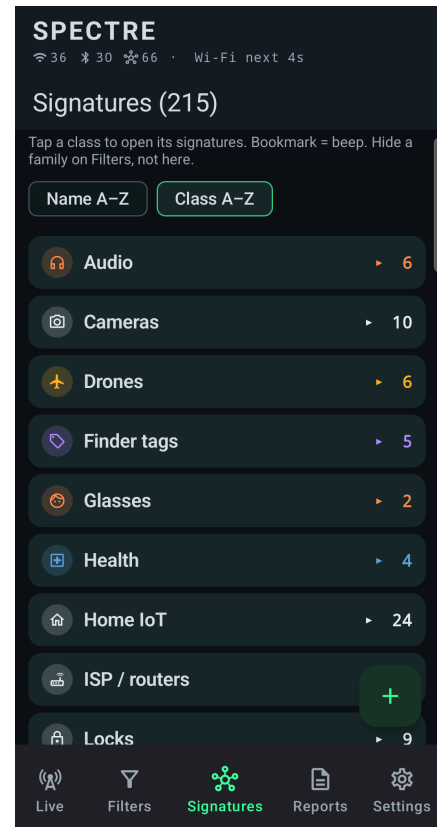


Fig. 7 — Signatures. Title is Signatures (N). Class A–Z starts collapsed; tap a class to open its rows. Name A–Z is the alphabet. Bookmark = watch (beep and/or spoken class). Hide a family on Filters. Privacy mode is on: last three MAC octets show as **.*.*.*

9.2.1 Create from Signature candidates

When several unmatched radios share one ID (the same H20- glob, the same vendor IE, the same stable OUI), do not long-press one MAC. Detail's Signature family card is the one-radio version of that test. Reports → Signature candidates (§5.6.1, §11.5) drafts the **shared** rule and omits the MAC pin. Save still lands in your catalog as a custom row — same editor, same Save. That is how you add a family without cataloging a house AP's BSSID. It only mines **unmatched** radios. A floor of stock iBeacons will not list here; for a store UUID on radios that already chip iBeacon, use §9.2.

9.3 Manual create and edit

The Signatures title is the catalog total (stock plus any you added). Tap + to create a blank signature. In the editor you set name, notes, Extra attention, AND vs match-any (OR), cluster flags, color, and the rule list. Each rule has an on/off switch: off keeps it, but it does not match. BLE signatures also have **Decode fields** (None or N fields) under the rules — §9.6. Wi-Fi-only rows hide that control.

Rule kinds: OUI, MAC prefix, name contains, name glob, service UUID, manufacturer ID, manufacturer data, radio kind, hidden SSID, vendor IE OUI. Enter manufacturer IDs as hex (for example 0x004C).

On a Wi-Fi row the MAC **is** the BSSID. A MAC-prefix rule with the full twelve hex digits is that one access point. Fewer digits (B4:1E:52) is the vendor block — same matcher as an OUI rule. Hidden SSID matches any AP whose scan result has an empty name; pair it with a MAC prefix (AND) to pin one hidden box. Stock Android never reveals the secret SSID string.

There is no matching on/off switch. Signatures always label when their rules hit. Hide a family on Filters (class Hide these, or Hide selected signatures). Display → signature names hides the chip on the Live display without dropping the radio (the decode hexagon goes with it). The bookmark requests an alert when that signature hits (and a silent shade card if that Settings switch is on).

Open a signature and tap Delete signature at the bottom to remove it (confirm first). Built-in rows can be deleted; Restore default signatures brings the stock set back. Settings → Export signatures writes the whole catalog to a JSON pack you can share or keep as a backup. Import signatures adds new rows and extra rules; it does not delete anything. Same id or the same match rules are skipped. A colliding name is imported as "Name (imported)". Restore defaults still wipes customs — export first if you want them back.

9.4 Practices that hold up in the field

- Start tight (full MAC or manufacturer data + company ID), then loosen if the target rotates addresses.
- Do not OR a lone RADIO_KIND rule.
- Do not treat a Silicon Labs or LiteOn OUI as unique. Pair it with a name, UUID, or require min-peers.
- Prefer 16-bit UUIDs in the 0xFDxx / vendor range over common GAP UUIDs.
- For a family, use match-any and several OUIs plus a name glob. For one radio, keep the MAC rule and turn match-any off if you also add other clauses.
- Hide Unknown Signature on Filters when you do not want generic ESP_/ANDROID-/DIRECT-/UNIT- names on the Live display.
- Mute a noisy method on one signature with the rule switch (e.g. LiteOn OUIs on Flock) instead of deleting the rule.
- After editing, glance at the Live display. If half the cafe just inherited your new name, the rule is too broad — undo immediately.

9.5 Using the pre-loaded catalog

On first launch, Spectre loads the stock signature catalog. Restore default signatures & presets loads it again. Every stock row is on: when its rules hit, the radio is labeled.

To take a noisy family off the Live display, use Filters — Hide these for a whole class, or Hide selected for one signature. Matching still runs. The log, Debrief, and bookmarks are unchanged.

Each signature has a **class** (Finder tags, Surveillance, Drones, Vehicle, and so on). Filters → Show only / Hide these and the Trackers, Surveillance, Drones, and Vehicle presets use class. Chip color on the Live display is separate; you can change it in the editor. Custom signatures default to class Other.

The Signatures tab and the Hide selected / Show only selected lists are sorted A–Z, including custom rows.

Stock chip colors follow class, so Live display and radar mean something before you open a row. A named match uses that signature's color. Unmatched radios use the RSSI palette: ≥ -55 green, -55 to -70 amber, -70 to -85 orange, else red. An upgrade re-applies stock colors to built-in rows once. Custom signatures are left alone.

Filter class	Stock signatures (abbreviated)
Finder tags	Apple AirTags, Samsung SmartTags, Tile, Chipolo, Pebblebee / moto tag
Retail beacons	iBeacon, Target Atrius basket, Minew, Estimote, Kontakt.io
Signage	Retail LED sign, Electronic shelf label
Wearables	Garmin, Fitbit, Oura, Pokemon GO Plus, Fieldy, Plaud Note
Surveillance	Flock, Raven, Penguin, Pigvision, FS Ext Battery, Genetec, Rekor, Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua, UniFi Protect, ASSA ABLOY, SALTO, dormakaba, Paxton
Drones	Remote ID, DJI, Skydio, Autel, Parrot, HOVERAir
Pentest	Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, Porkchop, Hobby BLE serial
Public safety	Axon, WatchGuard Video, Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc. Used in LE, not exclusive to it — government, municipal, and other corporate fleets likely run some of the same kit.
Vehicle	Tesla, Tesla tsTPMS, Rivian, Ford, Honda, Hyundai, Toyota, Nissan, Subaru, BMW, Volkswagen, Porsche, Jaguar Land Rover, BYD, Chevrolet hotspot, GM hotspot, Audi MMI, Mercedes MBUX, Uconnect, CarPlay, CARLINK, Motive, PeopleNet, Samsara, AUMOVIO, Winegard, Goodyear TPMS, Schrader TPMS, Pacific TPMS, Huf, FOBOT TPMS
Glasses	Ray-Ban / Meta glasses, Snap Spectacles
Audio	Apple audio, Sony, Bose, JBL / Harman, Sonos, Shokz
Cameras	Wyze, Ring, Arlo, eufy Security, Nest, Tapo, Reolink, GoPro, Osmo, Insta360
Thermostats	Nest Thermostat, ecobee, Sensi, Honeywell Home
Locks	August, Schlage, Nuki, Lockly, Kevo, Master Lock, igloohome, Tedee, Kwikset
Health	Honeywell Xenon HC, Omron, Withings, Dexcom
Home IoT	Nest Weave, Tuya, Govee, Haiku Fan, myQ, Hatch, Orbit B-hyve, Samsung appliance, EcoWater, Amazon, Logitech, HP, Epson, LG webOS TV, Roku, Nespresso, RadiaCode, Ruuvi, Blue Maestro, SensorPush, SnapAV

Filter class	Stock signatures (abbreviated)
ISP / routers	UniFi, UniFi AP, Meraki, Cisco, Aruba, Ruckus, Ruijie, Fortinet, Mist, Sophos, Extreme, Edgecore, WatchGuard AP, Mojo, NETGEAR, TP-Link, ASUS, Linksys, Eero, Google Wifi, Huawei, Plume, D-Link, DWnet, Belkin, Xfinity, Spectrum, AT&T, Verizon, Starlink, GL.iNet, MikroTik, EnGenius, Zyxel, Peplink, OpenWrt, Arris, T-Mobile, HUMAX, Sagemcom, Arcadyan, Askey, Calix, Nokia, AirTies, Tenda, WAVLINK, Sercomm, Luxul, CenturyLink, Adtran, Cambium, TRENDnet, Cudy, Vantiva, Hitron, Actiontec, Buffalo, Grandstream, Inseego, Franklin, Synology
Mesh	Meshtastic, Helium
Phones / PCs	Apple Device, Fast Pair, Google, Microsoft Device, Phone hotspot
Other	Unknown Signature

Color	Class	Stock signatures
Red	Pentest / cheap serial	Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, Porkchop, Hobby BLE serial
Amber	Surveillance and drones (same chip color; class splits them)	Flock, Raven, Penguin, Pigvision, FS Ext Battery, Genetec, Rekor, Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua, UniFi Protect, ASSA ABLOY, SALTO, dormakaba, Paxton, Remote ID, DJI, Skydio, Autel, Parrot, HOVERAir
Purple	Phones / Find My tags	Apple Device, Apple AirTags, Chipolo, Fast Pair, Google (Pixel / 0x00E0), Phone hotspot
Cyan	Wearable trackers	Samsung SmartTags, Tile, Pebblebee / moto tag, Garmin, Fitbit, Oura, Pokemon GO Plus, Fieldy, Plaud Note, iBeacon, Target Atrius basket, Minew, Estimote, Kontakt.io
Green	Mesh / LoRa	Meshtastic, Helium
Orange	Glasses and audio (same chip color; class splits them)	Ray-Ban / Meta glasses, Snap Spectacles, Apple audio, Sony, Bose, JBL / Harman, Sonos, Shokz
Teal	Public safety and vehicle (same chip color; class splits them)	Axon, WatchGuard Video, Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc, Tesla, Tesla tsTPMS, Rivian, Ford, Honda, Hyundai, Toyota, Nissan, Subaru, BMW, Volkswagen, Porsche, Jaguar Land Rover, BYD, Chevrolet hotspot, Mercedes MBUX, Uconnect, CarPlay, CARLINK, Motive, Samsara, Winegard, Goodyear / Schrader / Pacific / Huf / FOBO TPMS
Blue	Health	Honeywell Xenon HC, Omron, Withings, Dexcom
Silver	Cameras / PCs / home IoT / home Wi-Fi / retail signage / catch-all	GoPro, Osmo, Insta360, eufy, Wyze, Ring, Arlo, Nest, Tapo, Reolink, Microsoft Device, Amazon, Starlink, Logitech, HP, Epson, LG webOS TV, Nespresso, RadiaCode, Nest Thermostat, Nest Weave, ecobee, Sensi, Honeywell Home, Tuya, Govee, Haiku Fan, myQ, Hatch, Orbit B-hyve, August, Schlage, Nuki, Lockly, Kevo, Master Lock, igloohome, Tedee, Kwikset, Ruuvi, Blue Maestro, SensorPush, SnapAV, Retail LED sign, Electronic shelf label, UniFi, UniFi AP, Meraki, Cisco, Aruba, Ruckus, Fortinet, Mist, Sophos, Extreme, Edgecore, WatchGuard AP, Mojo, NETGEAR/TP-Link/ASUS/Linksys/Eero/Google Wifi/D-Link/Belkin/Xfinity/Spectrum/AT&T/Verizon/GL.iNet/MikroTik/EnGenius/Zyxel/Peplink/OpenWrt/Arris, Unknown Signature

Extra attention is a separate field on the signature. It is empty on almost every row. Editor Notes stay technical (OUIs, UUIDs) and do not show on the Live display or detail. If Extra attention is filled, a matching radio gets a “!” on the list, the full sentence on detail, and a line in Debrief / AI Export.

The stock catalog fills Extra attention on Hobby BLE serial, Axon, WatchGuard Video, Ray-Ban / Meta glasses, Snap Spectacles, Fieldy, Plaud Note, Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, Porkchop,

Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc, and the roadside / public camera + ALPR rows: Flock Safety Cameras, Penguin, Pigvision, FS Ext Battery, Genetec AutoVu, Rekor, Motorola Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua. Those rows ship with the bookmark on, so a new match beeps. That is body-cam, Meta / Snap glasses, recording wearables, pentest kit, public-safety vehicle APs, and public cameras / plate readers — not Tesla, not headphones, not UniFi Protect, not access-control locks. Flock LiteOn / Espressif OUIs can be noisy; unbookmark that row if it is. The Public safety rows are used in LE; they are not exclusive to it (government, municipal, and other corporate fleets likely run some of the same kit). Unbookmark any row you do not want to hear.

A “!” is a pattern, not identity and not proof of recording or an attack. Meta company IDs also match Quest headsets. Fieldy / Plaud Note are conversation recorders, not proof someone is recording you. PineAP clones look like ordinary SSIDs. You can type Extra attention on any other signature, built-in or custom.

Unknown Signature only matches ESP_*, ANDROID-, DIRECT-, and UNIT- names. Hide it on Filters if those generic-dev-board names are local noise. Do not bookmark it.

Home and ISP Wi-Fi rows (NETGEAR, TP-Link, Xfinity, Starlink, and the rest of that class) are Wi-Fi only. Most also match the vendor’s IEEE OUI on the BSSID, so a renamed SSID still hits. Virtual BSSIDs (guest / mesh / extra SSID) that set the locally-administered bit still hit if the universal OUI is in the catalog. Apple / Google / Samsung prefixes are not recovered, so a random phone hotspot does not become a router. A matching vendor IE still hits. Hide a family on Filters if factory-named gateways paint Live display.

Spectrum also matches Spectrum Mobile / Spectrum Free Trial hotspots and has no Charter OUI (OEM boxes). **Chevrolet hotspot** (myChevrolet*), **Mercedes MBUX** (MBUX*), and **Motive** (Motive * / Motive_*) are factory in-car / fleet SSIDs — not dealer or café names.

UniFi AP is factory SSIDs plus Ubiquiti IEEE OUIs (ISP / routers). Other Ubiquiti AP radios (airMAX, AmpliFi, UISP) that beacon can hit. A second **UniFi** row matches advertised names UniFi / Ubiquiti / UAP-* on either radio (same class). **UniFi Protect** is BLE names UVC G3/G4/G6 Instant (Protect cameras in setup), not AP BSSIDs — that row stays Surveillance.

Meraki is Meraki* plus Cisco Meraki IEEE OUIs (not Cisco Systems). **Cisco** is Wi-Fi Cisco* (Aironet / Catalyst / Business / RV) plus old Aironet tsunami and Cisco Systems IEEE OUIs. It is not a Cisco substring — that would hit San Francisco. Spectre sees AP beacons only, so a Cisco BSSID is an AP. Meraki and Cisco-Linksys keep their own OUI lists. Both sit in ISP / routers, not Surveillance. Meraki APs often carry a Cisco vendor IE (00:00:0C CCX / AP-name); Spectre keeps Meraki only on those beacons.

Aruba is Aruba* / SetMeUp* / InstantOn* (not a lone instant word; not Hewlett Packard Enterprise OUIs). **Ruckus** is Ruckus* / Configure.Me* plus Ruckus Wireless OUIs. **Fortinet** is Fortinet* / FortiAP* / FAP-config* plus Fortinet OUIs. Those sit in ISP / routers with UniFi AP, Meraki, Cisco, Mist, Sophos, Extreme, Edgecore, WatchGuard AP, and Mojo — campus access points, not cameras.

MikroTik (Routerboard.com OUIs), **EnGenius**, **Zyxel**, and **Peplink** / Pepwave add vendor OUIs. **OpenWrt** and **GL.iNet** stay name-only (no unique IEEE OUI; GL.iNet boards use chip-module prefixes). **Arris** / SURFboard has the Arris OUI; CommScope’s mixed block is not used. **Google Wifi** is not Pixel BLE, not Nest-* cameras, and not Google Inc OUIs (phones / hotspots).

Brand BLE rows in the stock catalog include Tesla, Rivian, Google, Sony, Bose, Garmin, Amazon, Fitbit, Oura, Logitech, HP, Epson, JBL / Harman, Sonos, Shokz, GoPro, Osmo, Insta360, DJI, Microsoft Device, Apple Device, Apple audio, Fast Pair, Tuya, Govee, Haiku Fan, myQ, LG webOS TV, Nespresso, and RadiaCode. Exact match IDs are in Appendix B. A few cautions: Google is company 0x00E0 / Pixel / Chromecast. Fast Pair is a separate Phones / PCs row (next paragraph). Apple Device is Continuity types, not AirPods (0x07). iPhones also send Find My 0x12; Spectre drops the AirTags chip when Continuity is on the same radio. Bose includes UUID FEBE as well as FE21 / 0x009E. Tuya is 0x07D0 / FD50 / TUYA*, not the two-letter TY. Govee identity is name-only; once it matches, Decode fields read H5074/H5075 (0xEC88) and H510x/H517x (0x0001) temp / humidity / battery (\$9.6). Shokz is OpenRun / OpenFit names, not Battery 0x180F. DJI is 0x08AA / DJI* (BLE and Wi-Fi). Osmo Action / Pocket / 360 / Nano are the Osmo row (0x08AA model IDs 0x0006–0x0022 plus OsmoAction* names), not DJI. Insta360 is Arashi

Vision 0x10D7 plus X3 / Ace Pro / GO 3 names. In-flight drones also hit **Remote ID** (ASTM UUID 0xFFFA). Skydio / Autel / Parrot ANAFI-Bebop / HOVERAir are name rows; Autel default-ssid and Parrot company 0x0043 are not used.

Fast Pair is not “Android.” It is Google’s tap-to-pair UUID (FE2C), a Phones / PCs row of its own. iPhones fill that class because they constantly advertise Continuity (Apple Device). Most pocket Androids advertise little Spectre can name, so they stay unmatched unnamed BLE. Google is Pixel / Chromecast only. Phone hotspot only while the phone is an access point with a factory SSID. When Fast Pair does appear, the advertisement is one of two shapes. **Pairing mode** is three bytes — a model ID. Android nearby will offer a tap-to-pair card (Pixel Buds, Galaxy Buds, a phone in pairing). That is uncommon and worth a look. Live display labels the chip Fast Pair pairing and puts pair on the subtitle; if the same radio later sends the long payload, that mark stays for the rest of the session. The **long payload** is an account-key filter from a device already on someone’s account. In a mall or station that is the usual Fast Pair chip: plaza noise, not a new pairing. Filters → Hide Fast Pair account-key hides those chips when Fast Pair is the only match. A Pixel that also matched Google still shows. Hide selected Fast Pair, or Hide phones, drops pairing-mode as well.

Public safety vs vehicle. Public safety is Axon / WatchGuard Video (body-worn and in-car) plus public-safety vehicle APs: Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc. Many of those radios are used in law enforcement; they are not exclusive to it. Government, municipal, and other corporate fleets likely run some of the same body-worn cameras and vehicle gateways. Pattern match, not that agency or that officer. Vehicle is Tesla (including Cybertruck phone-key names S...C), Rivian, Ford 0x0723, Honda 0x0915, Hyundai 0x0826, Toyota 0x0977, Nissan 0x0BA6, Subaru 0x0A10, BMW 0x05EB, Volkswagen 0x011F / FE30/FE31, Porsche 0x0120, Jaguar Land Rover 0x020B, BYD 0x0C34, Chevrolet / GM hotspots, Mercedes 0x017C / MBUX*, Audi 0x010E / MMI, Motive, Samsara, and BLE TPMS (Goodyear 0x0B99, Schrader 0x0601, Pacific Industrial 0x0E32, Huf 0x070A, FOBO / Salutica 0x0127). Tesla tsTPMS is the name tsTPMS* only — company 0x022B stays on Tesla; UUID 0x1122 is not unique. Factory 315/433 MHz valve-stem TPMS does not appear. Cheap kits that spoof Nokia 0x0001 are not cataloged. Kia / Volvo / Lucid / Polestar have no unique SIG company ID in this catalog. A miss is common: many cars never put that company ID in an advertisement, and Classic Bluetooth is invisible.

Drones vs surveillance. Drones is Remote ID (UUID FFFA), DJI, Skydio, Autel, Parrot, HOVERAir. Osmo / Insta360 action cameras are Cameras, not Drones. Surveillance is poles, ALPR, and commercial readers — not aircraft and not campus access points. Same amber chip; class splits them.

Glasses vs audio vs cameras. Glasses is Ray-Ban / Meta, Snap Spectacles. Audio is Apple audio, Sony, Bose, JBL / Harman, Sonos, Shokz. Cameras is Wyze, Ring, Arlo, eufy, Nest, Tapo, Reolink, GoPro, Osmo, Insta360 — consumer and action cameras. Not Flock / UniFi Protect (Surveillance) and not Axon (Public safety).

Signage vs retail beacons. Signage is LED message displays and Bluetooth ESL 0x1857. Retail beacons is iBeacon / Target Atrius basket / Minew / Estimote / Kontakt.io. The advertised name on a LED sign is the sign text, not a product name. iBeacon is Apple 0x004C type 0x02/0x15 (any vendor can send it — mute in a mall). Target Atrius basket is iBeacon UUID 5993A94C-... plus service 0xB1BB (shopping-cart tags; not Acuity 0x0346 on those radios). Minew is IEEE OUI AC:23:3F; Estimote is 0x015D; Kontakt.io is 0x01FD. Not Eddystone FEAA (JBL, printers, and others use it).

Environmental sensors (Home IoT). Ruuvi 0x0499, Blue Maestro 0x0133, SensorPush custom UUIDs. Ruuvi and Blue Maestro ship Decode fields (temp / humidity / pressure on the advertisement). SensorPush stays identity-only.

Smart locks / access (BLE). ASSA ABLOY (0x012E / HID 0x0124 / Yale 0x0BDE / UUID FCBF / Seos / Yale*) — not Apple FCB2. August (0x01D1 / FE24). Schlage / Allegion (0x013B / FCF4). Nuki (a92ee* services). SALTO (0x0199). dormakaba (0x0C64). Paxton (0x0196). Lockly names. Kevo / Unikey (0x015E). Master Lock (0x014B). igloohome (0x05BA). Tedee (0x0725). Kwikset names. Commercial rows are amber; home locks are silver.

Smart thermostats (BLE). ecobee (company 0x07D6). Nest Thermostat (Nest Labs 0x01B5; not Nest cameras). Sensi (name only — not Emerson 0x04DF). Honeywell Home / Lyric / Amazon Smart Thermostat names (not Honeywell 0x0526 or Resideo 0x0B01). A miss is common: most wall units are Wi-Fi after setup and do not keep advertising BLE.

Health vs wearables vs Home IoT. Health is clinic and home-medical BLE: Honeywell Xenon healthcare scanners (Xenon_*HC* / CCB-U00-H*, not warehouse CCB-U00-G), Omron Healthcare 0x020E (cuffs and scales; not industrial OMRON 0x02D5), Withings scales / BPM Connect, Dexcom G6/G7. Garmin / Fitbit / Oura stay Wearables. Honeywell Home thermostats stay Thermostats. Pattern match, not a patient or that hospital.

If you get local false positives, hide the family on Filters or mute the noisy rules in the editor (for example a neighborhood of Espressif gadgets tripping Flock OUI 3C:71:BF). Restore default signatures & presets rebuilds the catalog (including stock Decode fields), stock bookmarks, and the default Settings switches (Keep screen on, Tag detections with GPS, Online place names, TAK / CoT off). It also wipes custom signatures and presets. Export first if you want a backup. Do not use Restore as an undo for a single rule.

To mute noisy methods on Flock (for example LiteOn OUIs) without deleting the row, open the signature and switch off those rules. See §7.6 and §7.6.1.

See Appendix B for the exact default rules.

9.6 Decode fields

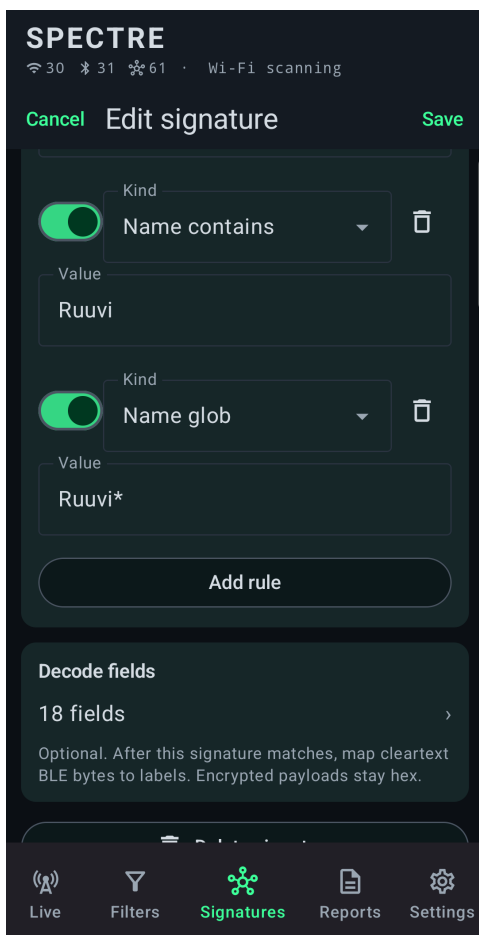


Fig. 7b — Signature editor (Ruuvi). Rules card, then Decode fields (18 fields). Tap the row. Encrypted ads stay hex. Privacy mode is on: last three MAC octets show as **:*.*.*.

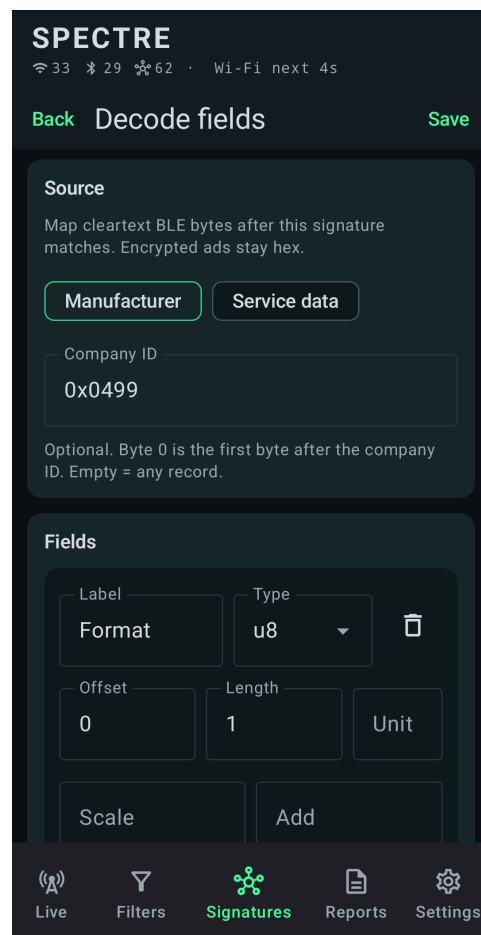


Fig. 7c — Decode fields. Source card (Manufacturer / Service data, company ID) then Fields cards (label, type, offset, length, unit, scale). Scroll for Only if, named values, Preview, Remove decode map. Privacy mode is on: last three MAC octets show as **:*.*.*.

After a signature matches a BLE advertiser, Spectre can map **cleartext** bytes in that advertisement to labels — temperature, model, Remote ID location, and so on. Identity match stays on the rule list: Decode fields do not help the matcher. The Live display does not parse the bytes (that path stays cheap). A small hexagon in the signature chip

— same color as the name — means that signature has a map; dual-chip radios mark only the mapped name(s). Display → Signature names off hides the chips, so the hexagon goes with them. Values fill in on device detail (Decoded fields), Share as text, AI Export for that radio, and Debrief's notable BLE lines.

This is BLE advertisements only: no pairing, no GATT connect, no crypto. Encrypted or unknown payloads stay hex, the same as Maker data already does. Apple Continuity, Fast Pair, Eddystone, and Microsoft stay on the built-in Known payloads / Maker data blocks; Decode fields is for catalog and custom rows those paths do not cover. Wi-Fi-only signatures hide the control.

You need two things, in this order: a signature whose rules hit the radio, and a field map for the payload you already understand from a vendor spec, a reverse-engineering note, or the hex on detail. The map does not print values on the Live display; the hexagon on that chip only means a map exists. If the radio is unmatched, create the signature first (§9.2 / §9.3), Save, then open Decode fields on that row.

9.6.1 From a payload you already understand

Worked path. You have a BLE radio on the Live display, the maker bytes (or service data) on detail, and a layout — byte 0 is format, bytes 1–2 are temperature, and so on. Spectre will not invent that layout; you type it.

- 1 Confirm identity. Open detail. Matched signatures lists the row that will carry the map. If none, Create signature from device (§9.2): keep a manufacturer ID / UUID / name rule that will still hit when the MAC rotates, then Save. A MAC-only pin is a poor home for a family map.
- 2 Read the hex you will slice. On detail, Maker data → Raw payload is the bytes **after** the Bluetooth company ID. Service data FEAF (or whatever UUID) is the bytes after that UUID. Count in hex pairs: the first pair is offset 0, the next is 1. Do not count the company ID or the UUID toward offset.
- 3 Signatures → that row → **Decode fields** (under Add rule). Pick Manufacturer or Service data to match the hex you counted. For manufacturer, set Company ID (0x0499, 0xEC88, ...) unless every matched radio has only one maker record. For service data, set the UUID (FFFA, FE6A, FEAF, FEED, or the full 128-bit form).
- 4 Add field for each value you care about. Label is what detail shows. Type, offset, length, endian, unit. Add field starts the next card at the previous offset + length. Save on the Decode fields bar writes the map; Back without Save discards.
- 5 Check Preview at the bottom if a matching radio is on the air. Then open that radio on detail: Decoded fields should match the spec. A miss with raw hex still on the page usually means offset 0 included the company ID, the wrong source, or an Only if that this packet failed.

Example (Ruuvi Data Format 5, company 0x0499). Manufacturer. Company ID 0x0499. Field Format: u8, offset 0, length 1. Field Temperature: i16, offset 1, length 2, endian BE, scale 0.005, unit °C, Only if offset 0 length 1 equals 05. Humidity is u16 BE at offset 3, scale 0.0025, unit %, same Only if. Format 3 packets skip those fields because byte 0 is 03, not 05 — that is the point of the gate. Stock Ruuvi already ships this map. Copy the idea for a sensor that is not in the catalog.

9.6.2 Source: manufacturer vs service data

Manufacturer reads AD type 0xFF. Spectre strips the two-byte company ID before your map runs, so byte 0 is the first payload byte after 0x004C / 0x0499 / 0xEC88 / ... — the same slice as detail's Raw payload. Company ID on this screen is a filter among maker records on that radio (empty = the first record). A Govee light and a Govee hygrometer can both match the Govee name rule; the hygrometer map reads 0xEC88 (H5074/H5075) and 0x0001 (H510x). A light with no matching payload simply does not decode.

Service data reads one service-data UUID. Byte 0 is the first byte of that service's data, not of the whole advertisement. UUID is required (16-bit FEAA / FFFA or the dashed 128-bit form). Several service-data blocks can sit on one radio; only the UUID you typed is parsed.

One signature, one source. If a product puts sensors in manufacturer data and a serial in service data, pick the slice you care about, or make a second signature for the other UUID (dual-label). You cannot point one map at both sources.

9.6.3 Field cards

Each card is one value. A field that does not fit — offset past the end, gate failed, empty UTF-8 — is skipped silently, and the other fields still show. That is how one map covers two packet formats.

Control	What it does
Label	Text on detail / Debrief / Preview. Changing it slugs the ID until you set a custom ID under More.
Type	How to read the slice. See the type table below. Defaults the length.
Offset	Start byte in the source payload. 0 is the first byte after the company ID (manufacturer) or the first service-data byte.
Length	Byte count. Type fills a default (u8=1, u16/i16=2, u24=3, u32/i32/f32=4, mac=6). utf8 / hex / bits start at 1 until you set them.
Unit	Appended after the number (°C, %, hPa, mV, dBm). Empty = number only.
Endian	LE (default) or BE. Hidden for u8 / i8 / utf8 / hex / bool. Applies to the integer/float/bits word, including bits.
Bit offset / width	Type bits only. After the slice is read as an unsigned integer in that endian, Spectre shifts from the least-significant bit. Width 1–32.
Scale / Add	Numeric types only. After the integer/float read: modulo (if set), then × scale, then + add. Temperature often needs scale 0.01 or 0.005. Pressure that is stored as (hPa – 500)×100 uses scale 0.01 and add 500.
Only if...	Skip this field unless a gate holds (§9.6.4).
Named values...	Map a raw number or hex to a word (§9.6.4).
More	ID (stable key; leave the slug unless you export maps or want TAK advertised-position pins). TAK / CoT looks up ids, not labels: latitude + longitude (optional alt_geo) pin the gadget; op_lat / op_lon are the Remote ID pilot and are not the aircraft pin. §5.8.3. Modulo is remainder before scale — Govee packed humidity is modulo 1000.

Type	Bytes (default)	What you get
u8 / i8	1	Unsigned 0–255 or signed –128...127. No endian.
u16 / i16	2	16-bit int. Set BE when the spec says big-endian (Ruuvi, Blue Maestro).
u24	3	Unsigned 24-bit. Govee H5075 packs temp in a 24-bit BE int.
u32 / i32	4	32-bit int. Remote ID lat/lon are i32 LE × 1e–7 degrees.
f32	4	IEEE-754 float. Rare in ads.
bits	from bit offset+width	Unsigned bitfield inside the slice. Use named values for 0/1 flags (asleep/awake).
bool	1	Any non-zero byte → yes; all zeros → no.
utf8	you set	Text, trailing NULs and spaces stripped. Remote ID UAS ID / Self ID.
hex	you set	Spaced hex dump of the slice. Tile's rotating private id.
mac	6	AA:BB:CC:DD:EE:FF from the first six bytes (Ruuvi payload MAC).

Math order is fixed: read the integer or float, then remainder (modulo), then multiply (scale), then add. Empty scale, add, or modulo leaves that step off. Named values run on the **raw** integer (or the UTF-8 / hex string), not on the

scaled number — map 65 to HERO13 Black, not 0.005×something.

9.6.4 Only if and named values

Only if... is how one signature handles two layouts. Each field can have its own gate. If the gate fails, that field is omitted; the rest still run.

- **equals** — bytes at Offset / Length must match Hex. Ruuvi Format 5: offset 0, length 1, hex 05. Remote ID location: the message-type nibble is packed with protocol 2, so the header byte is 12 for location (type 1, proto 2).
- **not equals** — skip when those bytes are Hex (a reserved/invalid marker).
- **mask** — every 1-bit in Hex must be set in the payload slice: (payload AND Hex) equals Hex. Use it for flag bytes when you only care that a bit is on.
- **payload length** (When → length) — the whole source payload must be exactly Bytes long. Offset/Hex are ignored. Govee H5074 is 7 bytes; H5075 is 5 or 6. Same company ID, different length, different fields.

Hex is raw bytes, no 0x, no spaces required (05, 0215). Length of Hex must match Length of the gate or the field never shows.

Named values... is a table of Raw → Show as. Raw is the integer as decimal (65) or hex (0x41), or the UTF-8 / hex string for those types. Unlisted values still show as the number (or scaled number). GoPro / Osmo / Nest Weave / Remote ID message type ship these tables. Add value for each known id; Remove clears the table.

9.6.5 Preview, detail, and reports

- **Preview** (bottom of Decode fields) uses a radio this signature already matches that is on the air now. You see the source hex and the parsed rows. No matching radio: Save anyway. A matching radio with empty hex: wrong source or that advertisement has no manufacturer/service bytes. Parsed empty: offset/length/gate — the usual miss is counting the company ID as offset 0.
- **Device detail** → **Decoded fields** is the same parse on the current advertisement. Several matched signatures with maps: each row is prefixed with the signature name. A map that exists but produced nothing prints a short note (Govee lights usually only send a name); raw hex stays below.
- **Live display chip** shows a hexagon when the signature has a map — not the parsed values. Radar labels do not carry the hexagon. TAK / CoT (§5.8) also reads the numeric values for ids latitude / longitude (and alt_geo) to pin advertised position; that is not shown on the Live display.
- **Share as text / AI Export** on detail include Decoded fields. Reports → Debrief notable BLE prints label: value under those radios.
- Save on Decode fields writes the map onto this signature (stock or custom). Back without Save discards the editor. **Remove decode map** (confirm) clears the whole map; empty fields + Save also stores None. Export / Import signatures includes the map. Restore default signatures & presets reloads stock maps and wipes custom rows — export first.

9.6.6 Stock maps

These rows ship a map (catalog upgrade fills them onto built-in rows that did not have one). You can edit or clear any of them. Fitbit, Find My tags, and other connect-or-crypto ads are identity-only — older Fitbit advertisements still do not carry steps.

Signature	Source	What detail can show
Ruuvi	Mfr 0x0499	Format 5: temp / humidity / pressure / accel / battery / TX / movement / sequence / MAC in payload. Format 3: humidity / pressure / accel / battery (Format 3 temperature is sign-magnitude, not a plain int).
Remote ID	Service FFFA	Open Drone ID app code, counter, message type. Protocol 2: Basic ID, location (field ids latitude / longitude / alt_geo), Self ID, System (op_lat / op_lon = pilot, not the aircraft pin), Operator ID. TAK Payload location uses those ids. Not a tail number.
Blue Maestro	Mfr 0x0133	Tempo Disc battery, log interval, stored logs, temperature, humidity.
GoPro	Mfr 0xF202	Schema, processor awake/asleep, Wi-Fi AP, pairing, model name, media offload.
Osmo / DJI	Mfr 0x08AA	Model id (Osmo Action / Pocket / 360 and some aircraft). Osmo and DJI rows share the map; identity rules still split cameras from drones.
Govee	Mfr 0xEC88 / 0x0001	H5074 / H5075 / H510x temperature, humidity, battery — after the name match. Lights with a Govee name may not fit those layouts; then the map does not apply.
Kontakt.io	Service FE6A	Location packet: battery, TX, channel, moving.
Estimote	Mfr 0x015D	Frame type (Nearable / Telemetry). Packed sensor bytes are not expanded.
Nest Weave	Service FEAF	Weave device-identification block: vendor (Nest Labs / Yale), product (Protect / thermostat / cam / Guard / Detect when the enum hits), pairing, 64-bit Weave device id. A 2-byte FEAF payload is the product id alone.
Tuya	Mfr 0x07D0	Bound flag and protocol version. UUID bytes stay encrypted.
Tile	Service FEED	Rotating private id (8 bytes of hex). Not a serial and not a stable identity.

A decoded value is still a pattern. Temperature, model, or a Remote ID location is what that advertisement contained, not proof of a serial, an owner, or that a drone is overhead. A miss (map did not apply) is common: different firmware, a short scan response, or a family that encrypts the rest of the payload. Raw bytes stay on the page.

9.6.7 Practices that hold up

- Get identity tight before you decode. A map on a rule that also hits lights, TVs, or every 0x004C radio will either stay empty or show nonsense.
- Count offsets on detail's Raw payload, not on a USB sniffer dump that still includes the company ID / UUID / AD length.
- Prefer Company ID filled in. Empty = first maker record; dual-company ads (Apple + vendor) will decode the wrong slice.
- Gate on a format byte or payload length when the family has more than one layout. Do not overlap two temperature fields without gates — you will show both or neither.
- BE when the spec says big-endian. LE is the editor default and is wrong for Ruuvi / Blue Maestro / Govee H5075 packing.

- Named values are for enums and flags, not for scaling. Scale 0.01 is not a named value of “°C”.
- Do not expect GATT. If the interesting number only appears after a connect (modern Fitbit steps, many locks), Spectre will never see it.
- Open a stock row (Ruuvi, Remote ID, Govee) and read its cards before you type a new map. The editor is the spec.
- If you want TAK to pin advertised position, set the field IDs to latitude and longitude (scale as the spec). Labels can be anything. `op_lat` / `op_lon` are not the pin. §5.8.3.
- After Save, open a live radio. If half the values are missing, the advertisement is a different format or a scan response — add Only if, or accept that this packet is short.

10. Alerts & Sensitivity

10.1 Watchlist

A watch target is either a device key (KIND:MAC) or a signature ID. Spectre alerts once when that radio first appears this session, and again only if it leaves (gone) and later returns. Sitting matches do not repeat. If New detections only is on, an alert fires only when that row would actually appear on the Live display. Already-here and learning-window Wi-Fi never alert while they are hidden.

- Device: open detail → bookmark icon to be notified when that radio appears. Prefills a name (advertised name or type guess). Settings → Named radios lists those MACs: rename, Alert on/off, remove one, Clear all. Signature watches are not on that list. A rotated BLE address stays until you delete it.
- Signature: the bookmark requests an alert when a new match appears. Stock bookmarks on first launch / Restore: Axon, WatchGuard Video, Ray-Ban / Meta glasses, Snap Spectacles, Fieldy, Plaud Note, Hobby BLE serial, Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, Porkchop, Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc, plus roadside / public camera + ALPR (Flock, Penguin, Pigvision, FS Ext Battery, Genetec AutoVu, Rekor, Motorola Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua).
- Master switch: Settings → Watchlist alerts. Off suppresses beep, voice, vibration, flash, and jump. Beep and Voice are independent (pip, spoken phrase, or both). Jump works with any of those. Settings → System notification (off by default) posts a silent shade card; skip it in the field.

10.2 Beep, voice, flash, and optional shade card

The default alert is in-app only. Beep (on by default) is a synthesized ~1 kHz double pip on media volume. Voice (off by default) speaks a short phrase. They are independent: pip only, spoken phrase only, or pip then phrase. Either cue also flashes the row for one second and can jump Live display to that row. Test alert plays whatever combination is on. On By class the matching class and signature open first so the flash is visible, and the jump keeps those headers on screen when the radio is not far down the outline. On radar the blip pings with two expanding rings and a bright core — that ping is the watchlist hit, not the pip itself, so voice-only still pings. Vibration is a short double pulse. Settings → System notification (off by default) posts a silent shade card on channel spectre_watch_v3 so the scan loop does not build a notification on every new hit. The separate spectre_scan channel is the low-importance “Spectre scanning” status and is not an alert.

10.2.1 Voice on watched signature

Voice is the second watchlist cue. It ships off. Turn it on under Settings → Voice on watched signature. Watchlist alerts still gates it — master off means no pip, no speech, no flash, no jump.

Settings → **What to say** picks the phrase (chips, enabled while Voice is on). Default is **Class** — the same bucket as the Live display glyph. A bookmarked AirTag, Chipolo, or Tile all say “finder tags.” Axon says “public safety.” ISP / routers says “I S P routers.” Use that in a pocket or on a drive when you need the kind. **Signature** says the catalog row instead (“Apple AirTags,” “Axon”). **Class + signature** says both (“finder tags, Apple AirTags”). Slashes in a name are spoken as a pause (Ray-Ban / Meta glasses → “Ray-Ban Meta glasses”).

A **named radio** (detail bookmark, Settings → Named radios) always speaks the watch name — the prefilled advertised name or type guess, or the custom name you typed on Named radios (“Peter’s Mesh Node”). Class / Signature / Both does not apply. An unmatched radio still speaks that name, not “unmatched.” A **signature bookmark** still follows What to say. If two signatures match the same radio, a signature bookmark speaks that family, not whichever chip landed first.

Pip and voice still mix three ways: pip only (the default), spoken phrase only, or pip then phrase about 400 ms after the double pip. Use voice-only when you cannot look. Use pip only on a sit when speech would be noise. Use both

when you want a heads-up then the words. Test alert plays the same mix — with Voice on and Class selected you hear “finder tags”; Signature says “Apple AirTags”; Class + signature says both. Jump to new watched detection works with beep, voice, or both; it is not tied to the pip. Hunt’s geiger tick never speaks.

Speech is on-device text-to-speech (no internet, no Spectre server). If a phrase is already being spoken, later hits in that burst are skipped so names do not overlap. A phone with no TTS language pack still pips if Beep is on. Raise media volume; voice uses the same stream as the pip.

10.3 Scan intensity

Mode	Use when	Cost
High performance	Walking, drive-by, first five minutes on site. Ships on. Full phone checklist: \$4.5.	Highest drain and heat. BLE low-latency with a ~70 s recycle. Wi-Fi about every 30 s (stock). For signed APs at speed, add Faster Wi-Fi AP scans (§10.3.1).
Balanced	Default for a one-hour sit.	Acceptable catch rate; Wi-Fi about every 40 s (stock).
Battery saver	All-day bag carry or overnight in a room.	You will miss short BLE bursts. Wi-Fi about every 55 s (stock). Raise stale if the list flickers to gone.

Stale window (Settings, 15–180 s, default 45) is independent of intensity. On-screen linger is the longer of Stale and Brief hold, so Brief hold 60 s keeps a radio visible a full minute after the last packet. A saver scan plus a short stale window will still mark slow advertisers gone unless Brief hold is raised. For tags, High performance or stale 90–120 s is safer.

10.3.1 Faster Wi-Fi AP scans

A second Settings switch, under scan intensity. Default off. It does not replace High / Balanced / Saver for BLE. It only changes how often Spectre asks Android for a new Wi-Fi AP list.

Why it exists. Catalog signatures (OUI on the BSSID or vendor IE, factory SSID globs such as IBR* / AirLink* / UniFi*) run on what the last scan batch contained. They do not search the air between batches. On a sit, the same APs return. On a drive, an interesting AP — a Cradlepoint in a fleet vehicle, an AirLink gateway, a UniFi or Cisco roadside box, a hidden SSID that still exposes its MAC — may only be in range for one short pass. At 45 mph you roll about 600 m between stock High-performance batches and about 160 m between faster batches; at 65 mph those figures are about 870 m and 230 m (§7.1.1). Faster scanning is more chances to hear that BSSID while it is loud enough to appear in scan results, so the signature can chip, the Extra attention “!” can show, and a stock bookmark can alert. Missing the scan window means missing the signature, even if the radio was real and in the catalog.

OS gate — Spectre checks before the switch will turn on. Android 11+ exposes whether Wi-Fi scan throttling is enabled. Spectre reads that flag when you try to turn Faster Wi-Fi AP scans on, and again when Settings resumes. If the OS is still throttling (the factory default), the Spectre switch stays off and a dialog tells you to use Developer options; Open developer options is offered. Spectre cannot write that OS flag. Android 10 cannot be queried, so the Spectre switch stays off there.

How to enable, in order. (1) Settings → About phone → tap Build number until the phone says you are a developer. (2) Settings → Developer options → Wi-Fi scan throttling → Off. (3) Return to Spectre → Settings → Faster Wi-Fi AP scans → On. Keep screen on and the scan notification still apply. Tag detections with GPS should stay on if you care where that hear was.

While it is in effect. About every 8 s Spectre calls `startScan()` and skips the four-per-two-minutes quota. The header’s Wi-Fi next Ns count is shorter. If the OEM still refuses scans, fail/backoff is unchanged (Wi-Fi waiting on OS, up to 45 s). If you turn OS throttling back on, faster mode is not used even if the Spectre switch is still saved on; the

header can read Wi-Fi fast scan needs Developer options. Turn the Spectre switch off when the drive is over — battery and heat are higher than High performance alone.

What it will not do. Not a continuous Wi-Fi stream. Not Hunt (still BLE). Not associated clients. Not a guarantee you will hear every fleet AP — body block, other vehicles, 5 GHz range, and a renamed randomized BSSID without a vendor IE still miss. Pattern match is still not that agency or that car. Use it as a drive-time AP net for signatures you already care about, then look with your eyes.

10.4 Alert configuration practices

- Stock bookmarks already watch body-cam, camera glasses, recording wearables (Fieldy / Plaud Note), pentest, public-safety vehicle APs, and roadside / public camera + ALPR. Unbookmark a family you do not want to hear (Meta IDs also hit Quest; Flock LiteOn / Espressif OUIs are noisy).
- Watch the signature, not twenty MACs, when the question is “is any Flock/Raven/AirTag here.”
- Watch the device when you have already isolated one radio and need to know if it returns after you leave and come back.
- Do not watch Unknown Signature. It will fire constantly in any populated area.
- Confirm the first alert visually. A single BLE packet can be a passerby.
- The shade card is silent by design. Raise media volume and tap Test alert (pip, spoken class, or both — whatever is on). Check POST_NOTIFICATIONS only if you also want the card.

11. Logging

11.1 What is written

Logging can be turned off in Settings. When it is on, observations append under `files/logs/` as `spectre-NNN.csv` or `spectre-NNN.jsonl`. Pick the format in Settings. Default rotation is 1024 KB; the index cycles through 000–011 (12 files). In a flood, Spectre writes new radios and every 25th repeat (capped per batch) so the log mutex does not stall the Live display. Export concatenates the current parts and strips extra CSV headers. If Tag detections with GPS is on and the phone has a fix, each written row includes `lat` and `lon` — the operator phone at hear-time, not the other radio. Those fields are empty when tagging is off or there is no fix yet. New rows also append `vendor_ie` (Wi-Fi vendor-IE OUIs, up to eight, pipe-separated). That column is last so older 17-column parts in a concatenated export still line up: `lat/lon` stay put; `vendor_ie` is empty on those rows. BLE rows leave it empty. Signature candidates uses it; WPA / RSN / P2P / Qualcomm chip IEs are logged but not clustered as families. The `fleets` column is the signature names at **write time** — re-match from the other columns if the catalog has changed.

11.2 CSV columns

Column	Meaning
timestamp	Unix epoch milliseconds (device clock).
iso	UTC timestamp yyyy-MM-dd'T'HH:mm:ss.SSS'Z'.
kind	WIFI or BLE.
mac	Normalized address, colon-separated hex.
name	SSID or BLE local name. Commas/newlines stripped. Empty if unknown or hidden.
rss	Last received signal strength, dBm.
channel	Wi-Fi channel derived from MHz; 0 for BLE.
freq	MHz. BLE is recorded as 2402 as a band marker, not a precise advertising channel.
oui	First three octets.
vendor	IEEE MA-L/M/S name for a universal MAC, else Bluetooth company name when that is all we have. Randomized BLE addresses usually empty.
fleets	Matched signature names joined with +, at write time. Column name is still fleets for log compatibility. Signature candidates re-matches; do not trust this column after a catalog change.
mfg	Manufacturer company ID as hex, or empty.
uuids	Service UUIDs joined with .
flags	RAND and/or HIDDEN.
raw	Manufacturer data hex, else raw advertisement hex, truncated to 80 characters.
lat	Operator latitude (WGS84 decimal degrees, 6 dp) when Tag detections with GPS is on and a fix exists. Empty otherwise. Phone GPS at hear-time, not the other radio.
lon	Operator longitude, same rules as lat.
vendor_ie	Last column. Wi-Fi vendor-IE OUIs, pipe-separated, up to eight (for example C8:3A:6B 00:50:F2). Empty on BLE, on older 17-column rows, and when the scan reported none. Signature candidates clusters product IEs; 00:50:F2 (WPA), 00:0F:AC (RSN), 50:6F:9A (P2P), and 8C:FD:F0 (Qualcomm chip) are logged but not used as a family ID.

Header row:

```
timestamp,iso,kind,mac,name,rssi,channel,freq,oui,vendor,fleets,mfg,uuids,flags,raw,lat,lon,vendor_
ie
```

Older files omit the last column. A Share log that concatenates mixed parts writes the 18-column header once; rows from before this feature have 17 fields, so vendor_ie reads empty and lat/lon stay in place.

11.3 JSON Lines

Each line is one object. Same facts as the CSV row. JSONL is easier to feed to jq or a notebook; CSV is easier in a spreadsheet. JSON uses rand / hidden booleans instead of a flags field, and mfg as an integer (CSV stores that company ID as hex). Older JSONL lines omit vendor_ie, rand, and hidden; the parser treats them as empty / inferred from the MAC.

Field	Meaning
ts	Unix epoch milliseconds (same as CSV timestamp).
iso	UTC timestamp.
kind	WIFI or BLE.
mac	Normalized address.
name	SSID or BLE local name.
rssi	Last RSSI, dBm.
channel	Wi-Fi channel; 0 for BLE.
freq	MHz. BLE recorded as 2402 as a band marker.
oui	First three octets.
vendor	IEEE or Bluetooth company name, or null.
fleets	Matched signature names joined with +, at write time. Re-match if the catalog changed.
mfg	Manufacturer company ID as integer, or null. (CSV stores the same ID as hex.)
uuids	Service UUIDs, comma-separated.
raw	Manufacturer or advertisement hex, up to 160 characters.
vendor_ie	Wi-Fi vendor-IE OUIs, pipe-separated, up to eight. Empty on older files and BLE. Same meaning as the CSV column.
rand	JSON only. Randomized address. CSV uses flags RAND instead.
hidden	JSON only. Hidden SSID. CSV uses flags HIDDEN instead.
lat	Operator latitude, or JSON null if tagging is off or there is no fix. Phone GPS at hear-time.
lon	Operator longitude, or JSON null. Same rules as lat.

11.4 Debrief and AI Export

Experimental sit report — not a finding. Experimental software. Use at your own risk. Spectre does not guarantee that trackers, cameras, tags, or any other device will be found. Each phone has radio and OEM limits that this app cannot overcome. Do not use Spectre in any situation where safety is in question, including real-life threatening situations. Debrief “possible tail” language and AI Export analysis are hypotheses from 15 minutes of in-memory radios. They are not identity, not complete, and not advice about whether you are safe. Do not hand a Debrief or an AI chat to anyone as proof. The AI Export prompt opens with this same disclaimer and tells the model to repeat it.

Reports has two Debrief buttons that share the same sit report and differ only in format. Neither is a chat prompt. Both are built from the last 15 minutes in the live map (about 400 radios; how that map fills and drops on a drive is §11.4.1), not the rotating on-disk log, not from whatever Live display view is open, and not from the current filter. Radar vs list, Moving with you, Signatures only, Hide selected, and New detections only change the Live display only. Radios keep scanning while the share sheet opens. Treat either share as operationally sensitive: neighbor SSIDs, MACs, and operator GPS when tagging is on.

- **Debrief (text)** — text/plain on the system share sheet. Paste into notes, chat, or a logbook.
- **Debrief (PDF)** — letter-size (612×792 pt) typeset copy via Android PdfDocument. Night-green SPECTRE / FIELD DEBRIEF header, numbered sections, amber Possible trackers with you / Possible tail (finder tags) plus Retail beacons with you / Wearables with you when those lists are non-empty, amber Extra attention callout for each matched special note, green takeaway box, page N / M, footer “Off Grid Pete LLC · operationally sensitive.” Cached under cache/debrief/spectre-debrief-YYYYMMDD-HHMMSS.pdf and shared as application/pdf through FileProvider app.spectre.files. Open in Drive, Files, email, or a PDF viewer.

11.4.1 What Debrief actually sees

Debrief is a sit report of the **live map in RAM**, not a replay of the log. Tap the button and Spectre takes the radios still in memory plus the operator GPS path, then keeps those whose first or last hear is in the last 15 minutes. Filters and the Live display view (radar / list / timeline / hybrid) do not change that snapshot. The rotating CSV/JSONL is a different file — that is Share log / Save log, and that is what Signature candidates reads (§11.5).

Two clocks, both short, and a size cap:

- **15-minute report window.** A radio last heard 16 minutes ago is not in this Debrief, even if the log still has the packets.
- **RAM eviction.** Unnamed radios (no signature match) drop about 3 minutes after the last packet. Named / signature-matched radios stay up to 15 minutes after last heard.
- **Crowd cap.** The live map holds about 400 radios (hard ceiling 900) so a street of rotating BLE MACs cannot grow forever. Oldest unnamed drop first. Radios still inside Brief hold are not dropped by that cap. If the set still overflows the hard ceiling, even named rows can go.

On a drive those caps bite. City streets are dense: phones rotating BLE, shop APs, other cars. A few kilometers of traffic can fill the 400. The café or intersection at the start of the trip is then gone from the next Debrief — unnamed BLE may already have been gone after three minutes. That is expected. It is not a failed scan, and it is not the log deleting itself.

What **does** stay is anything still advertising along the path. A bag tag, a car TPMS, a planted tracker, your own AirTag or watch — they keep getting packets, so last-heard stays fresh. If they match a signature they get the 15-minute linger instead of three, and they are not first in line for the crowd cap. Radios moving with you therefore show up in Debrief after Debrief. Roadside clutter you only passed does not. Filters → Moving with you is not required for that; Debrief’s co-travel callouts use the same GPS trails whether that switch is on or off.

How to work a long trip

- Tap Debrief (text or PDF) about every 10–15 minutes, or at each stop, and **keep those shares**. Each file is that window, not the whole drive.
- Do not wait until you get home to tap Debrief once. That last 15 minutes is the last few kilometers, not the departure neighborhood.
- Leave logging on. Share log / Save log is the hour-long file. Debrief will never reconstruct it from RAM.
- You do not need to sit still. A rolling Debrief still writes distance, Where you were, and co-travel for radios that stayed with the car.
- A thin inventory of unnamed BLE after a highway run is normal. Check Possible trackers / Possible tail / Vehicle / Extra attention first — those are the rows that survived on purpose.

AI Export uses the same memory snapshot (plus a 5-minute slice). Same caps, same advice: paste more than once on a long trip if you want the chat to see more than the last stretch.

The report body is the same in both formats: generated time, 15-minute window, scan settings, GPS-tag on/off, overall distance traveled (along-track path length and straight-line span when tagging is on), then (1) executive summary, (2) **Where you were** (stays vs transit along the operator path — coordinates once per stay, not on every radio line; street names on those stays if Online place names ran), (3) tracking assessment, then optional amber callouts **Possible trackers with you** and **Possible tail** (finder tags) when those lists are non-empty, plus **Retail beacons with you** and **Wearables with you** when those classes stayed with the path, then environment, Wi-Fi, BLE (notable BLE also prints Decode fields when a map applied), signature hits, persistence, Extra attention when a matched signature has that field filled (full caution text on each hit; amber highlighted boxes on the PDF; pattern match, not a skimmer detector), anomalies, privacy, recommended actions, and a one-line takeaway. **Anomalies** is not a second inventory: Extra attention, Signature hits, and tracking callouts already list named matches. Anomalies only keeps cues that are not those sections (Fast Pair in pairing mode, a very loud unnamed radio, a high randomized-BLE count). AirTag / SmartTag / Tile rosters and Extra attention reprints are omitted.

If Settings → Tag detections with GPS is on and you have moved about 45 m or more, Debrief tests whether radios co-traveled with your GPS path, split by class. The tracking assessment itself is short (did the test run, how far you went). Only radios that **stayed with the path** get their own amber callouts. House tags and other radios you only passed are omitted — they are not tracking you. **Possible trackers with you** (amber): finder tags — AirTag / Find My, SmartTag, Tile, Chipolo, Pebblebee, and loud pocket Apple BLE. Stayed with the path for the sit. That can be your kit or one planted in the car/bag/on you before you started. Do not dismiss it as yours; account for each MAC. A pocket iPhone that shows as Apple BLE / Continuity is included when it stays loud (peak ≥ −55 dBm, never below −70 dBm, at least two GPS stamps, heard in the last three minutes). **Possible tail** (separate amber callout): finder tags first heard after the sit started, then stayed with the path. That can mean someone started following you (their phone or tag), or a device was added during the trip. A late-join radio is only a possible tail if three extra gates also pass: the radio's GPS trail covers about half your path, at least two-thirds of those stamps are −75 dBm or louder (same idea as Filters → Moving with you), and the last stamp is not 12 dB below the loudest. A house Find My heard only on a sidewalk arc, or that faded as you walked a loop, is omitted — not a tail. **Retail beacons with you** (amber, separate): iBeacon, Target Atrius basket, Minew, Estimote, Kontakt.io that stayed with the path. Store beacons are usually fixtures — they do not typically move with you. If one did, account for a test tag, a badge, or a short overlap with a fixture. Not a Find My tail. **Wearables with you** (amber, separate): Garmin, Fitbit, Oura that stayed with the path. Usually your own watch/ring or someone walking with you. Not typically a planted tracker. Find My and iPhone addresses rotate; each MAC is this session, not a unique ID. GPS tagging off, or a sit in one place, cannot run this test — the report says so. Car drive-bys still need three GPS stamps on that radio (unchanged). This is a heuristic, not a legal finding and not identity. You do not need Moving with you on, and you do not need to be looking at the list, for these callouts to appear — Debrief reads the last 15 minutes whether the Live display is on radar, list, timeline, or hybrid.

When GPS tagging is on, **Where you were** splits the path into stays (~40 m clusters) and transits. Each stay lists UTC time, dwell, lat/lon (operator phone), optional street name, and the loud radios GPS-stamped at that stay. Sitting still is one stay. A walk to a new block becomes stay → transit → stay. Coordinates are not dumped on every AP/BLE line. Settings → **Online place names** (on by default) reverse-geocodes stay centroids via the system geocoder when the phone has internet. Offline: a note, coords still print, no error dialog. Phone GPS ≠ pole or tag location.

Reports → **AI Export** is a paste-ready prompt for a chat. Same 5/15-minute window as Debrief, not the rotating log. The prompt **includes the onboard Debrief verbatim** so the model has the same sit report you already filed, then the working inventories and precomputed statistics. The model is told **not to rewrite Debrief**. Its job is an addendum the phone cannot write: rates and mixes (5 vs 15 min, RSSI bands, signature-family share, turnover), stress-test of tracking callouts, depth on Extra attention hits, and what stock Android still cannot see. Do not dump MAC lists into the answer.

The prompt contains:

- Opens with the same hobby / as-is disclaimer as first-run and Debrief (hypotheses, local law, not for safety) and instructs the model to repeat it.
- Job: analyst addendum, not a second Debrief. Output headings are at the top of the paste so a size cap does not drop the instructions.
- Onboard Debrief, verbatim (same 15-minute sit report as Debrief text/PDF).
- Collection constraints (Wi-Fi = APs only; BLE advertisers; randomized MACs are not identity; signature/OUI matches are hypotheses; GPS is the operator phone; street names only if lookup ran).
- Where you were and GPS co-travel (possible trackers with you / possible tail only; radios you passed are omitted).
- Counts for 15 min and 5 min, plus precomputed statistics: randomized-BLE percent, arrivals per minute, persistent vs gone, RSSI bands for Wi-Fi and BLE, signature-family counts.
- Channel utilization, mesh/hotspot flags, top-5 strongest unique (dwell + RSSI Δ), time-window co-location (± 10 s of window start and now) — distinct from GPS co-travel.
- Inventories for calculation: every signature hit, full Wi-Fi, hidden SSIDs, notable BLE with looks-like / payload, persistence, Extra attention, machine flags (Fast Pair pairing-mode, loud unnamed, high randomized BLE — not a tracker roster).
- Required output: disclaimer; what Debrief already established (short); statistical picture with numbers; stays/path; tracking addendum (do not invent a tail); Extra attention depth; what onboard could not resolve; Spectre next steps; a takeaway that adds one insight the onboard takeaway does not already say. No safety advice.

The inventory is the same live map as Debrief (§11.4.1): about 400 radios, unnamed BLE older than about three minutes already gone, 15-minute window. Share-sheet size is capped (~90k characters). Treat the paste as operationally sensitive: neighbor SSIDs and MACs.

Device detail has its own **AI Export** and **Share as text** (§5.5). Those are for the open radio only: a dump of that page, or a prompt that asks a chat to decode OUI / company / UUIDs and say what the product most likely is. They do not replace Reports → AI Export (the sit). Do not paste either into a public chat without redaction. The model is still bound by the experimental disclaimer and must not give safety advice or name a person.

11.5 Signature candidates

Reports → Signature candidates is a log miner, not a sit report. It does not use the 15-minute RAM map. It concatenates the rotating parts under `files/logs/`, collapses rows to unique kind+MAC, runs the current catalog through SignatureEngine, and clusters unmatched radios that share a unique on-air ID. Operator-facing cards and Create signature: §5.6.1. How a custom row is saved: §9.2.1. Device detail runs the same ID quality on **one** radio (Signature family card, §5.5) using the live set plus this log.

What it reads from each unique radio (CSV names; JSON equivalents in §11.3):

- **kind, mac, name, flags (RAND / HIDDEN)** — identity. Randomized MACs with no other ID are skipped. House-like names (guest / xfinity / a single dictionary word) are skipped unless another unique ID remains.
- **vendor** — IEEE / company name. Chip-module vendors (Espressif, MediaTek, AMPAK, Qualcomm, UGSI, Realtek) and phone-house OUIs (Google, Apple, Samsung Electronics, Microsoft) are not used as an OUI-only family.
- **oui** — stable BSSID prefix. An OUI family needs three or more distinct MACs, or two with at least two non-house names, so one house AP's virtual BSSIDs do not become a catalog row.
- **vendor_ie** — Wi-Fi vendor-IE OUIs. This is the Roku-class ID when the BSSID is randomized. Only on rows written after this column shipped. Protocol IEs are logged but not clustered.
- **mfg, raw, uuids** — BLE manufacturer company + first data byte, and 16-bit service UUIDs. Broad company IDs (Apple 0x004C, Google, Samsung 0x0075, Microsoft) and generic UUIDs (Fast Pair FEF3/FCF1, Apple FCB2, battery 180F, ...) are not clustered.
- **fleets** — ignored. Re-match uses the catalog on the phone now.
- **lat, lon, rssi, timestamp** — not used for clustering. Frequency is distinct MACs, not packet count.

A family is two or more distinct radios sharing one of those IDs. Overlapping clusters (the same APs matching both a name glob and an OUI) merge into one card with extra rules. Cap is 20 families. Create signature drafts those rules with the radio kind set (Wi-Fi glob does not label BLE). Logging off, a cleared log, or only noise → empty list. Vendor-IE families need new packets after the column shipped; name globs and stable OUIs still work on older parts.

11.6 Share log, save, and clear

On Reports, Share log / Save / Reset sit under the sit-report buttons. Logging on/off, CSV vs JSONL, and rotate size stay on Settings. Share log uses the system share sheet. Save to storage uses the Storage Access Framework — pick internal storage or an SD card. Reset / clear log deletes rotated files on the phone. Export and save show a progress dialog; live logging is paused while the copy runs so the UI does not lock. Clear skips queued appends and opens a fresh file. FileProvider authority is `app.spectre.files`. There is no automatic off-device sync. Debrief (text or PDF) is the sit summary; Share log is the after-action file; Signature candidates mines that same file on the phone. If tagging was on, that file includes operator lat/lon on each new row. New Wi-Fi rows also include `vendor_ie`.

11.7 Reading history

- Sort by iso, then group by mac+kind. A new group is a new radio (or a rotated BLE address).
- Plot rssi against iso for approach/depart. Ignore single-sample spikes.
- Filter signatures != empty for a signatures-only after-action review — this survives whatever filter was on screen at the time.
- A WIFI row with empty name and flags HIDDEN is a hidden SSID AP; use mac/oui, not the name column.
- If two MACs share an OUI and appear within the same minute, check whether Unknown Signature or a cluster rule would have joined them live.
- lat/lon are the phone. Plot them as your track, not as the AP or tag's coordinates. Empty cells mean tagging was off or no fix at that write.
- vendor_ie is pipe-separated OUIs. Empty on older rows and BLE. Group by that column (skipping 00:50:F2 / 00:0F:AC / 50:6F:9A / 8C:FD:F0) when you want product IEs the BSSID OUI missed.
- fleets is write-time. After a catalog change, re-match from name / oui / vendor_ie / mfg / uuids, or use Reports → Signature candidates.

11.8 Storage

Twelve rotated files at 1 MB is about 12 MB worst case, plus config.json. Uninstalling the app deletes logs. Copy out before a Restore defaults or a reinstall. The clock is the phone clock; set it correctly before a timed sit. Tag detections with GPS requests live GPS and network updates while scanning (stale last-known older than 30 s is ignored) and writes that fix onto the in-memory sighting (detail, Moving with you, Debrief) and onto each new log row (lat/lon). New Wi-Fi rows also store vendor_ie. If Location is not high-accuracy, the path can stay 0 m while you drive. That stamp is hear-time correlation, not a survey-grade track. A Share log with tagging on contains operator coordinates — treat the file as sensitive.

12. Field Strategies

These are playbooks for a question you can actually ask the radios. Each one names the controls, what the Live display should look like, and what you must not claim. A signature chip means a pattern matched a public broadcast. It is not a person, a vehicle, or a serial number. Confirm with RSSI trend, presence over time, and your eyes. Filters and Display change only what the Live display shows; the log still records everything. Debrief's tracking section is always the last 15 minutes in memory, not whatever view or filter you have on now.

Run **one** playbook at a time until you can read the result. Stacking Moving with you + New detections only + Signatures only is allowed, but an empty list then has three explanations. Chapter 13 is radio hygiene (battery, screen, crowds).

Experimental — not for safety-critical use. Experimental software. Use at your own risk. Spectre does not guarantee that trackers, cameras, tags, or any other device will be found. Each phone has radio and OEM limits that this app cannot overcome. Do not use Spectre in any situation where safety is in question, including real-life threatening situations. Playbooks such as “Am I being followed?” are radio exercises, not a way to decide whether you are in danger.

12.1 How to read a playbook

- **Arrive first.** High performance, All traffic, Strength list, scan notification present. Note the loudest five APs so “home” noise has a face. If the list is already unreadable, open Display and thin the row (Subtitle None, extras off) before you start hiding radios.
- **Then narrow.** Apply the filter or signature set for the question. If the list goes empty, undo the last clause before assuming the area is clean.
- **Pause to inspect.** Open detail from the frozen row. What this looks like, flags, and payloads beat a glance at a chip.
- **Leave logging on** unless the plaza is melting the disk. You can always hide rows on the Live display and still have the file.
- **Write the sit.** Filters and the Live display view change only what you see. Debrief and AI Export see the 15-minute memory (cap ~400; §11.4.1) — including whether a tracker-like radio stayed with you. On a drive, tap Debrief more than once; Share log is the full file. Which share to tap is §12.12.

12.2 Am I being followed?

Not a safety test. This playbook cannot tell you that you are or are not being followed in real life. It only reports loud radios that stay with your phone's GPS path. Quiet tags, rotating Find My addresses, a phone that missed ads, and your own bag tag all break the test. If you believe you are in danger, leave and get help — do not wait on Spectre. See the Notice page.

Question. Is a loud radio staying with me as I walk or drive — not a house AP I meet at the door?

Setup.

- 1 Tag detections with GPS and Keep screen on are on by default. Location → high accuracy. Scanning must be running so live GPS updates can start.
- 2 Walk or drive until Filters shows a path of about 50 m. If it stays 0 m, the phone is not giving a live fix. Sitting still does not grow the path. After a sit, Live display → Start over if you want a fresh follow test.

- 3 Filters → Moving with you (the preset, or the switch under Radios to show). The switch clears Signatures only and class Show only so unmatched radios can co-travel.
- 4 Optional: Display → Sort → New at bottom so new co-travelers append instead of jumping the list.
- 5 When a row appears that is not your kit: Pause, open detail, bookmark if you want a beep on return.

What you should see. Only loud radios (most GPS-stamped samples about -75 dBm or stronger) that cover a large share of *your* path in both distance and time. A tag in your bag or car will match — that is the confidence check that the filter is working. House APs that only appear when you arrive should stay hidden. Passing cars on a highway will still come and go; a radio that is actually with you should stay. The “still here” window is not a fixed 50 m circle: it grows with how fast you have been moving, and it is longer for Wi-Fi (slow scans) than for BLE, so a cup-holder tag does not blink off between advertisements. Walking still uses a tight house-length. The checklist and the speed table are §8.5.

What it is not. Not direction finding. Not the other device's GPS — it is your phone's fix at hear-time. Find My / Offline Finding MAC rotation will not stitch a tail that changes address every minute. A radio that is quiet, weak, or only heard at one end of the path will not qualify.

Write it down. After ~45 m, Reports → Debrief (text) or Debrief (PDF). Tracking assessment is a short “did the test run.” Amber callouts list only radios that stayed with you, split by class: **Possible trackers with you / Possible tail** (finder tags), **Retail beacons with you** (iBeacon / Target Atrius basket / Minew / Estimote / Kontakt.io — fixtures; a Target basket you pushed will co-travel), **Wearables with you** (Garmin / Fitbit / Oura — usually own kit). House tags you passed are omitted. Plus overall distance — for the last 15 minutes in memory, whether you leave Moving with you on, hide the family, or switch to radar / timeline / hybrid / By class. Filters and view do not shrink Debrief. Carry an AirTag or iPhone: it should land in Possible trackers with you if it stayed loud on you (about -55 to -70 dBm). Find My MACs rotate, so you will see this session's address, not one ID for the hour. A neighborhood loop is a special case: Debrief will not call a house Find My a tail unless that radio covered about half your path, stayed loud (-75 dBm on most GPS stamps), and did not fade 12 dB from its loudest. Car drive-bys still need three GPS stamps. Optional Online place names if you want streets. AI Export if you want a chat to read those callouts plus the inventory — tell it “apply §12.2; do not dismiss whole-sit radios as yours; do not list radios I only passed; do not treat a retail beacon as a Find My tail.” A longer drive: tap Debrief again at the next stop (§11.4.1) — radios that stayed with you will still be there; the first few kilometers of unnamed roadside radios will not. Not a legal finding and not identity.

Own kit first. If you carry an AirTag, SmartTag, Tile, or a tracker in the car, Moving with you will show it. Hide that family or treat it as a known row before you call anything else a tail.

12.3 Has someone new entered the space?

Question. What arrived after I sat down — a phone, a tag, a new AP — that was not already in the room?

Setup.

- 1 Sit. High performance until the first Wi-Fi batch lands (header: Wi-Fi next Ns, then a jump in the AP count).
- 2 Filters → New detections only. the Live display shows New only · learning sitting Wi-Fi, then New only · N hidden.
- 3 When the room is the baseline, Mark seen (bar above the tabs). Pause then Mark seen absorbs the frozen picture, not radios that arrived after the freeze.
- 4 Brief hold 30 or 60 s so a slow advertiser does not pop in and out.
- 5 Reset seen if you change rooms and want that space to count as new. Clearing the log does not reset already-seen.

What you should see. Only radios that were not in the already-seen set. BLE can show as new during the Wi-Fi learning window. After Mark seen, the next new row is the next arrival. A watchlist alert fires only if that row would

actually appear on the Live display.

What it is not. Randomized BLE addresses look new on every rotation — an unnamed LE that “just arrived” may be a phone that changed MAC. Sitting Wi-Fi is hidden through the next scan on purpose. This filter does not identify who walked in; it tells you a radio the session had not already absorbed is now on the air.

Write it down. New detections only does not shrink Debrief or AI Export — those ignore the Live display filter. Use Debrief persistence / first-seen, or AI Export’s “first appeared in this window” table, to record arrivals. Share log if you need every written row later.

12.4 Find only this family

Question. I only care about trackers, or only cameras, or only one custom signature.

Setup.

- 1 Tap preset Trackers, or Filters → Show only plus the class chips (Cameras, Drones, Surveillance, Locks, Pentest, ...). Matching stays on; the Live display just thins.
- 2 If a family is local false positives, hide it on Filters (Hide these for a class, or Hide selected for one family), or mute noisy rules inside the editor (LiteOn OUIs on Flock — §7.6).
- 3 Bookmark the signature if you want a beep when a new match appears.
- 4 Radar or Hybrid for “how close”; list for triage; detail for the decode.

What you should see. Unmatched consumer noise drops away. Chips on the remaining rows are the families still on the Live display. If the list is empty, nothing matching is in earshot — or you hid that family on Filters. Matching still labels; it does not go quiet.

What it is not. Signatures only does not search harder. It hides unmatched radios. Stock catalog rows (Ring, Nest, Hikvision, ...) are mostly names — apartment cameras and coincidental SSIDs will label. Hide a family on Filters if that sit is noise. See §7.6.1 before treating a camera chip as a pole.

Write it down. Debrief “Signature hits” and camera/tracker flags. AI Export lists every signed radio in the window with looks-like / payload — useful when you want a second read on whether a family is local noise. Do not paste that into a public chat without redaction.

12.5 Get rid of the noise, keep the field

Question. I want the neighborhood, minus the families I already know are clutter.

Setup.

- 1 Start All traffic so unmatched radios stay.
- 2 Filters → Hide these, then the Finder tags chip (or preset Hide trackers). That is the whole class. To hide only your bag AirTag, use Hide selected signatures and turn that one row on.
- 3 Useful combo: Signatures only + Hide these Finder tags — named hits, minus bag-tag clutter, cameras still show.

What you should see. The rest of the field stays. Hidden classes drop off the Live display until you turn Hide these off. Class picks stay if you switch modes; Reset filter is what forgets them.

What it is not. Hiding Finder tags does not mean no AirTags are present — they are still matched and logged. Watchlist will not beep for a row the current Live display filter would keep off the list.

Write it down. Debrief and AI Export still list hidden families if they were in the live map. That is the point of the log: the Live display can be quiet; the file is not. Use Share log when you later need the clutter you hid.

12.6 Is this my own tag?

Bag-check, not accusation. Turn on Moving with you after ~50 m. If the only tracker-like row is the one you put in the bag or the car, the filter is doing its job. Bookmark that MAC or hide the family so it stops competing with a real candidate.

If you do **not** carry a tag and a loud tracker-like radio still covers the whole path: Pause, detail, What this looks like, RSSI trend. Share as text or detail AI Export if you want a second read on *that* radio. Then Reports → Debrief (text or PDF) — “Possible trackers with you” (yours or planted before you started — account for it) vs “Possible tail” (first heard after this sit started). Those callouts are the last 15 minutes in memory even if Moving with you is off and even if the Live display is on another view. Sit-level AI Export if you want a chat to stress-test those callouts with rates and mixes (it embeds Debrief; tell it not to rewrite). All of these are heuristics. Find My rotation still will not stitch.

12.7 Camera / ALPR sit

No chip does not mean no camera. Quiet and cellular-first poles are often invisible to this phone. Do not use a blank Live display to decide that a location is safe to photograph, protest, or drive. Experimental observation only. See §7.6.1 and the Notice page.

Setup. Preset Surveillance (or enable Flock / Raven / the camera names you care about, then Signatures only). High performance. Walk the block. Roadside / public camera + ALPR rows (Flock, Penguin, Pigvision, FS Ext Battery, Genetec AutoVu, Rekor, Motorola Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua) beep and get Extra attention “!” — the card says what that family is used for. When a chip appears, look with your eyes, then write what you saw — not what the chip said.

Honest limits. Newer poles are often cellular-first and RF-quiet on Wi-Fi/BLE; no chip does not mean no camera. Spectre cannot hear LTE, cannot see associated clients, and cannot promiscuously capture. IEEE B4:1E:52 plus a Flock-* SSID is high confidence; LiteOn / Espressif / Raspberry Pi OUIs are low unless a name or UUID corroborates. Phone GPS is where *you* were, not where the pole is. Full limits: §7.6.1.

Write it down. Look first, then Debrief (PDF is handy to file). Signature hits and camera flags are cues, not a pole ID. AI Export can list every camera-like row with payload decode — tell the model “§7.6.1; quiet poles exist; do not place a camera at the GPS pin.” Turn Online place names off if you do not want streets of *your* path in the paste.

12.8 Hunting a tracker

No guarantee of a find. Tags advertise slowly, rotate addresses, and may be outside what this handset hears. Absence of a row is not proof there is no tracker. Hunt is relative loudness, not meters and not a compass. Do not use this to decide personal safety. See the Notice page.

Find the row first. Preset Trackers (BLE, Show only Finder tags; unmatched stay hidden and Signatures only is dimmed). Battery saver is usually enough — tags advertise slowly. Brief hold 60 s and Stale after 90–120 s if they flicker. Timeline. Bookmark the family. Hide selected on your own tags if they drown the list. AirTag / Find My addresses rotate: you will see a new MAC, not one identity for the hour. A loud row that stays with you on a walk belongs in §12.2, not here — Debrief still writes that last-15-minute co-travel whether Hunt or Trackers is on the screen. When you have **one** BLE candidate, open its detail → **Hunt** (Wi-Fi has no Hunt button). Screen stays on. Hold the phone the same way the whole time. Reset this hunt after you change grip.

If you do not know which radio. Filter Trackers (Finder tags class). Strength list, High performance. Hide selected on kit you already own (keys, bag AirTag) or those will always win the top row. Then *place the phone* at spots a tag could hide — you are using the handset as a sniffer probe, not walking the street. On a car: each wheel well, inside the bumper, under the rocker, hitch, spare, cabin (console, under seats, visors). Dwell 10–20 s at each spot so a slow

advertiser can speak; tags do not ping every second. If a tracker-family row jumps to the top or RSSI climbs a lot versus the last spot, that is the candidate. Open that row → Hunt. Repeat the same grid with Hunt's Closer/Further, then the body-block turn if you still need a heading. If Hunt goes Gone, the MAC rotated — back to the Live display, pick the new top tracker row, Hunt again. A quiet sweep is not proof the car is clean: the tag may be asleep, out of this phone's earshot, or not in the tracker catalog. The car's metal will shadow; a peak at one wheel can be a bounce. Same grid works on a bag, coat, or stroller: park the phone in each pocket/compartment in turn.

Walk it down. Slow steps, one direction at a time. Watch Closer / Further / About the same and the loudest-this-hunt number — not the Live display radar angle (that is a MAC hash, not north). If Closer, keep that heading. If Further, stop, turn, try another way. About the same means wait or change heading; do not convert dBm to feet. Very Close (about -45 dBm or louder) means look around — usually in-hand, pocket, or the same bag; still not meters. Quiet can be a wall, not the wrong way. If Gone from the Live display, the address likely rotated — back to the Live display, pick the new row if it is still the same family, Hunt again.

Body-block for a possible bearing. The phone has no direction finding. Your torso damps BLE from one side, the same idea as holding an antenna close to the body and turning for a peak. Stand still. Hold the phone against the center of your chest (or the same pocket every time) so your body shadows one hemisphere. Turn slowly through a full circle. Watch the big RSSI and loudest-this-hunt. If the signal gets stronger on one heading, that is a *candidate* bearing — often the direction of least body in the path, i.e. roughly toward the radio. Repeat the circle. Treat it as tens of degrees, not a compass rose. Then Reset this hunt and walk that heading using Closer/Further. Reflections and metal can fake a peak (see structures below).

Structures lie. A wall, floor, or door can drop RSSI 10–20 dB with no change in true range. Metal (cars, cabinets, elevators, foil-backed insulation) reflects and shadows — the loudest body-block heading can be a bounce off a van. People and water absorb. Concrete and rebar scatter. Glass is kinder than brick. Two floors of a house can fake Further. Covering the antenna with a fist is another wall. Do not treat one peak as a map pin; walk, turn, and see whether Closer survives the next opening. More on the Hunt page itself: §12.13.

Write it down. Detail → Share as text or AI Export for that radio. If the question was a planted tag that stayed with you, Reports → Debrief still has the last-15-minute tracking assessment (§12.2) — Hunt, Trackers filter, and radar vs list do not change that report. Sit-level AI Export lists Find My / Fast Pair flags. Share log for a later spreadsheet of MACs (they will rotate).

12.9 Follow an SSID or OUI lead

Setup. Filters → Name / MAC contains, or OUI / vendor contains. AND. Both radios on. Do **not** enable Signatures only or you will hide an unmatched but relevant AP. Strength list. Long-press → Create signature if it keeps appearing.

A BSSID that repeats across visits is a fixture. A randomized BLE MAC that repeats is probably the same session, not a long-lived identity. Debrief Wi-Fi inventory and AI Export's full AP list are the sit snapshot; Share log is what you compare on the next visit.

12.10 Walk, sit, or drive

You are...	Use
Sweeping a new block	All traffic, High performance, Strength list or Hybrid. Learn the noise before you filter.
Walking — who is with me	§12.2 Moving with you on the Live display; Debrief still writes the last-15-minute tracking line even if you leave that filter or change view. Need GPS tagging and ~50 m of path. Walking uses a tight ~75 m still-here window (§8.5).
Driving — who is with me	Same filter. "Still here" grows with speed so a car tag does not flash between advertisements; passing roadside radios still fail the trail-moved gates. §8.5, §12.2.

You are...	Use
Sitting a room — who just arrived	§12.3 New detections only. Mark seen after the room is the baseline. Reset seen when you change rooms.
ATAK overlay (Remote ID / Extra attention)	§5.8 configure, §12.15 sit. Settings → TAK / CoT feed. Extra attention + Payload location. Same Wi-Fi LAN. Privacy mode off.
Long sit / parked vehicle	Balanced or Saver. Timeline. RSSI floor –80 if unreadable. Logging on. Watchlist, not a stare.
Drive, then arrive home	Moving with you should drop house APs you only meet at the door. If they still show, the path was too short or GPS only stamped at the destination.
Long drive (several km)	Do not wait for one Debrief at the end. Tap Debrief every 10–15 min or at stops and keep the shares (§11.4.1). City RF can fill the ~400 live-set cap in a few kilometers. Radios moving with you stay in each report; roadside unnamed BLE will not. Share log is the hour-long file.
List too fast to tap	Pause. Then detail. Resume. Or Display → Subtitle None so more of the list fits without Pause.
Fit the row to the job	§5.3. Display is look (Title/Subtitle, extras). Filters are who. Plaza: Subtitle None. Copy MAC: Title → MAC. Channel sit: Frequency on.
Write the sit	§12.12. Debrief (text or PDF) for a sit report; AI Export for a chat prompt; Share log for the file. Long drive: Debrief each stop (§11.4.1).
Walk toward one BLE / a tracker chip	§12.8 (tracker) or §12.13 (Hunt page). Walk for Closer/Further; body-block turn for a candidate bearing. Not meters.
Don't know which tracker	§12.8. Filter Trackers, hide own kit, dwell the phone at each hide (wheels, bumpers, cabin). Top row → Hunt. Repeat. Not a clean bill.
A “!” / card reader / pentest kit	§12.14. Extra attention on Hobby BLE serial (not a skimmer detector) and Pineapple / Flipper / Pwnagotchi / Marauder / Porkchop. Open detail, look with your eyes. A miss is not a clean bill.

12.11 The list is unreadable

A busy list is not the same as the wrong radios. Show less on each row before you start hiding devices. Display changes how much of each radio you see; Filters change who is on the list. Logging is not affected either way.

- Pause if you cannot tap a row. The radios keep scanning.
- Live display → Display (tune). Set Subtitle to None for one identity line and more rows on screen; rand/gone move onto the title. Turn off RSSI bars, Frequency, first/last, and (if chips are noise) Signature names. Prefer Strength list over Hybrid in a plaza.
- Title already defaults to MAC. If type guesses (“Apple, Inc. · ...”) crowd the subtitle, set Subtitle to Advertised name or None.
- Still too many radios: then Filters. Raise the RSSI floor (preset Strong signal, –70 dBm) or the slider –80. Signatures only, Hide these Finder tags, or Hide selected on one family.
- Do not stack every filter. Undo the last clause if the list goes empty.
- Turn logging off only if you do not need the file. Crowd cap is 400; unnamed BLE older than ~3 minutes may already be gone from the Live display but still in the log if it was written.

12.12 After the sit — reports and AI Export

Debrief, sit-level AI Export, Signature candidates, Share log, Save, and Reset / clear log are on the **Reports** tab. GPS tagging, Online place names, and logging on/off stay on Settings. Two more shares live on the **device-detail** page (§5.5): Share as text and AI Export for *that radio only*. Sit-level Debrief and AI Export are built from the last 15 minutes in the live map (about 400 radios; hard ceiling 900) — not from whatever filter is on the Live display, not from radar vs list vs timeline vs hybrid, and not from the rotating on-disk log. How that memory fills and drops on a drive is §11.4.1: tap Debrief more than once; radios moving with you stay; roadside clutter does not. A planted-tracker co-travel line is in that sit report whenever tagging was on and you moved, even if Moving with you is off. Signature candidates is the opposite window: it reads the rotating log for unmatched families (§5.6.1, §11.5). Radios keep scanning while the share sheet opens. Treat every share as operationally sensitive: neighbor SSIDs, MACs, and — if tagging or place names were on — your path and streets.

Share	What it is	Use it when
Debrief (text)	Sit report, plain text. Distance, Where you were, tracking assessment (last 15 min, ignores Live display view/filter), environment, inventories, actions, takeaway. Reports tab.	After a walk: was something with me? Notes, Signal/SMS, a logbook. Same words as the PDF.
Debrief (PDF)	Same sit report, letter-size typeset (SPECTRE header, numbered sections, amber Possible trackers with you / Possible tail callouts when those lists are non-empty, amber Extra attention callout per special note, takeaway box). Tracking does not care which Live display view or filter was on.	Hand to someone, file the sit, print. Easier to read than the text dump.
AI Export (Reports)	Sit-level analyst <i>prompt</i> : onboard Debrief verbatim, plus 5/15-minute working data and precomputed stats. Asks for an addendum (rates, mixes, tracking stress-test, Extra attention depth) — not a rewrite of Debrief.	Paste into a chat when you want numbers and competing hypotheses the phone report cannot write. Not a legal memo.
Signature candidates	Log miner on Reports. Re-matches the rotating log, lists unmatched families that share a unique on-air ID (2+ radios). Create signature is a draft with the shared rule, no MAC pin. Save returns to the list and re-runs it. Offline.	After a sit with logging on: recurring unmatched globs / vendor IEs / OUIs worth a custom signature. Not every unknown radio. §5.6.1, §9.2.1, §11.5.
Share log / Save	Rotating CSV or JSONL of what was written to disk this session (including vendor_ie on new Wi-Fi rows). Reports tab, under the sit-report buttons.	After-action file, spreadsheet, revisit correlation. Use this when you need rows the 15-minute memory already dropped.
Share as text (detail)	Plain dump of the open radio's detail page.	Notes, a ticket, or to keep one MAC/payload without the whole sit.
AI Export (detail)	Prompt about one radio: dump plus registry/format decode job. Same disclaimer as sit-level AI Export.	"What is this AP / tag / chip?" Do not use it as a following test — that is Reports → Debrief after you move.
Hunt (detail, BLE)	Closer/further needle, rings around YOU, hunt sparkline. Very Close at about -45 dBm or louder. Beep/Vibrate geiger tick at the bottom. Optional body-block heading. Structures shadow and reflect.	Walk toward one BLE in a room, bag, or car. Not meters. Not a following test. §12.13.

How they pair with the playbooks

- **§12.2 Followed?** Tag GPS, walk about 50 m, then Debrief. Read Possible trackers with you and Possible tail (the last 15 minutes in memory — Live display view and Moving with you do not change it). On a longer walk or drive, tap Debrief again at the next stop (§11.4.1) — a radio that stayed with you will still be there. AI Export if you want a chat to stress-test those callouts with rates and mixes — tell it not to rewrite Debrief, not to dismiss whole-sit radios as yours, and not to list radios you only passed.
- **§12.3 New in the room?** Debrief persistence / first-seen, or AI Export arrival table. The New detections filter does not change the report.
- **§12.4 / §12.7 Families and cameras.** Debrief signature hits. AI Export for every signed row plus payload decode. Keep §7.6.1 in the prompt for camera sits.
- **§12.5 Hidden clutter.** Still in Debrief/AI Export/log. the Live display was quiet; the file was not.
- **§12.6 Own tag?** Debrief “Possible trackers with you” (yours or planted — account for it) vs “Possible tail.” Same verdicts in AI Export.
- **§12.8 Tracker hunt.** Filter Trackers. If you do not know which row: dwell the phone at hide spots (car wheels, bumpers, cabin) until one pops to the top, then Hunt. Walk / body-block to narrow. Structures shadow. Not a clean bill.
- **§12.9 SSID/OUI lead.** Debrief or AI Export Wi-Fi inventory for the sit; Share log to compare BSSIDs next visit. Signature candidates if the same unmatched glob or vendor IE keeps showing up.
- **Catalog gap?** Open an unmatched radio: Signature family on detail is the one-radio check (Strong / Possible / This radio only). Logging on, then Reports → Signature candidates for the sit-wide list. Recurring unmatched name globs / vendor IEs / stable OUIs. Create signature, Save, then the Live display should label the next hear. Not a house SSID and not a chip-module OUI.
- **§12.14 Extra attention / pentest kit / card-reader caution.** Debrief Extra attention section and PDF amber callouts. Detail Share / AI Export quote EXTRA ATTENTION. Tell a chat “pattern, not a skimmer detector, not proof of an attack.”

Online place names (on by default) applies to both Debrief and AI Export. Offline: a note, no error, no streets. Distance traveled does not need internet. Leave it off if you do not want addresses in a chat paste.

Compare OUIs and names across visits in a spreadsheet. A repeating BSSID is a fixture. Treat Debrief “possible tail” language — and anything an AI writes from AI Export — as a cue to look again, not as a conclusion.

12.13 Hunt — walk toward a BLE radio

Relative loudness, not a tracker. Hunt is a fox-hunt needle for one BLE advertiser. It does not measure meters, does not give a compass bearing, and does not identify a person. Randomized addresses can vanish mid-hunt. Absence of Closer is not proof the radio is gone. Do not use this page to decide personal safety. See the Notice page.

When to use it. You already have **one** BLE row you care about (a tag chip, a loud unnamed LE, a speaker). Open detail → Hunt. Wi-Fi APs have no Hunt button: the OS only batches them about every 30 s (about 8 s with Faster Wi-Fi AP scans in effect — still not Hunt). Room, bag, car, one floor are the realistic envelope. A parking garage through walls will lie.

Setup. High performance. Hold the phone the same way the whole hunt (same hand, same tilt). Screen stays on on this page. Reset this hunt after you change grip or after a body-block turn so the walk is not compared to the spin. Logging and GPS tagging can stay on; in a dense plaza they add work — turn them off if the needle feels late.

Walk. Slow steps. Watch Closer / Further / About the same (~3 dB over a few seconds), the rings around YOU (in = louder, out = quieter), and the loudest-this-hunt number. Beep / Vibrate sit under Reset and Back (off until you turn

them on): a geiger tick on last-heard RSSI — faster when louder, silent when Quiet or Gone. Raise media volume if you chose Beep and hear nothing. Very Close (about -45 dBm or louder) means look here — pocket, bag, in-hand — not a tape measure. Pause when Quiet; it may be a wall, not the wrong way. If Gone from the Live display, the MAC likely rotated or left — back to the Live display, pick the new row if it is the same family, start Hunt again. Do not convert dBm to feet. Fig. 5b.



Fig. 12 — Hunt while walking. Rings around YOU: inward on Closer, outward on Further. Beep / Vibrate at the bottom. Privacy mode is on: last three MAC octets show as **:*.**.*.

Body-block heading (optional). The handset has no DF. Your torso is a lossy shield, the same trick as holding a whip close to the body and turning for a peak. Stand still. Hold the phone against the center of your chest (or the same pocket every time). Turn slowly in place through a full circle. Watch loudest-this-hunt and the big RSSI — not the Live display radar angle (that is a MAC hash, not north). The heading where it peaked is often the direction of least body in the path, i.e. roughly toward the radio. Reflections, metal, and a second source can fake a peak. Repeat the circle. Treat the heading as tens of degrees, not a compass rose. Then Reset this hunt and walk that way using Closer/Further. This is not UWB Precision Finding and not a bearing you should write as a fact.

Structures change the needle. Hunt reads one path to the radio, not a vacuum. A wall, floor, or door can drop RSSI 10–20 dB with no change in true range — Quiet or Further may mean you walked behind masonry, not away from the tag. Metal is worse: car bodies, filing cabinets, appliances, foil-backed insulation, and elevator cars reflect and shadow. The loudest heading on a body-block turn can be a bounce off a van, not the source. Water and people absorb (a crowd between you and a tag looks like distance). Concrete and rebar (garages, basements) scatter 2.4 GHz; a peak on the sparkline at a doorway is often a slot through the structure. Glass is kinder than brick. Two floors of a house are enough to fake Further. Hold the same grip: covering the antenna with a fist is another “wall.” Do not treat a single peak as a map pin; walk, turn, and see whether Closer survives the next opening.

Must not conclude. Distance. A person. That you are safe because Hunt went Quiet. That two Hunts on rotating MACs were the same tag unless the payload still matches. That the loudest heading is the true bearing when you are next to metal or a wall.

Write it down. Detail → Share as text or AI Export for that radio. If the question was following you, Reports → Debrief still has the last-15-minute tracking assessment (§12.2) even from Hunt, and even if the Live display is on radar. Not for locating a bag tag in a room.

12.14 Extra attention — card readers and pentest kit

Not a skimmer detector, not proof of an attack. A “!” on the Live display means a matched signature has Extra attention filled. That is a pattern on a public broadcast, not a person, not a planted overlay, and not a finding that a crime is in progress. The same hobby BLE modules show up on printers, cars, and DIY. Pentest firmware can be renamed or quiet. A miss is not a clean bill. Look with your eyes. Spectre does not connect and does not try PINs. See the Notice page.

Question. Is there a radio here that the catalog tagged for Extra attention — cheap BLE serial names next to a card reader, or a default pentest AP / Flipper / handshake collector — and what should I actually do with that mark?

Setup.

- 1 Signatures tab: Hobby BLE serial, Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, and Porkchop have Extra attention filled. Hide a row on Filters if that name is local noise.
- 2 Live display: All traffic, Strength list (or Hybrid). A “!” on the row is the Extra attention mark — not the editor Notes field.
- 3 Optional: Filters → Signatures only after you have seen the room, if you only want those families. Do not start filtered or you will miss unmatched radios.
- 4 Tap the “!” row. Read Extra attention on detail. Bookmark if you want a beep when that MAC returns. BLE only: Hunt if you need closer/further (§12.13).

What you should see. The “!” only appears when a *matched* signature has Extra attention text. Detail shows the full caution. Debrief and AI Export quote it. The PDF draws an amber callout per hit. Empty Extra attention = no mark.

Family	What Spectre can hear	Honest limit
Hobby BLE serial	BLE names HMSoft / HM-10 / CC41 / AT-09 / JDY / BT05 / ESP32 BLE. Extra attention: same boards have been used in some pump/ATM overlays.	Not Classic HC-05/HC-06 (Spectre does not see Classic). Same modules on printers, cars, DIY. Loud next to a card reader: look with your eyes. Not proof of a skimmer. A miss is not clean (name changed, Classic, or cellular).
Hak5 Pineapple	Setup SSID Pineapple_XXXX / Hak5 / WiFi Pineapple.	Admin/management AP. Not Alfa OUI 00:C0:CA. PineAP clones look like ordinary café SSIDs. Renamed or clone-only: miss.
Flipper Zero	OUI 0C:FA:22; BLE name Flipper*.	Newer units use that IEEE OUI. Custom firmware can change name and MAC. Bluetooth off: miss. Not proof of an attack.
Pwnagotchi	Classic BSSID de:ad:be:ef:de:ad; name pwnagotchi.	Handshake-collector beacon. Custom MAC/name: miss.
Marauder / Deauther	Default names MarauderAP / Marauder / Deauther.	ESP32 Marauder or Spacehuhn-style defaults. Same boards are DIY. Renamed: miss.

Family	What Spectre can hear	Honest limit
Porkchop	SSID/name PORKCHOP; BACON fake-AP vendor IE 50:52:4B.	M5PORKCHOP Cardputer or CYD port. Not Espressif OUI. BLE spam (Apple/Android lookalikes) is not matched. Passive-only or no AP: miss.

What it is not. Not a card-skimmer scanner. Not a pentest-kit detector that sees USB-only Hak5, renamed firmware, or quiet radios. Body-worn / glasses / recording-wearable Extra attention (Axon, WatchGuard Video, Ray-Ban / Meta, Snap Spectacles, Fieldy, Plaud Note) is the same “!” mark — those families are §9.5 / the appendix, not this playbook. Spectre never connects. Do not treat a “!” as identity or as a reason to touch someone else’s gear.

Write it down. Detail → Share as text or AI Export for that one radio (the dump starts with EXTRA ATTENTION). Sit: Reports → Debrief (PDF if you want the amber callouts) or AI Export — tell the model “§12.14; Extra attention is a pattern, not a skimmer detector, not proof of an attack.” Filters do not shrink Debrief. Do not paste into a public chat without redaction.

12.15 Overlay on ATAK (TAK / CoT)

Coordinates on the LAN. This sit puts full MACs and coordinates onto whatever is listening at the host:port you set. Privacy mode must be off or Spectre will not send. Multicast stays on this Wi-Fi LAN. Not a safety picture, not DF, not a Remote ID interceptor. §5.8.

Question. Can I see Extra attention radios and in-flight Remote ID on the same map as the rest of the team, without building a Remote ID plugin or teaching ATAK a Spectre format?

Setup.

- 1 Phone and ATAK (or WinTAK / iTAK) on the same Wi-Fi. Confirm the AP does not isolate clients. If multicast never arrives, you will switch Host to the ATAK device’s IPv4.
- 2 ATAK already listens to UDP 239.2.3.1:6969 for SA. No Spectre plugin. For a TAK server, have that server’s unicast host and CoT input port ready.
- 3 Spectre → Settings → Privacy mode Off. Tag detections with GPS On (high-accuracy Location) if you want heard-here pins; Remote ID advertised position does not need it.
- 4 Settings → TAK / CoT feed On. Port 10011 (ATAK CIV). For ATAK CIV on this phone, set Host to the phone’s Wi-Fi IPv4. SA multicast is 239.2.3.1:6969; many TAK servers use 8087. Extra attention On, Payload location On, Watchlist Off, All signatures Off.
- 5 Live display: All traffic, High performance. Do not filter to Drones only if you also want body-cam / glasses — the feed ignores Live display filters, but you still need those radios to match.

What you should see. Within a few seconds of a qualifying hear, ATAK plots a marker. Extra attention (Axon, glasses, Flipper, Pineapple, ...) sits at *your* GPS — walk and it follows you while that radio is in earshot. Remote ID Location sits at the *advertised* aircraft lat/lon (CoT type unknown UAV); the next Basic ID packet does not clear it. Callsign is the signature name. Remarks say advertised position or heard here. A radio that leaves goes stale on ATAK after about two minutes.

If this...	Then...
Nothing on the map	Privacy mode, master off, wrong LAN, or no Extra attention / Remote ID on the air yet. Open the Live display and confirm a “!” or a Remote ID chip. Try unicast to the ATAK IP if multicast is filtered.
Pins sit on me, not the drone	That radio has no advertised lat/lon (or Payload location is off). Heard-here is working. Remote ID needs a Location message (protocol 2) on BLE FFFA.

If this...	Then...
Drone pin is kilometers away	Expected for advertised position. Spectre did not DF it. The aircraft encoded that WGS84.
Café APs filled the map	All signatures is on. Turn it off. Extra attention + Payload location is the field default.
Marker vanished after ~2 min	The radio left earshot or scanning stopped. CoT stale, not a delete. Start scanning again if you still want it.
I want one bag tag on the map	Watchlist On, bookmark that MAC (Alert on). Extra attention can stay on. Do not use All signatures.
Custom sensor with lat/lon in the ad	Decode fields: ids latitude and longitude (scale as the spec). Payload location On. No TAK checkbox. §5.8.3, §9.6.

What it is not. Not a team SA substitute (Spectre is not your self-marker). Not Wi-Fi NAN Remote ID. Not encrypted ads. Not a range ring. Filters, Pause, and Display do not change the feed. Stop on the scan notification stops sending.

Write it down. ATAK is the live overlay. Spectre still has Debrief / Share log for the sit file. The CoT remarks are not a report. If you share a screen of the map, remember the MAC is full — Privacy mode was off to publish.

13. Field Hygiene

Chapter 12 is the playbook. This chapter is how to keep the phone collecting while you run those strategies.

13.1 Detection quality

- Keep the phone out of a Faraday-ish bag and off a metal dashboard.
- Body blocking is real: a BLE tag behind you will look 10–20 dB weaker.
- 5 GHz APs have shorter range; losing them while 2.4 GHz remains is geometry, not “they turned off.”
- If Wi-Fi counts freeze, the OS is throttling or Location dropped. Settings shows the throttle hint on API 30+.
- Airplane mode with Wi-Fi and Bluetooth manually re-enabled can reduce cellular noise but is not required and may confuse some OEMs.

13.2 Battery, screen, and background

Two jobs. A sit or drive where you want every scan the OS will give: §4.5 (High performance, Keep screen on, Unrestricted background, optional Faster Wi-Fi) and accept the drain. This section is the all-day / pocket compromise after that picture is stable.

- Start High performance (§4.5). Drop to Balanced once the picture is stable if you need the battery. Battery saver is overnight / bag carry — you will miss short BLE bursts.
- **Keep screen on** (Settings, on by default) holds the display while Spectre is in front so Samsung does not park BLE on screen-off. Turn it off when you pocket the phone.
- **Allow background usage** opens Spectre’s Battery page (that switch). **Unrestricted battery** opens the same page; select Unrestricted. Some phones (Samsung among them) do not open onto that choice — tap Allow background usage to click through and select it. Background usage lets the scan run when the app is not in front; Unrestricted stops the OEM freezing it to save battery. Spectre’s switches follow those Android grants. They do not keep the screen on and do not lift Wi-Fi or BLE scan quotas. Samsung: also do not sleep Spectre under Background usage limits. §4.5.3.
- Dark theme is default; it does not save much on LCDs.
- A cheap USB battery pack is more useful than arguing with the saver slider if you need low-latency BLE all day.

13.3 Crowded plazas

A saturated street will produce tens of thousands of BLE ads and thousands of rotated MACs. If the list ever went empty and the phone felt stuck, that was the heap hitting the 256 MB cap and Samsung then parking the scanner. Current builds cap the live set, batch ingest, and drop overflow ads, so you will still see a busy list without seeing every random address. Prefer Strength list over Hybrid if the UI feels heavy, and turn logging off if you do not need the file.

13.4 Ethical and legal considerations

Spectre only processes signals that devices choose to broadcast to every listener in range. That does not make every use lawful or appropriate. You are responsible for local law on wireless interception, stalking, and recording. In particular:

- Do not use matches to harass, follow, or dox people.
- Do not claim a LiteOn OUI is “a Flock camera” in a report without corroboration.
- Do not attempt to access, disable, or tamper with infrastructure you observe.
- Logs can contain MAC addresses of neighbors’ phones and TVs. Treat exports as sensitive.
- If you are on property where radio monitoring is restricted, leave.
- Do not use Spectre where safety is in question. Experimental use only. See the Notice page.

Experimental — not for safety-critical use. Experimental software. Use at your own risk. Spectre does not guarantee that trackers, cameras, tags, or any other device will be found. Each phone has radio and OEM limits that this app cannot overcome. Do not use Spectre in any situation where safety is in question, including real-life threatening situations.

Operational security. Spectre does not hide the fact that the phone’s Wi-Fi and Bluetooth are on. A scan notification is visible on the lock screen unless you change the channel. The app does not provide anonymity.

14. Appendix

A. Glossary

Term	Definition
AP	Wi-Fi access point. Every Spectre Wi-Fi row is one of these (router, hotspot, mesh, soft-AP). Associated clients are not reported. On the Live display it is the small Wi-Fi icon at the start of the subtitle, not the circle and not a two-letter tag. Detail spells "Wi-Fi access point." Subtitle None drops the icon with the second line. §1.1, §5.4.
LE	Bluetooth Low Energy advertiser. Headphones, phones, tags, unnamed BLE rows. Spectre listens; it does not pair for detection. On the Live display it is the small Bluetooth icon at the start of the subtitle, not the circle and not a two-letter tag. Detail spells "BLE advertiser." §1.1, §5.4.
Class glyph	Circle on a Live display row, Filters class chips, and By class headers. First matching signature class, or ? if unmatched. The glanceable mark; radio kind is the Wi-Fi / Bluetooth icon on the subtitle. §1.1, §5.4, §9.5.
BSSID	The AP's MAC address in a scan result. A full MAC-prefix rule is that BSSID.
Company ID	16-bit Bluetooth SIG manufacturer identifier in AD type 0xFF. Looked up offline (~4012 names).
CoD	Bluetooth Class of Device. 24-bit major / minor / service-class bitfield, decoded when advertised or provided by the stack.
BLE	Bluetooth Low Energy. Advertisements are connectionless broadcast packets.
Classic Bluetooth	BR/EDR (headsets, file-send, HC-05/HC-06 serial). Spectre does not run Classic inquiry. A dual-mode BLE advertiser may claim Classic in flags or CoD; that is not a Classic scan. §2.2, §3.5, §7.2.
Signature	A named bundle of match rules plus optional cluster flags.
Decode fields	Optional BLE cleartext map on a signature (Signatures → row → Decode fields). After the rules hit, device detail / Share / AI Export / Debrief notable BLE parse manufacturer or service-data bytes into labels (temp, model, Remote ID, ...). TAK / CoT also reads numeric ids latitude / longitude for advertised-position pins. Not a matcher. Not pairing or GATT. Encrypted ads stay hex. the Live display does not parse these; a hexagon on that signature's chip means a map exists (§5.4). Byte 0 is after the company ID (manufacturer) or the first service-data byte. How to build a map: §9.6.1–§9.6.5. Stock maps: §9.6.6. TAK ids: §5.8.3.
Decode hexagon	Small hexagon inside a signature name chip (same color as the name) when that signature has a Decode fields map. Dual-chip radios mark only the mapped name(s). Catalog check, not a parse of this packet. Hidden when Display → Signature names is off. Extra attention "!" and the phosphor alerted bell are separate chips. Same mark on the Signatures list, By class signature rows, and detail Decoded fields. §5.4, §9.6.
Signature family (detail)	Card on device detail, above Create signature from device. Same on-air ID rules as Signature candidates, for this radio: Strong family, Possible family, This radio only, or Already tagged. Counts distinct MACs in the log and on the air now. Verdict only — Create from device still pins this MAC. Already tagged is not a veto: a second UUID/OUI signature can dual-label (iBeacon + store). Candidates skip tagged radios. §5.5, §9.2, §9.2.1.
Named radios	Settings list of one-MAC custom names and optional alerts (detail Save name, or the bookmark icon). Rename, Alert on/off, remove one, or Clear all. Does not include signature bookmarks. Privacy mode masks MAC tails. Orphans (gone or rotated) stay until you delete them. Filters → Named radios only. §5.5, §5.7, §10.1.
Privacy mode	Settings switch, off by default. Masks the last three octets of MACs on the screen and in Debrief / AI Export / detail Share (AA:BB:CC:***:***). GPS last-fix and sit-report coordinates show as masked; street names omitted. Logs, matching, filters, Hunt math, Moving with you, and saved signatures stay full. Pauses a TAK / CoT feed so full MACs and coordinates are not sent. §5.7, §5.8.

Term	Definition
TAK / CoT feed	Settings switch, off by default. UDP Cursor-on-Target markers to ATAK / WinTAK / iTAK. Default port 10011 (ATAK CIV); host 239.2.3.1 (SA multicast). Same-phone ATAK CIV: this phone's Wi-Fi IPv4 and 10011. Other ports: 6969 (SA multicast), 8087 (many TAK servers). Pins Extra attention at operator GPS (heard here) and advertised decode lat/lon (Payload location — stock Remote ID, or any map whose field ids are latitude/longitude). Privacy mode pauses it. Not DF, not a Remote ID plugin, not the Live display. §5.8, §12.15.
Night mode	Settings switch under TAK / CoT, off by default. Red-on-black field display: text, chips, RSSI, Hunt, Extra attention. Background stays dark. Overrides Dark theme while on. Phone brightness is unchanged. Fig. 6a, §5.7.
Heard here (TAK)	CoT pin at this phone's GPS at last hear. The other radio is in earshot, not on that point. Needs GPS tagging and a live fix. Extra attention uses this unless a payload lat/lon exists.
Advertised position (TAK)	CoT pin from decode field ids latitude / longitude (optional alt_geo). Stock Remote ID Location. Sticky across ASTM message types. op_lat / op_lon are the pilot and are not this pin. GPS tagging can be off.
Payload location	TAK What-to-send chip, on by default when you turn the feed on. Selects radios with sticky advertised lat/lon. Required for stock Remote ID (no Extra attention mark).
Reports	Bottom tab. Debrief (text/PDF), AI Export, Signature candidates, Share log, Save log, Reset / clear log. Config for GPS, place names, and logging on/off stays on Settings. §5.6.
Signature candidates	Reports action. Re-matches the rotating log against the current catalog, then lists unmatched families that share a unique on-air ID on two or more radios. Randomized addresses and house-like names are skipped. Create signature opens the editor as a draft (shared rule, no MAC pin). Save returns to the list and re-runs it. Offline. §5.6.1, §9.2.1, §11.5.
vendor_ie (log)	Last CSV column / JSON field on new Wi-Fi rows: pipe-separated vendor-IE OUIs, up to eight. Empty on BLE and on older 17-column rows. Signature candidates uses product IEs; WPA/RSN/P2P/Qualcomm chip IEs are logged but not clustered. §11.2, §11.5.
Alerted (list)	Phosphor notification pip on a Live display row (list, hybrid, timeline, By class) after a watchlist alert this session. Lasts until you leave Spectre. Distinct from Extra attention "!" and from the one-second flash. Newest alert ranks by the same event. On radar the same radios keep a phosphor ring after the ping. §5.3, §5.4, §6.1.
Extra attention	Optional field on a signature, separate from editor Notes. If it is not empty, a match gets a "!" on the Live display, an Extra attention card on detail, and a line in Debrief / AI Export (amber PDF callout). Empty = no mark. The "!" is its own chip, not the decode hexagon and not the phosphor alerted bell. Stock fills it on Hobby BLE serial, Axon, WatchGuard Video, Ray-Ban / Meta glasses, Snap Spectacles, Fieldy, Plaud Note, Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, Porkchop, Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc, and roadside / public camera + ALPR (Flock, Penguin, Pigvision, FS Ext Battery, Genetec AutoVu, Rekor, Motorola Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua) — those rows also ship with the bookmark on. Pattern match, not identity, not a safety finding. §5.5, §9.3, §9.5, §12.14.
Hobby BLE serial	Catalog signature (on). BLE advertised names for cheap UART modules (HMSOft, JDY, CC41, AT-09, BT05, ESP32 BLE). Not Classic HC-05/HC-06. Extra attention cautions that the same boards have been used in some pump/ATM overlays; look with your eyes if it is loud next to a card reader. Not proof. Turn the row off if those names are local noise.
Axon	Catalog signature (on). IEEE OUI 00:25:DF plus Axon Body / Fleet / Dock names. Extra attention: body-worn, in-car, dock, or TASER. Public safety class — used in LE, not exclusive to it. Not that officer. Quiet LTE units will not appear.
WatchGuard Video	Catalog signature (on). IEEE OUI 00:1D:96 (WatchGuard Video, not the firewall company). Extra attention: body-worn / in-car. Now Motorola. Public safety class — used in LE, not exclusive to it.
Ray-Ban / Meta glasses	Catalog signature (on). BLE company IDs 0x01AB / 0x058E / 0x0D53 and Ray-Ban names. Extra attention. Quest and other Meta wearables can match the same IDs.
Snap Spectacles	Catalog signature (on). BLE company ID 0x03C2 plus Spectacles names. Extra attention. Not proof of recording.

Term	Definition
Hak5 Pineapple	Catalog signature (on). Setup SSID Pineapple_XXXX. Extra attention: admin AP, not every cloned café SSID.
Flipper Zero	Catalog signature (on). OUI 0C:FA:22 and BLE name Flipper*. Extra attention. Custom firmware can hide it.
Pwnagotchi	Catalog signature (on). Classic BSSID de:ad:be:ef:de:ad. Extra attention.
Marauder / Deauther	Catalog signature (on). MarauderAP / Deauther default names. Extra attention. DIY boards match too.
Porkchop	Catalog signature (on). SSID/name PORKCHOP; BACON fake-AP vendor IE 50:52:4B. Extra attention. Cardputer / CYD firmware, not every ESP32.
Moving with you	Filter: loud radios (about -75 dBm or stronger on the trail) with a GPS trail along your path, still being heard. “Still here” is not a fixed radius: allowed distance = the larger of 50 m or (recent speed × 15 s BLE / 50 s Wi-Fi), plus 25 m of GPS slack. Walking holds a house-length; highway hold is a handful of advertisements so a car tag does not blink off between packets. Passing radios still fail the trail-moved gates. Needs live tagging (not stale last-known) and ~45 m of path. Bag/car tag yes; house AP on arrival no; a second iPhone usually no (BLE MAC rotation). The switch starts a follow test on all radios (clears Signatures only / Show only / Named radios only; Hide these stays). The preset replaces the whole filter. Live display → Start over clears path and trails. Debrief still writes the same co-travel assessment for the last 15 minutes even if this filter is off and even if the Live display is not on the list. §8.5.
Start over	Live display button while Moving with you is on. Clears the operator GPS path and radio GPS trails. List and log stay. Path meter returns to 0 m.
Live display	The first bottom tab (labeled Live on the phone). The on-screen picture of radios: radar, list, timeline, hybrid, or By class. Filters change who appears here; Display (tune) changes how each row looks. Not “live vs recorded” — Debrief, the log, and the TAK feed are separate. §5.1–5.3.
By class	Live display view (Display → By class). Outline of the filtered set: every class A–Z by name → signatures A–Z → radios → detail. Unmatched last. Show all (default) keeps empty classes; Collapse empty hides zeros. Those chips scroll with the list. Class headers use the same glyphs as Live display rows (unmatched = ?). Counts are radios, not packets. Radio rows follow Display (bars, chips, Frequency, first/last). Dual-chip radios sit in each class they matched. A watchlist hit opens that class and signature so the row can flash, and the jump keeps those headers on screen when the radio is close enough. Not a Report. §6.5.
Display (Live display)	Tune / chevron on the Live display. How the list looks for this mission: View, Sort, Brief hold, Title line, Subtitle line, bars, signature names, Frequency, first/last. View includes By class (outline of the filtered set). The circle on each row is a class glyph, not radio kind — that is the Wi-Fi / Bluetooth icon on the subtitle. Filters hide radios; Display hides fields. Not on Settings. §5.3, §6.5.
Sort (Display)	Order of the list, hybrid, and timeline. Strongest signal; Strongest averaged over 30 s (the default); Newest heard; Newest alert; Newest arrival; New at bottom; Name A–Z (Title line); Signatures first. Does not hide radios. Radar still plots by RSSI radius. §5.3.
Preset (Filters)	Chip at the top of Filters. A tap replaces the whole filter, not Display. Short stock set: All traffic, Wi-Fi only, BLE only, Strong signal, Moving with you, Trackers, Hide trackers, Hide phones — plus chips you saved. Class Show only (Cameras, Drones, ...) is the chips further down; Save current as... if you want that sit as a preset. Show-only presets imply Signatures only (switch dimmed). Long-press any chip to delete it. Stock chips you remove stay gone until Restore default signatures & presets. Reset filter clears this tab; it is not undo. §8.4.

Term	Definition
Signature class	Bucket on every signature: Finder tags, Retail beacons, Signage, Wearables, Surveillance, Drones, Pentest, Public safety, Vehicle, Glasses, Audio, Cameras, Thermostats, Locks, Health, Home IoT, ISP / routers, Mesh, Phones / PCs, Other. Public safety is Axon / WatchGuard Video and public-safety vehicle APs (Cradlepoint, AirLink, Compex, Novatel, Utility Inc). Those radios are used in LE; they are not exclusive to it — government, municipal, and other corporate fleets likely run some of the same kit. Health is clinic / home-medical BLE (Honeywell Xenon HC scanners, Omron cuffs, Withings scales, Dexcom). Cameras is consumer / action cameras, not poles (Surveillance) and not Axon. Glasses is Meta / Snap. Audio is AirPods / Sony / Bose / JBL / Sonos. Color is the Live display chip; class is the filter. Custom rows default to Other. An old Body-worn class folded into Wearables. §8.3, §9.5.
Signature pack	JSON file from Settings → Export signatures (spectre-signatures-YYYYMMDD.json). Stock plus your edits. No logs, GPS, filters, or watchlist. Import skips the same id or the same match rules, merges extra rules onto a stock row, and renames a colliding name to “Name (imported)”. Restore defaults still wipes customs. §5.7, §9.3.
Show only / Hide these	Filters → signature classes. Show only keeps radios matching the class chips you pick (unmatched stay hidden; Signatures only is implied and the switch is dimmed). Hide these drops those classes and leaves unmatched. Live display only — matching, log, and Debrief still see them. Show only with no class picked leaves the Live display unchanged. §8.1–8.4.
Title line	Display → first line of each list, hybrid, or timeline row. Advertised name, Name + type, or MAC (default). Stays bold even when it is a MAC. Name A–Z sorts by this line (so the default is MAC order). Radar labels ignore it.
Subtitle line	Display → second line. Always starts with a small Wi-Fi icon (access point) or Bluetooth icon (BLE advertiser). Then Advertised name, Name + type (default), or MAC, plus rand/pair/gone. pair is Fast Pair pairing-mode seen this session. None hides the line so more rows fit; crumbs move onto the title. The kind icon does not move onto the title. Unnamed BLE on this line is unnamed (the icon already marks LE). Vendor is not on this line.
Brief hold	Display → how long a radio stays after the last packet (Off, 10, 30, or 60 s). Last RSSI, rank, and radar ring are frozen; they do not decay. Linger is the longer of this and Stale after.
Gone / stale	Last packet older than the longer of Stale after and Brief hold. Held Wi-Fi / BLE recycle do not count. Still listed until that linger; dim on radar only when marked gone.
Keep screen on	Settings switch, on by default. Holds the display while Spectre is visible. Turn off when you pocket the phone. Not the same as unrestricted background. Max-collection sit: §4.5.
Max collection (§4.5)	Checklist to hear as often as stock Android allows: system Location / Wi-Fi / Bluetooth, every Spectre permission (Precise Location, Nearby Wi-Fi, Bluetooth scan+connect, Notifications), Unrestricted battery / do not sleep the app, Keep screen on while you watch, High performance, optional Faster Wi-Fi AP scans. Battery and heat are the cost. Does not lift monitor-mode / client / Classic limits.
Faster Wi-Fi AP scans	Settings switch, off by default. About 8 s AP batches instead of ~30/40/55 s. Spectre reads the OS Wi-Fi scan-throttle flag (Android 11+) and will not turn this on until Developer options → Wi-Fi scan throttling is Off. Purpose: more chances to hear a signed AP (OUI / factory SSID) while it is in range — important on a drive. Not Hunt, not clients, not continuous RF. More battery and heat. §7.1.1, §10.3.1.
Manufacturer data	BLE AD type 0xFF: 16-bit company ID plus vendor bytes.
Signatures only	Filter that hides devices with no signature match. Dimmed while class Show only or Show only selected signatures is narrowing the Live display. §8.1–8.3.
Named radios only	Filter that hides radios without a custom name. Alert can be off. Not Signatures only. Random / privacy MACs stay pinned to that address. §5.5, §8.1.
Hide Fast Pair account-key	Filter that hides Fast Pair-only radios whose payload is the longer account-key filter, not the 3-byte pairing-mode model ID. Pairing-mode stays (chip Fast Pair pairing, subtitle pair). Dual-chip radios stay. Hide selected Fast Pair drops both. Always AND. §8.1, §9.5.

Term	Definition
Show only selected signatures	Filters collapsing list, grouped Class A–Z like the Signatures tab. Only radios matching the signatures you pick stay on the Live display. Empty list = no extra include. Separate picks from Hide selected. §8.1–8.3.
Hide selected signatures	Filter switch for one family. When on, matching devices drop off the Live display. Picks stay stored if you turn the switch off. Empty list hides nothing. Prefer class Hide these when the whole Finder tags / ISP bucket is clutter.
Pause (Live display)	Freezes the Live display. Radios and the log keep running. Filters and Settings still update; the new filter applies when you run the Live display again. Detail opened from a paused row is that snapshot, even if the radio has since gone. Tap the Live tab again (the control then reads Live) to run again. Mark seen while paused uses the frozen list.
Mark seen	Live display button, above the tabs, only while New detections only is on. Adds what is on the Live display to already-seen. While paused, uses the frozen list.
Reset seen	Live display button, above the tabs, only while New detections only is on. Clears already-seen to zero so those radios can show as new. Does not clear the log.
Debrief (text)	Reports share of the field sit report as plain text. Opens with DISCLAIMER (hobby / as-is; hypotheses not identity; local law; 15-minute live set). Last 15 minutes in the live map (about 400 radios, not the log). Ignores Live display view and Filters. Where you were (stay/transit + operator lat/lon). Tracking assessment is short; amber callouts for Possible trackers with you / Possible tail (finder tags), Retail beacons with you, Wearables with you. Radios you passed are omitted. On a drive, tap more than once — unnamed roadside BLE drops after ~3 min; radios moving with you stay. §11.4.1.
Debrief (PDF)	Same sit report, letter-size typeset PDF. Disclaimer at the top, then SPECTRE header, numbered sections, amber finder-tag / retail-beacon / wearable co-travel callouts, takeaway. Same 15-minute RAM window as the text share; view and filter do not change it. Long trip: tap more than once (§11.4.1). Share as application/pdf.
Online place names	Settings switch, on by default. Debrief and AI Export reverse-geocode GPS stamps via the system geocoder when online. Offline: a note in the report/prompt, no error dialog. Turn off if you do not want streets of your path.
New at bottom	Display → Sort. First-seen order, oldest at top; new radios append; gone radios drop. List follows the bottom unless you scroll up.
unnamed LE	BLE Advertised name / Name + type when there is no advertised name and no useful decode. On the subtitle the Bluetooth icon already marks LE, so the body is unnamed (not “unnamed LE” twice). Title Advertised name still shows unnamed LE. The title is the MAC unless you change it.
AI Export	Two buttons. Reports: sit-level prompt (onboard Debrief + working data; asks a chat for statistical addendum, not a rewrite). Device detail: prompt about that one radio (dump + registry decode). Both paste into a chat, both open with the experimental disclaimer, both are hypotheses — not identity. Treat as sensitive.
Signature color (stock)	Built-in rows share a palette by class: red pentest, amber cameras/ALPR/UniFi Protect/DJI, purple phones/Find My, cyan wearable trackers, green mesh, orange glasses and audio, teal public safety and vehicle, silver home IoT/ISP Wi-Fi (including UniFi AP)/retail signage/Unknown. Unmatched radios use RSSI color (≥–55 green, –55 to –70 amber, –70 to –85 orange, else red). Change any row. §9.5.
UniFi AP	Wi-Fi-only catalog signature (ISP / routers). Factory SSIDs UniFi* / UAP-* / UBNT* plus Ubiquiti IEEE OUIs on the BSSID or a vendor IE. Virtual BSSIDs miss the MAC OUI but still hit on a Ubiquiti vendor IE. A separate UniFi row is name-only on either radio. UniFi Protect cameras stay Surveillance. §9.5, Appendix B.
Tag detections with GPS	Settings switch, on by default. Current GPS/network updates while scanning; stamps each hear (detail, Moving with you, Debrief, log lat/lon, heard-here TAK pins). Last-known older than 30 s ignored. Operator phone at hear-time, not the other radio. Advertised TAK pins (Remote ID) do not need this. High-accuracy Location or the path stays 0. §5.7, §5.8.

Term	Definition
Share as text	Device-detail button. Plain dump of the open radio (identity, signal, decode, session). Not the sit report and not the rotating log.
Voice (watchlist)	Settings → Voice on watched signature, off by default. What to say (signature watches): Class, Signature, or Class + signature. Default Class. A named radio speaks its watch name instead, including a custom name from Named radios. Independent of Beep. Not Hunt. Overlapping speech is dropped. On-device TTS. Test alert plays the signature mix. Jump works with voice alone. §5.7, §10.2.1.
Hunt	Device-detail, BLE only. Full-screen closer/further from smoothed RSSI. Rings around YOU contract on Closer and expand on Further. Very Close at about −45 dBm or louder (look around; still not meters). Loudest-this-hunt + hunt sparkline. Beep / Vibrate at the bottom (off by default; faster as RSSI gets louder; silent when Quiet/Gone; not the watchlist chirp, never speaks). Optional body-block turn for a crude heading (§12.13). Walls, metal, people, and floors change RSSI without a change in range. Not distance, not DF. Wi-Fi omitted (OS scan throttle; Faster Wi-Fi AP scans still batch, not Hunt).
lat / lon (log)	CSV columns and JSON fields on each new log row when Tag detections with GPS is on and a fix exists. Operator phone at hear-time, not the other radio. Empty/null otherwise. Sit before vendor_ie in CSV so older files still parse.
Where you were	Debrief / AI Export section. Operator path split into stays (~40 m) and transits. Lat/lon once per stay, optional street name, loud radios heard there. Not lat/lon on every inventory line.
Fast Pair	Google tap-to-pair. Service UUID 0xFE2C. Catalog signature (on): Android phones and many buds. Three-byte model ID in pairing mode (Live display chip Fast Pair pairing); longer payloads are an account-key filter. Filters → Hide Fast Pair account-key drops account-key-only chips. Hide selected Fast Pair drops both. §8.1, §9.5.
iBeacon	Apple manufacturer layout 0x02/0x15: UUID + major + minor + calibrated TX.
Appearance	BLE GAP field: what the device claims to be (headphones, mouse, watch...). Used in the detail "What this looks like" guess.
New detections only	Filters switch. Mark seen / Reset seen are on the Live display above the tabs while the filter is on. Already-seen grows only on first turn-on (plus the next Wi-Fi scan) or Mark seen. Reset seen clears it to zero. New radios stay while heard, then at least Brief hold after the last packet. Live display hint: New only · N hidden.
OUI	Organizationally Unique Identifier — first 24 bits of a MAC.
Randomized MAC	Locally administered unicast address (U/L bit set). Common on phones and some tags.
RSSI	Received signal strength indicator, dBm. Larger (closer to 0) is stronger.
Sparkline	RSSI graph on Hybrid and detail. Packet order (newest right), fixed −30 to −100 dBm grid — not a clock and not auto-scaled.
Trend mark	>> > = < << next to RSSI. Recent packets getting stronger or weaker.
Service UUID	BLE identifier for a service. 16-bit aliases expand to the Bluetooth base UUID.
SSID	Wi-Fi network name. May be hidden (empty in the scan result).
Promiscuous / monitor mode	Wi-Fi chip receiving all frames on a channel. Stock Android apps cannot enable this. Spectre is startScan() AP beacons + BLE advertisements only.
Cellular-first / quiet radio	Newer camera/ALPR poles that backhaul on LTE/5G and leave Wi-Fi/BLE off except for install or maintenance. Expected to be invisible to Spectre most of the time.
Rule enable	Switch on each rule in the signature editor. Off keeps the rule in the list but it does not match.

B. Default pre-loaded signatures

All of the following use match-any unless noted. Confidence comments are operational, not legal findings. Exact OUI lists can be inspected in the in-app editor. Every stock signature always labels when its rules hit. Hide a noisy family on Filters (Hide these for a class, or Hide selected for one row). Chip colors are by class (§9.5). Extra attention and a stock bookmark ship on the body-cam, camera-glasses, recording-wearable, pentest, public-safety vehicle AP, and roadside / public camera + ALPR rows noted below. Consumer cameras, ISP gateways, and office mice will label when their factory names or OUIs are heard. The table is A–Z by signature name.

Signature	Primary rules	Confidence notes
Actiontec	Wi-Fi Actiontec* plus Actiontec IEEE OUIs	FiOS / Frontier gateways. Verizon names stay on Verizon.
Adtran	Wi-Fi Adtran* plus Adtran IEEE OUIs	Fiber gateways. Common CenturyLink / Lumen / Quantum Fiber OEM.
AirLink	Wi-Fi AirLink* plus Sierra Wireless / AirLink Communications IEEE OUIs. Extra attention filled. Stock bookmark.	Sierra Wireless AirLink vehicle gateways. Common in US police fleets; also government, municipal, and other corporate fleets. Hidden/renamed SSIDs still hit on BSSID. Not AirLink WiFi Networking 00:23:D3. Pattern match, not that agency. Beeps on a new match. Public safety class — used in LE, not exclusive to it.
AirTies	AirTies IEEE OUIs	ISP mesh extenders.
Amazon	Company 0x0171; names Echo / Amazon / Fire TV	Echo / Fire.
Apple AirTags	Name AirTag / Find My; mfg data 0x004C / 12; UUID FD44	Offline Finding. iPhones also send 0x12 — dropped when Continuity (Apple Device) is on the same radio unless the name is AirTag or UUID FD44. Not Continuity 0x10 and not AirPods (0x07).
Apple audio	0x004C / 07 Proximity Pairing; 0x004C / 09 AirPlay; names AirPods, Beats	AirPods / Beats / AirPlay. Not tags, not Nearby Info phones.
Apple Device	Apple 0x004C types 0x10 / 0x0F / 0x0B / 0x05 / 0x0C–0x0E / 0x08 / 0x0A; names iPhone, iPad, MacBook	Phone / tablet / Mac Continuity. OF 0x12 on the same radio is not a second AirTag chip. A street of iPhones will light this up.
Arcadyan	Arcadyan IEEE OUIs	ISP gateways / mesh. Verizon / T-Mobile OEM.
Arlo	Arlo / Arlo* / ARLO_VMB_*; Arlo Technology IEEE OUIs	Cameras and VMB base-station APs. Renamed SSID still hits on BSSID.
Arris	Wi-Fi Arris* / SURFboard* plus Arris and CommScope IEEE OUIs	Cable gateways. Ruckus keeps its own OUI list. ISP-issued names after provision often miss.
Aruba	Wi-Fi Aruba* / SetMeUp* / InstantOn* plus Hewlett Packard Enterprise IEEE OUIs	HPE Aruba Instant / Instant On. Not a lone instant word. Not HP Inc. printers. ISP / routers class.
Askey	Askey Computer IEEE OUIs	ISP / 5G gateways. T-Mobile OEM.
ASSA ABLOY	BLE 0x012E / HID 0x0124 / Yale 0x0BDE; UUID FCBF; Seos UUID 00009800-...; names Seos / Yale*	Locks, readers, Seos credentials. Phones on HID Mobile Access can hit Seos. Not Apple FCB2.
ASUS	Wi-Fi ASUS* plus ASUSTeK IEEE OUIs	Renamed SSID still hits on BSSID. Laptop hotspot can hit.

Signature	Primary rules	Confidence notes
AT&T	attwifi / ATT-WIFI* / ATT-GUEST* / ATT??????? plus AT&T and 2Wire IEEE OUIs	Pace-style factory names. Many boxes are Arris OEM.
Audi MMI	Wi-Fi Audi_MMI_*	Audi in-car hotspot.
August	BLE company 0x01D1; UUID FE24; names August*	August Home locks. Field L40A33A. Not ASSA 0x012E.
AUMOVIO	AUMOVIO IEEE OUIs	Ex-Continental vehicle Wi-Fi. Vehicle class.
Autel	Names Autel*	Autel Robotics drones. Not EVO* and not SSID default-ssid. Drones class.
Avigilon	Avigilon / Avigilon*. Extra attention filled. Stock bookmark.	Motorola cameras / LPR on municipal poles and commercial sites. Name-only. Beeps on a new match.
Axis	AXIS-* / Axis Camera. Extra attention filled. Stock bookmark.	Municipal / public CCTV poles. Name-only. Beeps on a new match.
Axon	OUI 00:25:DF (Axon Enterprise); names Axon Body / Fleet / Dock / BWCDEVICE / Axon*; UUID 0xFE6C (Axon Signal). Extra attention filled. Stock bookmark.	Public safety class — used in LE, not exclusive to it (government, municipal, and other corporate fleets likely run some of the same kit). Body-worn, in-car, dock, or TASER. Body 3/4 often advertise BLE on the public OUI. Not that officer. Not Axon Networks 00:58:28. ZTE Axon phones can hit the name. Beeps on a new match.
Belkin	Wi-Fi Belkin* plus Belkin IEEE OUIs	Some Linksys Velop land here.
Blue Maestro	BLE company 0x0133	Tempo Disc temp / humidity loggers. Home IoT. Decode fields: battery, interval, logs, temperature, humidity. §9.6.
BMW	Company 0x05EB; BLE name BMW; Wi-Fi BMW_*	Not “BMW of ...” dealer names.
Bose	Company 0x009E; UUIDs FE21 / FEBE; name Bose	Headphones / speakers. FEBE is the usual LE speaker/QC advertisement.
Buffalo	Wi-Fi Buffalo* / AirStation* plus BUFFALO.INC IEEE OUIs	AirStation / routers. Renamed SSID still hits on BSSID.
BYD	Company 0x0C34; BLE name BYD	Pattern match, not a model.
Calix	Calix IEEE OUIs	Fiber gateways (GigaSpire class).
Cambium	Wi-Fi Cambium* / cnPilot* / IgniteNet* plus Cambium and IgniteNet IEEE OUIs	WISP / cnPilot APs. Quantum Fiber factory names can also hit.
CARLINK	Wi-Fi CARLINK-????? (6 hex); Panasonic Automotive OUI CC:57:63; Zhuolian 68:8F:C9	CarPlay / Android Auto adapter or head-unit hotspot. Alps Alpine E0:2D:F0 is common on the same cluster but is not an OUI-only rule (Toyota / Lexus head units). Vendor IE 00:A0:40 (old Apple) is not cataloged. Vehicle class, not ISP.
CarPlay	Wi-Fi CarPlay* / name contains CarPlay	Alpine / head-unit in-car hotspot. Vehicle class, not ISP.
CenturyLink	Wi-Fi CenturyLink*	Factory gateway SSID.
Chevrolet hotspot	Wi-Fi myChevrolet*	GM in-car hotspot. Factory SSID; BSSID often randomized.
Chipolo	Name / glob Chipolo*	Find Hub / Find My tags. Name-only.

Signature	Primary rules	Confidence notes
Cisco	Wi-Fi Cisco*; tsunami; Cisco Systems and Cisco SPVTG IEEE OUIs	Not a Cisco substrings (Francisco). AP beacons only. Meraki / Cisco-Linksys have their own OUI lists. ISP / routers class.
Compex	Wi-Fi 114K-* plus Compex IEEE OUIs 04:F0:21 / 00:80:48 / 00:40:29. Extra attention filled. Stock bookmark.	Some public-safety vehicle APs. Same OUIs on other Compex radios. Government, municipal, and other corporate fleets likely run some of the same kit. Pattern match, not that agency. Beeps on a new match. Public safety class — used in LE, not exclusive to it.
Cradlepoint	Wi-Fi IBR* / IBR600* / IBR1100* / IBR1700* / R-series / Cradlepoint* plus CradlePoint IEEE OUIs 00:30:44 / 00:E0:1C. Extra attention filled. Stock bookmark.	Ericsson Cradlepoint vehicle routers. Common in US police fleets; also government, municipal, and other corporate fleets. Hidden/renamed SSIDs still hit on BSSID. Pattern match, not that agency. Beeps on a new match. Public safety class — used in LE, not exclusive to it.
Cudy	Wi-Fi Cudy* plus Cudy IEEE OUIs	Travel / home routers. Factory Cudy-XXXX.
D-Link	Wi-Fi D-Link* / DIR-* plus D-Link IEEE OUIs	Renamed SSID still hits on BSSID.
Dahua	Dahua / DAHUA / Dahua*. Extra attention filled. Stock bookmark.	Commercial CCTV and some public poles. Name-only. Beeps on a new match.
Dexcom	BLE Dexcom*	G6 / G7 glucose sensors. Pattern match, not a patient. Health class.
DJI	Company 0x08AA; DJI* on BLE and Wi-Fi	Drones / RC / setup AP. Osmo cameras are the Osmo row. OcuSync is not an AP. In-flight ASTM Remote ID is the Remote ID row. Drones class. Decode fields: 0x08AA model id (\$9.6).
dormakaba	BLE company 0x0C64; names dormakaba* / Saflok* / Oracode*	Hotel / commercial locks.
DWnet	DWnet Technologies IEEE OUIs	Consumer / SMB APs. Cloud SSIDs are house names. ISP / routers class.
ecobee	BLE company 0x07D6; names ecobee*	Premium BLE (setup / Spotify). Room sensors can hit the same ID.
EcoWater	Wi-Fi H2O- plus 12 characters	Water-softener setup AP (often the MAC with no colons). Not a bare H2O word. Home IoT.
Edgecore	Edgecore IEEE OUIs	Campus / open Wi-Fi APs. Cloud SSIDs are site names. ISP / routers class.
Eero	Wi-Fi eero* plus eero inc. IEEE OUIs	Not Amazon Technologies (Echo).
Electronic shelf label	BLE UUID 0x1857 (SIG ESL Service)	BT 5.4 PAwR ESL. Most Hanshow / SES-imagotag tags use a proprietary radio and will miss. Signage class.
EnGenius	Wi-Fi EnGenius* / EnMGMT* plus EnGenius IEEE OUIs	Cloud APs; EnMGMT* while unclaimed.
Epson	Wi-Fi / BLE names *EPSON-ET-* / *EPSON-WF-*	EcoTank and WorkForce. DIRECT-* also hits Unknown Signature. Not Seiko Epson 0x0040. Home IoT.

Signature	Primary rules	Confidence notes
Estimote	BLE company 0x015D; names Estimote*	Location beacons / stickers. Decode fields: frame type (Nearable / Telemetry). Packed sensors are not expanded. §9.6.
eufy Security	eufy / EufyCam / eufy*	Anker cameras and tags.
Extreme	Extreme Networks IEEE OUIs	Campus APs. Cloud SSIDs are site names. ISP / routers class.
Fast Pair	BLE UUID 0xFE2C	Android phones and many buds/speakers. Pairing-mode 3-byte model ID (Live display: Fast Pair pairing / pair crumb) or a longer account-key filter (plaza noise). Filters → Hide Fast Pair account-key drops account-key-only chips. Hide selected Fast Pair drops both. Not a person. Not every unnamed LE. Phones / PCs class.
Fieldy	BLE names Fieldy*. Extra attention filled. Stock bookmark.	Field Labs AI note-taker pendant. Not Arduino/ESP32 example UUID 4FAFC201-..., not company 0xFEFE. Not proof of recording. Beeps on a new match.
Fitbit	Company 0x018E; UUIDs FD62/FD63; name Fitbit	Wearable. Identity only — advertisements do not carry steps; modern dumps are GATT/encrypted. No Decode fields map.
Flipper Zero	OUI 0C:FA:22; BLE names Flipper*. Extra attention filled. Stock bookmark.	Flipper Devices IEEE (2024). Custom firmware can rename. Not proof of an attack. Beeps on a new match.
Flock Safety Cameras	OUI B4:1E:52; ~28 additional field OUIs (LiteOn / Espressif / related); names Flock, FLCK, CONDOR, FALCON, SPARROW; globs Flock-*, Flock-?????. Extra attention filled. Stock bookmark.	Roadside ALPR / camera pole. High only for B4:1E:52 or a Flock-* SSID. Other OUIs are component vendors — false positives expected. Beeps on a new match.
FOBO TPMS	BLE company 0x0127 (Salutica); names FOBO*	Motorcycle / car aftermarket BLE TPMS. Vehicle class.
Ford	Company 0x0723; BLE names Ford / Lincoln	Phone-as-key / infotainment when that ID is advertised. Not a dealer SSID.
Fortinet	Wi-Fi Fortinet* / FortiAP* / FAP-config* plus Fortinet IEEE OUIs	Renamed SSID still hits on BSSID. ISP / routers class.
Franklin	Wi-Fi RG3100* plus Franklin Technology Inc. IEEE OUI 50:FB:FF	5G / LTE home-internet gateway (often carrier-issued). Not Franklin Electric. Not Qualcomm chip IEs. ISP / routers class.
FS Ext Battery	Name FS Ext Battery; globs FS_*, FS Ext*; Silicon Labs OUIs 04:0D:84, 1C:34:F1, 38:5B:44, 94:34:69, B4:E3:F9, F0:82:C0, 58:8E:81, EC:1B:BD, 90:35:EA. Extra attention filled. Stock bookmark. Surveillance class.	Usually a Flock-style camera pack. Name is medium confidence. OUI-only hits are low confidence. Beeps on a new match.
Garmin	Company 0x0087; UUID FE1F; name Garmin	Watches / inReach.
Genetec AutoVu	Genetec, AutoVu. Extra attention filled. Stock bookmark.	Municipal / parking ALPR. Name-only. Beeps on a new match.
GL.iNet	GL-iNet* / GL-MT* / GL-AR* / GL-AXT* plus GL Technologies IEEE OUI 94:83:C4	Travel routers. Not a bare GL-*. Many boards still use chip-module prefixes.
GM hotspot	Wi-Fi myCadillac* / myGMC* / myBuick* / CADILLAC* / BUICK* / CHEVROLET*	GM in-car hotspots. myChevrolet* stays on Chevrolet hotspot.

Signature	Primary rules	Confidence notes
Goodyear TPMS	BLE company 0x0B99	Intelligent tire / SightLine-class BLE. Not 315/433 MHz factory stems. Randomized MAC; vendor is the company ID. Vehicle class.
Google	Company 0x00E0; names Pixel / Chromecast	Pixel / Chromecast. Fast Pair UUID FE2C is the Fast Pair row.
Google Wifi	Google Wifi / Nest Wifi	Not Pixel BLE. Not Nest-* cameras. Not Google Inc IEEE OUIs.
GoPro	UUIDs FEA5/FEA6; GoPro*	Action cameras. Cameras class, not Glasses. Decode fields: company 0xF202 schema / awake / Wi-Fi AP / pairing / model / offload (\$9.6).
Govee	BLE names Govee* / GBK_* / ihoment_* / GV5108* / GVH5*	Lights and sensors (H5075 / H510x hygrometers). Identity is name-only. Decode fields: 0xEC88 H5074/H5075 and 0x0001 H510x temp / humidity / battery after that match. Lights may not fit those layouts. \$9.6.
Grandstream	Wi-Fi Grandstream* / GWN* plus Grandstream IEEE OUIs	GWN APs. Glob is prefix, not a mid-name GWN word.
Haiku Fan	BLE UUID E0FC1000-1FB1-4168-96DF-B3F057A86E01; names Haiku Fan / Mammoth Fan	Big Ass Fans. Custom 128-bit service.
Hak5 Pineapple	Names Pineapple_XXXX / Hak5 / WiFi Pineapple. Extra attention filled. Stock bookmark.	Setup/management AP. Not Alfa OUI 00:C0:CA. PineAP clones look like ordinary SSIDs. Beeps on a new match.
Hatch	BLE company 0x0434; OUI C8:FA:9C; names Hatch Rest* / Restore* / Mini*	Hatch Baby sound machines. Not 180A/180F. Home IoT.
Helium	Helium / Helium*	LoRaWAN / Helium hotspot names.
Hikvision	Hikvision / HIKVISION / Hikvision*. Extra attention filled. Stock bookmark.	Commercial CCTV and some public poles. Name-only. Beeps on a new match.
Hitron	Wi-Fi Hitron* plus Hitron IEEE OUIs	Cable gateways. Common Xfinity OEM.
Hobby BLE serial	BLE names HMSoft / HM-10 / CC41 / AT-09 / JDY-08/10/16/31 / BT05 / MLT-BT05 / ESP32 (BLE only). Extra attention filled. Stock bookmark.	Cheap UART modules. Same boards appear in DIY and in some pump/ATM overlays. Not a skimmer detector. Not Classic HC-05/HC-06. Beeps on a new match. Hide on Filters if those names are local noise.
Honda	Company 0x0915; BLE names Honda / Acura	Not Sony Honda Mobility 0x0EDE.
Honeywell Home	BLE Honeywell Home* / Lyric Thermostat / Lyric T* / Amazon Smart Thermostat*	Resideo-built. Not Honeywell 0x0526 or Resideo 0x0B01 (too broad). T9/T10 sensors are 900 MHz.
Honeywell Xenon HC	BLE Xenon_*HC* / Xenon_CCB-U00-H* / 1962h* / 1952h* / 1902h*	Healthcare barcode scanner / charge-comm base. Not warehouse CCB-U00-G, not Honeywell Home. Health class.
HOVERAir	Wi-Fi Hover* / HoverX1_*; names HOVERAir*	Zero Zero Robotics flying cameras. Drones class.
HP	BLE company 0x0065; UUID FE78; ENVY*; Wi-Fi HP-Print*	Printers. Office / home noise.

Signature	Primary rules	Confidence notes
Huawei	Wi-Fi HUAWEI* / Huawei* plus Huawei IEEE OUIs	CPE and phones share those OUIs. A public-OUI phone hotspot also hits; randomized hotspots need the factory SSID. Not Honor. ISP / routers.
Huf	BLE company 0x070A	Huf Hülsebeck TPMS and vehicle access. Not only a valve stem. Vehicle class.
HUMAX	HUMAX IEEE OUIs	5G / cable gateways. Often T-Mobile Home Internet.
Hyundai	Company 0x0826; BLE names Hyundai / Genesis	When that company ID is advertised.
iBeacon	Apple 0x004C type 0x02 length 0x15; names *iBeacon*	Protocol, not a vendor. Dropped when a product signature already labeled the radio (Sony TV, Tesla phone-key). Minew / Estimote / Kontakt / Target Atrius basket still dual-label. Not Nearby Info 0x10 / AirTags 0x12 / AirPods 0x07. Not Eddystone FEAA.
igloohome	BLE company 0x05BA; names igloohome*	Keyboxes / locks. Not a bare igloo.
Inseego	Wi-Fi Inseego* plus Inseego Wireless IEEE OUIs	5G / MiFi hotspots. Not a bare MiFi*.
Insta360	Company 0x10D7; Insta360* / X3* / X4* / X5* / Ace Pro* / GO 3* / ONE X* / ONE RS*	Arashi Vision action / 360 cameras. Cameras class, not Surveillance.
Jaguar Land Rover	Company 0x020B; names Jaguar / Land Rover / Range Rover	When that company ID is advertised.
JBL / Harman	Company 0x0057; JBL / Harman	Audio.
Kevo	BLE Unikey 0x015E; names Unikey* / Kevo*	Kwikset Kevo.
Kontakt.io	BLE company 0x01FD; names Kontakt*	Kontakt Micro-Location beacons. Decode fields: UUID FE6A Location packet (battery / TX / channel / moving). \$9.6.
Kwikset	BLE names Kwikset*	Not Spectrum Brands 0x0356. Kevo is the Unikey row.
LG webOS TV	BLE names [LG] webOS* / webOS TV*; UUID FEB9	Living-room TVs. Not LG company 0x00C4. Unnamed FEB9 can be other LG radios. Home IoT.
Linksys	Wi-Fi Linksys* / Velop* plus Linksys IEEE OUIs	Some Velop use Belkin OUIs.
Lockly	BLE names LOCKLY*	Name only. Not Nordic 0x0059.
Logitech	Company 0x01DA; UUID FE61; Logitech / Logi	Mice / keyboards. Office noise.
Luxul	Luxul IEEE OUIs	SMB APs.
Marauder / Deauther	Names MarauderAP / Marauder / Deauther. Extra attention filled. Stock bookmark.	ESP32 Marauder or Spacehuhn-style defaults. Same boards are DIY. Beeps on a new match.
Master Lock	BLE company 0x014B; names Master Lock*	Bluetooth padlocks. Not a bare Master substring.
Meraki	Wi-Fi SSIDs Meraki* plus Cisco Meraki IEEE OUIs	Not Cisco Systems OUIs. Renamed SSID still hits on BSSID. Cisco vendor IE 00:00:0C on the same beacon stays Meraki only. ISP / routers class.
Mercedes MBUX	Wi-Fi MBUX*	In-car hotspot. Factory SSID.

Signature	Primary rules	Confidence notes
Meshtastic	Meshtastic / Meshtastic_*; UUID 6ba1b218...	LoRa mesh nodes. Strong name + service UUID.
Microsoft Device	Company ID 0x0006; names Surface, Xbox	Swift Pair / Nearby Sharing on Windows PCs.
MikroTik	Wi-Fi MikroTik* plus Routerboard.com IEEE OUIs	Renamed SSID still hits on BSSID.
Minew	IEEE OUI AC:23:3F; names Minew*	Shenzhen Minew beacons / sensors. Field AC:23:3F often also iBeacon or Eddystone.
Mist	Mist Systems IEEE OUIs	Juniper Mist campus APs. Cloud SSIDs are site names. ISP / routers class.
Mojo	Mojo Networks IEEE OUIs	Now Arista Cognitive Wi-Fi. No Mojo* SSID glob. ISP / routers class.
Motive	Wi-Fi Motive * / Motive_* / Motive Hotspot* / KeepTruckin*	KeepTruckin ELD / fleet hotspot. Not substring Motive (Locomotive). Vehicle class.
Motorola Vigilant	Vigilant, Vigilant Solutions, Motorola Vigilant. Extra attention filled. Stock bookmark.	LPR used by agencies and parking. Name-only when advertised. Beeps on a new match.
myQ	BLE company 0x0878 (Chamberlain); UUID 26D91A37-...; names MyQ-*	Garage door hubs.
Nespresso	BLE company 0x0225; Vertuo/Venus/Barista/Mini UUIDs; names Vertuo* / Venus_*	Coffee machines. Not a bare Venus word. Home IoT.
Nest	Nestcam, Nest Cam, Nest-Hello, Nest-*	Avoids a lone Nest word. Cameras, not the Nest Thermostat BLE row.
Nest Thermostat	BLE company 0x01B5 (Nest Labs); names Nest Thermostat*	Not Google 0x00E0. Nest Temperature Sensors on E / 3rd-gen can hit 0x01B5. Wall units often Wi-Fi-only after setup.
Nest Weave	BLE UUIDs 0xFEAF / 0xFEB0 (Nest Labs Weave-over-BLE)	Field N02QP is Protect 2nd gen (Topaz2, Weave product 0x0009). Same UUID on other Weave cameras/thermostats/sensors. Not Matter 0xFFFF6. Decode fields: FEAF identification block (vendor / product / pairing / device id). §9.6.
NETGEAR	Wi-Fi NETGEAR* / Orbi* plus NETGEAR IEEE OUIs	Renamed SSID still hits on BSSID.
Nissan	Company 0x0BA6; BLE names Nissan / Infiniti	When that company ID is advertised.
Nokia	Nokia Solutions and Networks IEEE OUIs	Gateways / small cells. Not Nokia phones.
Novatel Wireless	Inseego IEEE OUI 28:80:A2. Extra attention filled. Stock bookmark.	Ex-Novatel Wireless. Reported in public-safety vehicle AP work. Same prefix can also hit the Inseego row. Government, municipal, and other corporate fleets likely run some of the same kit. Pattern match, not that agency. Beeps on a new match. Public safety class — used in LE, not exclusive to it.
Nuki	Custom UUIDs a92ee000–a92ee300 / a92ae200; names Nuki*	Keyturner / Ultra / Opener. No SIG company ID.
Omron	BLE company 0x020E; names OMRON* / HEM-* / BLESmart_*	Omron Healthcare cuffs and scales. Not industrial OMRON 0x02D5. Health class.

Signature	Primary rules	Confidence notes
OpenWrt	Wi-Fi OpenWrt*	Factory OpenWrt SSID. No OpenWrt IEEE OUI.
Orbit B-hyve	Names bhyve* / B-hyve*; OUI 44:67:55; UUID FE32	Orbit hose timers. Not Pro-Mark company 0x047F alone. Home IoT.
Osmo	0x08AA model IDs 0x0006–0x0022; OsmoAction* / OsmoPocket* / Osmo360* / OsmoNano* / XtraEdgePro*	DJI Osmo Action / Pocket / 360 / Nano cameras. Not Osmo Mobile gimbals. Not DJI aircraft (those stay DJI). Cameras class. Decode fields: 0x08AA model id (\$9.6).
Oura	Company 0x02B2; name Oura	Ring.
Pacific TPMS	BLE company 0x0E32	Pacific Industrial OEM TPMS. Vehicle class.
Parrot	Names ANAFI* / Bebop*	Parrot drones. Not company 0x0043 (automotive). Disco* not used. Drones class.
Paxton	BLE company 0x0196; names Paxton* / Net2*	Readers / access.
Pebblebee / moto tag	Pebblebee*, moto tag / Moto Tag	Find Hub locators. Name-only.
Penguin	Name / glob Penguin*. Extra attention filled. Stock bookmark.	Flock-family / roadside camera provisioning name. Name-only. Low uniqueness. Beeps on a new match.
PeopleNet	Wi-Fi PNet* plus PeopleNet IEEE OUI 98:5D:46	Fleet ELD / truck hotspot. Not Motive. Vehicle class.
Peplink	Wi-Fi Peplink* / Pepwave* plus Peplink IEEE OUIs	Travel / branch routers. Not MAX-*.
Phone hotspot	Wi-Fi AndroidAP* / Galaxy-* / Galaxy * / Pixel-* / Pixel *	Factory personal-hotspot SSIDs. BSSID usually randomized. iPhone stays Apple Device. DIRECT-* stays Unknown Signature. Custom names do not match. Phones / PCs.
Pigvision	Name / glob Pigvision*. Extra attention filled. Stock bookmark.	Flock-family / roadside camera name. Name-only. Beeps on a new match.
Plaud Note	BLE names Plaud Note*. Extra attention filled. Stock bookmark.	Plaud Note / NotePin AI recorder. Not proof of recording. Beeps on a new match.
Plume	Wi-Fi Plume* / SuperPod* plus Plume Design IEEE OUI 60:B4:F7	SuperPod / HomePass mesh. House SSIDs still hit on that OUI. ISP-branded pods often use Arris / Sercomm / Hitron instead. ISP / routers.
Pokemon GO Plus	Names Pokemon GO Plus*; UUIDs 138C35B6 (Plus +) and 21c50462-... (original)	Nintendo clip-on. Not company 0x0553 or OUI 60:1A:C7 (Joy-Con / Switch). Wearable class.
Porkchop	Names PORKCHOP / M5PORKCHOP; vendor IE 50:52:4B. Extra attention filled. Stock bookmark.	M5PORKCHOP Cardputer or CYD port. CYD remote AP is named PORKCHOP. BACON fake APs brand 50:52:4B. Not Espressif OUI. BLE spam not matched. Beeps on a new match.
Porsche	Company 0x0120; BLE name Porsche; Wi-Fi Porsche_WLAN*	Not Volkswagen 0x011F.
Pwnagotchi	MAC DE:AD:BE:EF:DE:AD; name pwnagotchi. Extra attention filled. Stock bookmark.	Handshake-collector beacon. Beeps on a new match.
RadiaCode	BLE UUID E63215E5-7003-49D8-96B0-B024798FB901; names RadiaCode*	Radiation detectors (101/102/103/Zero). Field 0x77AC is not SIG assigned. Home IoT.

Signature	Primary rules	Confidence notes
Raven / ShotSpotter	Names RAVEN, ShotSpotter, SoundThinking; UUIDs 3100–3500; mfg 0x09C8; OUI D4:11:D6	UUID range and 0x09C8 are the stronger digital fingerprints.
Ray-Ban / Meta glasses	BLE company IDs 0x01AB, 0x058E, 0x0D53; names Ray-Ban / Meta View / Oakley Meta. Extra attention filled. Stock bookmark.	Often Ray-Ban Meta. Same IDs on Quest and other Meta wearables. Not proof of recording. Beeps on a new match.
Rekor	Rekor / Rekor*. Extra attention filled. Stock bookmark.	Highway / transit ALPR. Name-only. Beeps on a new match.
Remote ID	BLE UUID 0xFFFFA (ASTM F3411 / FAA Remote ID)	In-flight digital license plate. DJI, Skydio, Autel, Parrot, HOVERAir, Dronetag / Aerobits / BlueMark modules. Not FIDO FFF9 or Thread FFFB. Wi-Fi NAN often misses on stock Android. Not a tail number. Drones class. Decode fields: Open Drone ID app/counter/message; protocol 2 Basic ID / location / Self ID / System / Operator ID (§9.6).
Reolink	Reolink / Reolink*	Consumer cameras.
Retail LED sign	BLE UUID 56D63956-93E7-11EE-B9D1-0242AC120002	LED message displays. Advertised name is the sign text (FOOD, STORY BAORD), not a product name. Not Earda OUI F0:96:02. Signage class.
Ring	Ring-*, Ring Doorbell / Camera / Setup	Avoids a bare Ring substring.
Rivian	BLE company 0x0941; names Rivian*	Phone-as-key / camp speaker / sensors.
Roku	Wi-Fi Roku IEEE OUIs (BSSID or vendor IE C8:3A:6B); DIRECT-roku* / Roku-*	Streaming stick / Roku TV. Hidden Wi-Fi Direct remote APs hit on the vendor IE even with a randomized BSSID. Not WPS 00:50:F2 or P2P 50:6F:9A. Home IoT.
Ruckus	Wi-Fi Ruckus* / Configure.Me* plus Ruckus Wireless IEEE OUIs	Renamed SSID still hits on BSSID. ISP / routers class.
Ruijie	Wi-Fi @Reyee* / Reyee* / Ruijie* plus Ruijie Networks IEEE OUIs	Reyee campus / SMB APs. Renamed SSID still hits on BSSID. ISP / routers class.
Ruuvi	BLE company 0x0499; names Ruuvi*	Broadcast temp / humidity / pressure / motion tags. Home IoT. Decode fields: Data Format 5 (RAWv2) and Format 3 humidity/pressure/accel/battery. Format 3 temperature is sign-magnitude, not a plain int. §9.6.
Sagemcom	Sagemcom Broadband IEEE OUIs	ISP gateways (Comcast and others).
SALTO	BLE company 0x0199; names SALTO*	Commercial access.
Samsara	BLE company 0x0B6B; names Samsara*	Fleet ELD / trailer / gateway. Vehicle class.
Samsung appliance	Wi-Fi [fridge]* / [oven]* / [range]* / [cooktop]* / [refrigerator]*	Family Hub / range setup APs. Not SmartTags. Home IoT.
Samsung SmartTags	Names SmartTag / Smart Tag / Galaxy SmartTag; UUID FD5A; mfg 0x0075	FD5A is the usual SmartTag service.
Schlage	BLE Allegion 0x013B; UUID FCF4; names SCHLAGE*	Encode and other Allegion BLE.
Schrader TPMS	BLE company 0x0601	AirCheck BLE / trailer / RV. Not Nokia 0x0001 clones. Vehicle class.

Signature	Primary rules	Confidence notes
Sensi	BLE names Sensi*	Emerson / Copeland setup BLE. Not Emerson 0x04DF.
SensorPush	BLE UUIDs EF090000-...090AA9 / ...090AB0; names SensorPush*	HT / HTP temp / humidity. Custom 128-bit services. Home IoT.
Sercomm	Sercomm IEEE OUIs	ISP cable / fiber OEM gateways.
Shokz	BLE names OpenRun / LE-OpenRun / OpenFit / Shokz	Bone-conduction headphones. Not Battery 0x180F and not Qualcomm FD92. Audio class.
Skydio	Names Skydio*	US public-safety / enterprise drones. RID in flight is often Wi-Fi beacon (easy to miss) — UUID FFFA is the Remote ID row. Drones class.
Snap Spectacles	BLE company ID 0x03C2; Spectacles names. Extra attention filled. Stock bookmark.	Snap Spectacles or other Snap BLE. Not proof of recording. Beeps on a new match.
SnapAV	Wi-Fi Control4* / Wattbox* plus SnapAV IEEE OUIs	Control4 / Wattbox home-AV APs. Home IoT class.
Sonos	Company 0x05A7; UUID FE07; name Sonos	Speakers (S41 / S57 LE).
Sony	Company 0x012D; names Sony / WH-1000 / WF-1000	Headphones, TVs, cameras. Not Sony Ericsson 0x0056.
Sophos	Sophos IEEE OUIs	Firewall / AP. ISP / routers class.
Spectrum	SpectrumSetup* / MySpectrumWiFi* / Spectrum Mobile / Spectrum Free Trial	Charter gateway and mobile hotspots. No Charter IEEE OUI.
Starlink	Wi-Fi SSIDs STARLINK* / Starlink*; SpaceX OUI 00:26:12	BSSID often randomized; name is the usual hit.
Subaru	Company 0x0A10; BLE name Subaru	Not Starlink satellite internet.
Synology	Wi-Fi Synology* plus Synology IEEE OUIs	NAS / router APs.
T-Mobile	Wi-Fi TMOBILE-* / T-Mobile*	Home Internet / hotspot. Often HUMAX / Arcadyan / Askey OEM.
Tapo	Tapo / Tapo*	TP-Link cameras.
Target Atrius basket	Apple iBeacon UUID 5993A94C-7D97-4DF7-9ABF-E493BFD5D000; service 0xB1BB	Target shopping-basket / Atrius tags. Two stores: hundreds of unnamed radios, unique major/minor, TX 0xC3. Not Acuity company 0x0346 on those radios. Dual-labels with iBeacon. Retail beacons class.
Tedee	BLE company 0x0725; names Tedee*	Retrofit locks.
Tenda	Wi-Fi Tenda* plus Tenda IEEE OUIs	Consumer APs / routers.
Tesla	Company 0x022B; UUIDs FE96/FE97; names Tesla / Cybertruck; BLE glob S?????????????C (VIN SHA1 phone-key); Apple iBeacon UUID 74278BDA-...; Wi-Fi TeslaGW* / tesla-vehicle / TeslaWallConnector* / Cybertruck*	Phone-key ads are S + 16 hex + C. Tesla also uses iBeacon layout so iOS can find the car — Spectre labels Tesla, not iBeacon. Vehicle class.
Tesla tsTPMS	BLE names tsTPMS*	Tesla BLE tire sensors. Not company 0x022B on this row and not UUID 0x1122. Vehicle class.

Signature	Primary rules	Confidence notes
Tile Trackers	Name Tile; UUIDs FEED, FEDD; mfg 0x00C7	Older Tiles are noisier on name than on UUID. Decode fields: FEED 8-byte rotating private id (not a serial). §9.6.
Toyota	Company 0x0977; BLE names Toyota / Lexus; Wi-Fi TOYOTA* / LEXUS*	When that company ID or factory SSID is advertised.
TP-Link	Wi-Fi TP-Link* / TP-LINK* / Deco* plus TP-Link IEEE OUIs	Not Tapo names. Tapo cameras on a TP-Link OUI can also hit this row.
TRENDnet	Wi-Fi TRENDnet* plus TRENDnet IEEE OUIs	Consumer APs. Renamed SSID still hits on BSSID.
Tuya	BLE company 0x07D0; UUID FD50; names TUYA*	Plugs / lights / cameras / sensors. Not two-letter TY. Decode fields: bound flag and protocol version (UUID bytes encrypted). §9.6.
Uconnect	Wi-Fi Uconnect*	Stellantis in-car hotspot. Vehicle class.
UniFi	UniFi, Ubiquiti, UAP-* (either radio)	Name only. Use UniFi AP when you want BSSID matching. ISP / routers class.
UniFi AP	Wi-Fi UniFi* / UAP-* / UBNT* plus Ubiquiti IEEE OUIs (BSSID or vendor IE)	Campus / city / house APs. ISP / routers class, not Surveillance. Virtual BSSIDs miss the MAC OUI; a Ubiquiti vendor IE still hits.
UniFi Protect	BLE names UVC G3/G4/G6 Instant	Protect Instant cameras in BLE setup. Not AP BSSID OUIs. Custom 16-bit 252A/2529 not used. Surveillance class.
Unknown Signature	ESP_*, ANDROID-, DIRECT-, UNIT- names only	Catch-all for those names.
Utility Inc	Utility, Inc IEEE OUIs 00:09:BC / 00:16:ED. Extra attention filled. Stock bookmark.	Reported in public-safety vehicle AP work. Government, municipal, and other corporate fleets likely run some of the same kit. Pattern match, not that agency. Beeps on a new match. Public safety class — used in LE, not exclusive to it.
Vantiva	Wi-Fi Technicolor* / THOMSON* / Vantiva* plus Vantiva and Technicolor IEEE OUIs	Ex-Technicolor ISP gateways. ISP names after provision often miss.
Verizon	Verizon-* / Fios-* / MyVerizon* plus Verizon IEEE OUIs	Not Verizon Connect / Telematics. Many FiOS boxes are Actiontec OEM.
Verkada	Name / glob Verkada*. Extra attention filled. Stock bookmark.	Cloud cameras / LPR on buildings and some public sites. Name-only. Beeps on a new match.
Volkswagen	Company 0x011F; UUIDs FE30/FE31; names Volkswagen / VW; Wi-Fi My VW*	Skoda / SEAT / Porsche are separate IDs.
WatchGuard AP	WatchGuard Technologies IEEE OUIs 00:01:21 / 00:90:7F	Firewall / AP. Not WatchGuard Video 00:1D:96. ISP / routers class.
WatchGuard Video	OUI 00:1D:96; names WatchGuard / VISTA WiFi / VISTA XLT. Extra attention filled. Stock bookmark.	Public safety class — used in LE, not exclusive to it (government, municipal, and other corporate fleets likely run some of the same kit). WatchGuard Video (now Motorola) body-worn / in-car. Not WatchGuard firewall 00:01:21. Patrol units may stay quiet. Beeps on a new match.
WAVLINK	Wi-Fi WAVLINK* plus Winstars IEEE OUI 80:3F:5D	Consumer APs. ISP / routers class.

Signature	Primary rules	Confidence notes
Winegard	Wi-Fi Winegard* plus Winegard IEEE OUI 00:17:1A	RV / marine Wi-Fi. Vehicle class.
Withings	BLE Withings* / WBS0* / BPM Connect	Body scales and BPM Connect. Not Nokia phones or ISP gateways. Health class.
Wyze	Wyze / WyzeCam / Wyze*	Consumer cameras. Cameras class, not Surveillance.
Xfinity	xfinitywifi / XFSETUP* / Xfinity* plus Comcast IEEE OUIs	Most boxes are Arris / Hitron OEM.
Zyxel	Wi-Fi Zyxel* plus Zyxel IEEE OUIs	Renamed SSID still hits on BSSID.

Flock-related OUI inventory (Flock Safety Cameras, in addition to B4:1E:52)			
70:C9:4E	3C:91:80	D8:F3:BC	80:30:49
B8:35:32	14:5A:FC	74:4C:A1	08:3A:88
9C:2F:9D	C0:35:32	94:08:53	E4:AA:EA
F4:6A:DD	F8:A2:D6	24:B2:B9	00:F4:8D
D0:39:57	E8:D0:FC	E0:4F:43	B8:1E:A4
70:08:94	3C:71:BF	58:00:E3	5C:93:A2
64:6E:69	48:27:EA	A4:CF:12	82:6B:F2

C. Troubleshooting

If the Live display does not match the chapters above, start here. The log and Reports still run even when the list is filtered or quiet. A pattern match is not identity.

Symptom	Likely cause	Action
Permission gate on every launch	A required runtime permission was denied or reset by the OS.	Grant Location (Precise), Nearby Wi-Fi, Bluetooth scan/connect, Notifications. Turn system Location on. Full list: §4.5.2.
I want max hear / the phone gets hot	High performance, Keep screen on, and Faster Wi-Fi use the radios and the display hard.	Expected for a sit. Walk through §4.5. Plug in or use a pack. Drop to Balanced and turn Faster Wi-Fi off when you are done. Empty list is still not "safe."
Zero Wi-Fi rows, BLE works	Between OS scan windows, throttle, Wi-Fi off, or Location off.	Enable Wi-Fi and Location. Read the header: Wi-Fi next Ns or waiting on OS. Last APs should stay held; a new batch arrives about every 30 s on high performance, or ~8 s if Faster Wi-Fi AP scans is on and Developer options Wi-Fi scan throttling is off.
Zero BLE rows after it was working	Samsung parked the scanner.	Watch for BLE cycling or BLE parked - restarting in the header. Keep screen on while you watch. Allow background usage and Unrestricted battery if you leave the app. Toggle intensity only if it stays dead.
UI freeze then both lists empty in a crowd	A plaza of rotating BLE addresses filled memory.	The live set is capped at about 400 radios. Turn logging off if you do not need the file. Prefer Strength list over Hybrid in a dense crowd.
Radar looks empty, list does not	Gone radios are drawn dim on radar, or a Live display filter is hiding them.	Dim blips are radios past Stale / Brief hold. Check that the Live display filter is not Signatures only with no matches.

Symptom	Likely cause	Action
Everything is “gone”	Stale window shorter than the advertisement interval, or the service was killed.	Raise stale to 90–120 s, or raise intensity. Confirm the scan notification is still present. Exempt from battery optimization.
Service dies or BLE dies when the screen locks	OEM battery manager and/or Samsung screen-off BLE policy.	Settings → Keep screen on while watching. Settings → Allow background usage and Unrestricted battery. Leave Spectre in Recents (do not swipe it away) if you want it to keep scanning.
Scan / beeps continue after I kill the app	The scan service is still running.	Swipe Spectre out of Recents, or tap Stop on the scan notification. The Home button leaves scanning running.
Cannot delete a signature	Delete is on the editor page, not the list.	Open the signature and tap Delete signature at the bottom. Confirm. Restore defaults if you removed a built-in by mistake.
No Decode fields row on a signature	That row is Wi-Fi-only (hidden SSID / vendor IE / radio=Wi-Fi), or you are on the Live display not the editor.	Open a BLE signature (Ruuvi, Remote ID, Govee, ...). Decode fields sits under Add rule. §9.6.
Decoded fields missing on detail	The signature has no map, this advertisement is encrypted/short, or the gate (Only if) did not match.	Open the signature → Decode fields. A note on detail means a map exists but this packet did not fit. Usual miss: offset 0 counted the company ID. Encrypted Fitbit / Find My ads stay hex. §9.6.1 / §9.6.5.
Preview is empty but the hex looks right	Offset includes the company ID, wrong endian, or Only if hex does not match this packet.	Count pairs on detail Raw payload (first pair = 0). Try BE if the spec is big-endian. Check Only if length vs Hex length. Open stock Ruuvi and copy the card shape. §9.6.3–§9.6.4.
Custom name edit is gone on detail	The address is random / privacy (IEEE local bit or Android Random type).	Expected. The control is hidden so you do not name a MAC that will rotate. A name you already saved still shows. Bookmark can still watch this MAC.
I want a Cameras / Drones / Surveillance chip	Those class sits are not stock presets.	Filters → Show only, pick the class chip, Save current as... Restore default signatures & presets puts the short stock set back and wipes custom chips.
Filters chips look dead while Pause is on	Pause freezes only the Live display.	Expected. Filters and Settings still take taps. The frozen list does not change until you tap Live display on the Live display to run it again.
I deleted a preset but the Live display still looks the same	Delete removes the snapshot, not the filter that is on.	Expected. Apply another chip or tap Reset filter to change what the Live display shows.
Cannot find Title line / Subtitle line / Frequency under RSSI	Display is collapsed (chevron up), or you are looking on Settings.	Live display → tap the tune / chevron (top right). Title and Subtitle are dropdowns under Brief hold. Frequency is a switch; channel · MHz sit under the RSSI, not on the identity line.
List still shows “Apple, Inc. · AirTag...”	Display → Subtitle is Name + type, which uses the same type guess as detail.	IEEE vendor is on the detail page. To stop seeing the guess, set Subtitle to Advertised name or None.

Symptom	Likely cause	Action
A "!" on a Live display row	A matched signature has Extra attention filled (stock: Hobby BLE serial, Axon, WatchGuard Video, Ray-Ban / Meta glasses, Snap Spectacles, Fieldy, Plaud Note, Hak5 Pineapple, Flipper Zero, Pwnagotchi, Marauder / Deauther, Porkchop, Cradlepoint, AirLink, Compex, Novatel Wireless, Utility Inc, Flock, Penguin, Pigvision, FS Ext Battery, Genetec AutoVu, Rekor, Motorola Vigilant, Verkada, Avigilon, Axis, Hikvision, Dahua).	Open detail and read Extra attention. Pattern match, not identity. Meta company IDs also match Quest. Fieldy / Plaud Note are recording wearables, not proof someone is recording you. Camera / ALPR rows are roadside or public CCTV / plate readers, not that pole. Hide that family on Filters if it is local noise. Not the decode hexagon and not the phosphor alerted bell.
A bell on a Live display row	That radio already fired a watchlist alert this session (beep / voice / flash).	Expected. It stays until you leave Spectre. Extra attention is the red "!". Newest alert sorts by the same event. Radar keeps a phosphor ring on that blip after the ping.
Hexagon on a signature chip, no Decoded fields on detail	The hexagon means that signature has a map, not that this advertisement parsed. Govee lights share the Govee name with hygrometers; lights usually only send a name.	Open detail: a note means the map did not fit this packet. Hygrometers are H5074/H5075 (0xEC88) or H510x (0x0001). Display → Signature names off hides the hexagon. To drop a dummy map: Signatures → row → Decode fields → Remove decode map. \$5.4, \$9.6.
Phones / PCs is a wall of Fast Pair	Those are mostly already-paired account-key ads (buds or phones on an account), not someone pairing.	Filters → Hide Fast Pair account-key. Pairing-mode still shows (chip Fast Pair pairing). Hide selected Fast Pair if you want none of them. Pocket Androids without Fast Pair stay unmatched; Apple Device is the loud Continuity crowd. \$9.5.
I never see Android under Phones / PCs	Apple Continuity is always-on; most Android phones do not broadcast a stable "I am a phone" payload.	Expected. Fast Pair is the Android-shaped chip (often unnamed). Google is Pixel / Chromecast only. Phone hotspot only while the phone is an AP with a factory SSID. Quiet pocket Androids are unmatched BLE. \$9.5.
List is a wall of text / I want less on each row	Display extras are on, or Subtitle is still MAC.	\$5.3 / \$12.11: Subtitle None, bars/Frequency/first-last off. That does not hide radios. Then Filters if you still have the wrong set.
Cannot find Mark seen / Reset seen	Those buttons appear only while New detections only is on.	Turn on Filters → New detections only. Mark seen and Reset seen appear on the Live display, just above the tabs.
Cannot find Start over	It is on the Live display, not Filters.	Turn on Filters → Moving with you. Start over sits above the tabs. It clears the GPS path, not the live list.
Watchlist never fires	Alerts off, both Beep and Voice off, or the hit is hidden by New detections only / another filter.	Enable Watchlist alerts, then Beep, Voice, or both. Confirm the family is bookmarked and the row would show on the Live display. Tap Test alert; raise media volume. Jump works with any of those cues. System notification is optional and off by default.
Watchlist beeps but does not speak	Voice on watched signature is off, media volume is down, or the phone has no text-to-speech pack.	Settings → Voice on watched signature. Raise media volume. Tap Test alert — Class says "finder tags," Signature says "Apple AirTags," Class + signature says both (after the pip if Beep is also on). Hunt never speaks.

Symptom	Likely cause	Action
Trackers / Surveillance shows nothing	Show only that class, and no matching radios are on the air.	Empty Live display means none of that class is in earshot (a bag AirTag is the check for Trackers). Hide these on another class does not mute labels.
Signatures only switch does nothing under Show only	Show only already hides unmatched radios.	The switch is dimmed while Show only has a class or selected signatures picked. Turn Show only off to use Signatures only, or use Hide these if unmatched radios should stay.
Moving with you empty after Trackers / Show only	Class Show only or Signatures only was still on. Finder tags rotate MACs, so they often fail co-travel, and unmatched radios were hidden.	Tap the Moving with you preset, or turn the switch on. Either one clears Show only and Signatures only. Hide these stays if you were hiding a bag tag. The path still needs about 50 m.
I hid AirTags, then they came back	Class Hide these or Hide selected was turned off. Hiding only runs while that mode is on.	Turn Hide these (Finder tags) or Hide selected back on. Reset filter is what forgets the picks.
Pause, then detail says left range	The radio aged out after you opened detail from a running list.	Pause first, then tap the row. Detail uses the frozen snapshot.
Only one signature chip on radar	Radar labels the blip with the first match.	Expected. List / Hybrid / Timeline show up to three; detail lists all.
New signature labels half the cafe	Rule too broad (OUI of a common chip, or RADIO_KIND in OR).	Hide it on Filters, or delete a custom row. Tighten to MAC, name, UUID, or manufacturer data.
Create-from-device matches all BLE	The signature includes a Radio kind rule with match-any.	Delete that signature. Long-press the radio again and keep MAC, name, or UUID rules only. Do not add Radio kind alone.
Export share sheet empty / fails	No log lines yet, or the viewer cannot take a content URI.	Wait for a few observations. Share to Files or Drive, not to an app that rejects text/plain.
Debrief looks empty / thin	Live display has few radios in the last 15 minutes, unnamed BLE already evicted (~3 min), or a drive already filled the ~400 cap so earlier streets dropped.	Scan a few minutes first. Text and PDF use the same memory snapshot, not the on-disk log. On a trip, tap Debrief every 10–15 min or at stops (\$11.4.1) — radios moving with you stay in each file. Use Share log for the full hour.
Cannot find Share log / Save to SD / Debrief / Signature candidates / clear log	Those buttons are on the Reports tab.	Bottom bar → Reports. Signature candidates is under Catalog. The one-radio version is Signature family on device detail. Settings still has GPS, Online place names, and logging on/off.
Cannot find Debrief / AI Export	Those buttons are on the Reports tab.	Bottom bar → Reports. Settings still has Tag GPS, Online place names, and logging on/off. One-radio AI Export is on the device-detail page, not Reports.
Signature candidates is empty	Logging was off, the log is short, or leftover radios are randomized / house-like / one-off MACs.	Logging on in Settings, sit a while, then tap Signature candidates again. A family needs two distinct radios sharing a unique ID. One loud unmatched MAC is Create from device, not a family.
No vendor-IE families (Roku-class hidden APs)	Older log rows have no vendor_ie column.	Expected until new Wi-Fi packets are written after this build. Name globs and stable OUIs still mine from old parts. Share log of a new sit if you need IEs off-phone.
I Saved a candidate but it is still on the list	The draft did not match those radios (rule too tight, or you changed it), or Save did not finish.	Open Signatures and confirm the row is there. Save from a candidate draft returns to the list and re-runs it; that family should drop if the new rule hits. Cancel leaves the list unchanged.

Symptom	Likely cause	Action
Detail AI Export vs Reports AI Export	Two different prompts.	Detail = this radio (what is it?). Reports = onboard Debrief plus working data, asking for a statistical addendum (not a rewrite). Do not treat a one-radio decode as a following test.
Hunt Reset / Back to detail are hard to reach	The cue text grew and pushed the buttons off screen.	Reset, Back, then Beep / Vibrate sit at the bottom of Hunt. Cue and hint stay a fixed height so the dBm number does not jump.
Hunt says Further but I am walking toward it	A wall, car, crowd, or floor is in the path; or the phone grip changed.	Open a doorway or walk around metal and watch whether Closer returns. Reset this hunt after a body-block turn. Do not convert dBm to feet. §12.13.
Hunt went Quiet / Gone	No ads for ~8 s, or the BLE address rotated off the Live display.	Wait. If Gone, back to the Live display and pick the new row if it is still the same family. Randomized MACs do not stitch.
Where you were has coords but no street	Online place names is off, or on with no internet / no geocoder.	Expected. Stays still print lat/lon. Distance does not need internet. Turn the switch off if you want no lookup attempt.
Last fix or Debrief shows "masked" instead of lat/lon	Privacy mode is on.	Expected. Settings → Privacy mode hides GPS coordinates on the screen and in sit reports. Street names are omitted too. The log still has lat/lon. A TAK / CoT feed is paused. Turn Privacy mode off when you need the pin or the overlay.
The whole UI went red / I want green chips back	Night mode is on.	Settings → Night mode Off. Dark theme stays as you left it. Restore defaults also turns Night mode off. Fig. 6a.
ATAK map stays empty	TAK feed off, Privacy mode on, wrong LAN / multicast filtered, or no qualifying radio with a pin.	Settings → TAK / CoT feed On, Privacy mode Off. Extra attention and Payload location on. Same Wi-Fi as ATAK; if 239.2.3.1 never arrives, set Host to the ATAK IPv4. Confirm the Live display shows a "!" or Remote ID. Heard-here also needs GPS tagging and a live fix. §5.8, §12.15.
Remote ID is on the Live display but not on ATAK	Payload location chip off, no Location message yet, or 0,0 / invalid coords.	What to send → Payload location On. Wait for an ASTM Location message (type 1, protocol 2); Basic ID has no lat/lon but a previous Location sticks this session. 0,0 is rejected.
TAK pins sit on me, not on the other radio	Heard-here: that family has no advertised lat/lon.	Expected for Extra attention (Axon, glasses, Flipper, ...). Remote ID Location is advertised position. GPS tagging off stops heard-here only.
ATAK filled with café APs	All signatures is on.	Turn All signatures off. Field default is Extra attention + Payload location.
TAK marker vanished	CoT stale (~120 s) after the radio left or scanning stopped.	Expected. Spectre does not send delete events. Keep scanning to refresh. A rotated BLE MAC is a new uid.
GPS path stays 0 m while I drive / Moving with you says keep moving	The phone is not giving a live fix.	Turn on Tag detections with GPS and high-accuracy Location. Scanning must be running. Wait until the path meter is not 0, then walk or drive.
Second iPhone in the car did not show under Moving with you	iOS rotates the BLE address, so Spectre sees a new radio with an empty GPS trail.	Expected. Use a tag with a stable MAC (AirTag/Tile in the bag) as the confidence check. Phones will not stitch as one follower.

Symptom	Likely cause	Action
Moving with you lists house APs after I get home	The radio did not actually co-travel; it was only heard at the door.	A match must stay loud on the trail (about -75 dBm or stronger) with a GPS path that moved, and still be heard. House APs you meet on arrival should stay off. Bag and car tags should stay on.
Moving with you flashes on the highway	Passing radios only overlap for a few seconds, so they fail the trail-moved gates. A tag in the car should not flash: "still here" grows with speed (about 400 m of phone travel for BLE at 55 mph, longer for Wi-Fi) so a few quiet seconds are not a miss. §8.5.	Keep Tag detections with GPS on. Passing phones and roadside APs will still appear and leave — that is the filter working. A tag in the car should stay; if it does not, Start over after you have ~50 m of path and confirm the tag is still advertising.
LE color / "What this looks like" flips on and off	BLE advertisements rotate payloads (for example iBeacons, then Nearby).	Spectre keeps manufacturer records per company and type for that radio, so a hit should stay labeled when the payload rotates.
Debrief PDF will not open in the other app	The other app did not get read permission, or it rejects application/pdf.	Share to Files, Drive, or a PDF viewer.
Restore defaults wiped custom signatures	Restore rewrites config.json from the catalog.	Expected. Export signatures from Settings before Restore if you want a backup. Import that JSON to get customs back (stock ids merge extra rules; they are not cloned).
Import signatures says not a Spectre pack	The file is a log, a photo, or config.json, not an exported pack.	Export from Settings → Export signatures. The file starts with format spectre-signatures. Logs stay on Reports.
Import added nothing	Every row in the pack already matches an id or the same rules on this phone.	Expected for a second import of the same pack, or two phones on the same stock catalog. Only new customs and extra rules on stock rows are added.
Counts freeze	Process still up but scans failing, or you are looking at a stale filter.	Swipe the scan notification. If it is gone, relaunch. Reset filter. Reboot radios (Airplane 5 s).

D. Document control

Field	Value
Product	Spectre
Author	Off Grid Pete
Copyright	Copyright (c) 2026 Off Grid Pete LLC. All rights reserved.
Instagram	@OffGridPete
X	@OGridPete
Document	User Manual and Technical Documentation
Application ID	app.spectre
Software version	1.2.10 (versionCode 13), field build of 13 September 2026
Document version	1.8.140
Document date	13 September 2026
License	SPECTRE LICENSE (see LICENSE)
Platform	Android 10+ (minSdk 29), targetSdk 35

Field	Value
Classification	Unclassified. Operationally sensitive if filled with site logs.

E. About this document

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